Finiteness of planar central configurations of five bodies for all positive masses: On Smale's sixth problem

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Abstract

Combining local boundary analysis, algebraic curves and the computational statements specified in the appendix, we prove that the planar Newtonian five-body problem has finitely many central configurations up to rotation, translation and scaling for every choice of positive masses. This gives the five-body statement of Smale's sixth problem. We use the singular-sequence classification of Albouy and Kaloshin and resolve the corresponding boundary equations through weighted blow-up charts, rank calculations and local compatibility conditions. For the two remaining diagrams, we prove that the product Δ of all squared mutual distances vanishes. On a hypothetical solution curve containing a positive arc, a pole of a four-product therefore gives a zero of Δ. A pole of Δ elsewhere on the completed curve, together with $\Delta^4 = \prod_{i<j} p_{ij}$, then contradicts the same local vanishing result. The computational statements for Diagrams 2, 3 and 6 involve exact algebra, interval arithmetic and ball arithmetic.

Keywords: Central configurations, planar five-body problem, Smale's sixth problem, singular sequences, algebraic curves, computer-assisted proof

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1 Introduction

The finiteness of central configurations asks whether fixed positive masses admit only finitely many shapes generating relative equilibria of the Newtonian n-body problem. We study the planar five-body case. Our argument combines the singular-sequence classification of Albouy and Kaloshin with local analysis of the normalized equations and a product identity on an algebraic curve.

For masses $m_1, \dots, m_n > 0$, a collision-free configuration $q = (q_1, \dots, q_n) \in (\mathbb{R}^2)^n$ is a *central configuration* if

$$\lambda(q_k - c) = \sum_{l \neq k} m_l \frac{q_l - q_k}{|q_l - q_k|^3}, \qquad k = 1, \dots, n, \qquad c := \frac{\sum_{j=1}^n m_j q_j}{\sum_{j=1}^n m_j},$$

for some $\lambda \in \mathbb{R}$. We identify configurations related by translation, rotation and positive scaling. The question is whether the resulting set is finite for every fixed positive mass vector.

Background and development of the problem

Central configurations determine motions of constant shape: rigid rotation gives relative equilibria, while a time-dependent scale and rotation give homographic motions. Their classification therefore connects the geometry of the configuration space with the dynamics of the Newtonian equations; see Wintner [Win41]. This connection also underlies the stability analysis of elliptic relative equilibria, for example for the $(1+n)$-gon configurations studied by Hu, Long and Ou [HLO20].

The classical solutions of Euler and Lagrange give the five classes for three bodies. Moulton [Mou10] established the exact number $n!/2$ of collinear classes for arbitrary positive masses. The general finiteness question appears in Wintner [Win41] and became Problem 6 in Smale's list of mathematical problems for the twenty-first century [Sma98]. Hampton and Moeckel [HM06] proved finiteness for four positive masses. Albouy and Kaloshin [AK12] subsequently introduced an analysis of singular sequences that gives another proof for four bodies and a five-body theorem

with an algebraic exceptional set. The four-body theory was further extended to arbitrary nonzero real masses by Yu and Zhu [YZ26].

After fixing the moment of inertia, Shub [Shu71] proved that central configurations for fixed positive masses stay away from collisions. Using analytic continuation, Xia [Xia91] obtained exact counts for certain open sets of positive mass vectors. In a complementary direction, Hampton and Jensen [HJ15] proved that a fixed planar configuration of $n-1$ masses admits only finitely many extensions to a central configuration after a prescribed additional mass is added. Further questions on central configurations, symmetry, stability and singularities are collected in [ACS12].

For five bodies, the theorem of Albouy and Kaloshin states that there is a closed algebraic subset A of codimension two in the mass space such that

$$m \in \mathbb{R}^5_{>0} \setminus A \quad \Longrightarrow \quad \#\{\text{positive normalized central configurations at } m\} < \infty.$$

Their method associates one of sixteen two-coloured diagrams with each singular sequence of complex normalized solutions. The leading equations impose mass relations, and two singular limits on the same algebraic curve yield the codimension-two restriction. Jensen and Leykin [JL25] developed a tropical geometric method and completed the computation for $n \leq 5$, recovering generic finiteness. Chang and Chen [CC24, CC25] organized the singular-sequence method into symbolic procedures for determining diagrams, orders and mass relations, with applications to the planar six-body problem.

In Part I of their study of the planar six-body problem, the authors [CT26] establish a reduction from an infinite family of positive normalized central configurations to a singular germ with an unbounded seven-product. Their first-limit diagram analysis confines the mass vector to a proper algebraic exceptional set. This work extends the curve reduction and local singular-limit methods used here to the six-body setting.

Complementary approaches describe particular mass vectors and geometric classes. Lee and Santoprete [LS09] classified the isolated central configurations of five equal masses using homotopy continuation and interval verification. Moczurad and Zgliczyński [MZ19] obtained complete computer-assisted lists for $n = 5, 6, 7$ equal masses; their later work [MZ26] gives exact counts for selected five-body mass vectors on exceptional loci. Chen and Hsiao [CH18] derived geometric restrictions on strictly convex five-body central configurations, while Deng and Hampton [DH23] gave a computer-assisted proof of finiteness within the equilateral pentagonal class. These results develop the mass-specific and geometric aspects of the same classification problem.

The continuing interest in central configurations also encompasses their geometry, symmetry and uniqueness. Albouy, Fu and Sun [AFS08] proved that a convex four-body central configuration is symmetric across a diagonal if and only if the two masses on the other diagonal are equal. Santoprete [San21a, San21b] proved that each of the co-circular and trapezoidal four-body classes contains at most one central configuration for each cyclic ordering of the masses. Hampton [Ham05] constructed stacked planar five-body central configurations containing a Lagrange equilateral three-body subconfiguration. Chen and Hsiao [CH12] studied convex configurations that are not strictly convex. Sun, Xie and You [SXY23] gave a computer-assisted proof of uniqueness for convex four-body configurations with a prescribed order on a closed domain of positive mass vectors. Recent work examines four-body kite configurations [Rob25], symmetric concave four-body configurations with two pairs of equal masses [LX26], rhomboidal five-body configurations [CC26], and convex pentagons with three right interior angles [FGMS26]. Tang and Zhu [TZ24] derive nonexistence

criteria for centered co-circular configurations for several classes of unequal masses under power-law potentials. These geometric and symmetry questions describe how central configurations are organized within the solution set.

The local structure of this set is another active direction. Hampton [Ham24] introduced twist-vector coordinates on the tangent space to planar configurations and applied them to four-body central configurations; Albouy and Sun [AS25] study the eigenvalues associated with a central configuration, and Sun, Xie and You [SXY25] investigate degeneracy of planar central configurations. Natrup, Wang and Wang [NWW25] derive upper and lower bounds for the number of nondegenerate planar central configurations for homogeneous potentials. Together, these studies connect the defining equations and their linearization with the geometry and finiteness of their solutions.

The dynamics generated by central configurations provides a further source of interest. Quantitative linear stability of planar elliptic relative equilibria is studied by Hu, Ou and Sun [HOS25], while the $(1+7)$-gon case and spectral instability of regular n-gon elliptic relative equilibria are treated in [OS23, OW25]. Hu, Ou and Tang [HOT23] use index theory for spatial elliptic relative equilibria. Related variational questions include the Morse index of homothetic motions [OP25] and index theory for collision, parabolic and hyperbolic solutions [HOY21]. These developments connect central configurations with stability questions and variational methods in celestial mechanics.

Recent preprints of Tomar [Tom26] and Liu and Liu [LL26a] address finiteness for all positive masses in the planar five-body problem. Both use the singular-sequence framework of Albouy and Kaloshin. Tomar combines valuation balance with exact rational certificates and separate counts for special mass vectors. Liu and Liu combine a lower bound for the potential at physical configurations with asymptotic estimates and local analytic reductions. For central configurations of affine dimension $n-2$, Moeckel [Moe01] proved generic finiteness. In the spatial five-body problem, Hampton and Jensen [HJ11] obtained finiteness outside explicitly given exceptional mass conditions. For configurations of exact affine dimension three, Liu and Liu [LL26b] use the Dziobek equations and local analysis of zeros and poles on a compact complex curve.

Main result

We obtain the following statement. Positive normalized configurations are defined by the polynomial system in Section 2.

Theorem 1.1. *Assume the computational statements listed in Appendix A. For every $(m_1,\dots,m_5)\in\mathbb{R}^5_{>0}$ there are finitely many positive normalized central configurations of the planar five-body problem. Equivalently, the planar five-body problem has finitely many central configurations up to rotation, translation and scaling.*

The central feature of our proof is the use of the product of all squared mutual distances. Writing $r_{kl}^2 := z_{kl}w_{kl}$ in the complex coordinates of Section 2, we set

$$\Delta := \prod_{k<l} r_{kl}^2, \qquad p_{ij} := r_{ij}^2 r_{kl}^2 r_{lh}^2 r_{hk}^2, \qquad \{k,l,h\} = \{1,\dots,5\}\setminus\{i,j\}.$$

We use the four-products of Albouy and Kaloshin, recalled in Section 2. The product identity converts the local distance estimates into a global contradiction on the completed solution curve. We now explain this step and the local analysis that makes it possible.

Technical difficulties and contributions

The proof has three main components.

(1) **Representing every singular branch in local coordinates.** The diagrams record leading orders, while the equations also couple distances that vanish or diverge at different rates. We construct weighted blow-up charts for Diagrams 1–4 and 6–8 and prove that every corresponding Puiseux germ has a finite boundary limit in its chart. The rules and estimates of [AK12, §3] give the required order inequalities. This converts the analysis of singular sequences into equations and rank conditions at finite boundary points. We also derive the mass relations used in the local arguments.

(2) **Resolving degenerate boundary points.** At boundary points where the Jacobian loses rank, the first nonzero compatibility condition determines whether a solution branch can enter the interior. We compute these conditions through local expansions and Lyapunov–Schmidt reduction. The resulting arguments exclude Diagrams 1, 2, 4, 6 and 8 for complex germs. For Diagram 7, the inertia identity combines with an unbounded four-product to give $I \to 0$, whereas a curve through a real configuration satisfies $U = I = c > 0$. For Diagram 3, analytic continuation gives

$$\delta_{15} + 2\delta_{13} = 0$$

on the irreducible curve, which excludes a positive point. Thus identities on the whole curve connect the complex boundary analysis with the real configurations counted in Theorem 1.1.

(3) **Closing the argument with the product of distances.** At a boundary point P carrying a germ of Diagram 5 or 9, we prove

$$\operatorname{ord}_P \Delta > 0.$$

For Diagram 5 this follows from force-order estimates for the shrinking triangle and the two outer bodies, together with the nondegeneracy transverse to the triangle's critical manifold. For Diagram 9 it follows from the distance estimates of [AK12, §8.9]. On a hypothetical irreducible curve containing a positive arc, some four-product has a pole at a point P of the smooth projective completion. The diagram exclusions leave Diagrams 5 and 9, so $\operatorname{ord}_P \Delta > 0$. Since Δ is nonzero on the affine curve, it is then nonconstant and has a pole at another point Q. The identity

$$\Delta^4 = \prod_{i<j} p_{ij} \quad \Longrightarrow \quad 4 \operatorname{ord}_Q \Delta = \sum_{i<j} \operatorname{ord}_Q p_{ij} < 0$$

forces a four-product to have a pole at Q. Applying the same local results gives $\operatorname{ord}_Q \Delta > 0$, a contradiction. This argument uses the vanishing in Diagrams 5 and 9 directly to complete the proof for the exceptional masses.

The computational statements enter the analysis of Diagrams 2, 3 and 6 through exact polynomial identities, Taylor coefficients, and interval and ball arithmetic over the parameter sets. Appendix A records the statements and the associated program modules. The local arguments for the other diagrams are given explicitly in the text. We use the singular-sequence classification of [AK12, Fig. 11], its rules and estimates in [AK12, §3], and its four-product boundedness argument for Diagrams 10–16. Section 2 states these inputs, and Section 4 treats Diagram 9 through its distance estimates.

Organization

Section 2 introduces the normalized equations and the singular-sequence framework. Section 3 passes from an infinite set of positive configurations to an algebraic curve and a pole of a four-product. Section 4 states the local boundary results, proves the Diagram 9 estimate and deduces Theorem 1.1 from the product identity. Sections 5–11 establish the diagram statements. Appendix A collects the computational statements used in the proof.

2 The framework of Albouy and Kaloshin

We express normalized central configurations as a complex algebraic set and record the quantities preserved along its curves. The singular-sequence classification then reduces behaviour at infinity to finitely many diagrams, which we analyse in the following sections.

2.1 Normalized central configurations

Fix masses $m = (m_1, \dots, m_5) \in \mathbb{R}^5_{>0}$. Following [AK12, §2] we use the complex coordinates $z_k = x_k + iy_k$, $w_k = x_k - iy_k$ and inverse distances $\delta_{kl} = \delta_{lk}$, and write $z_{kl} = z_l - z_k$, $w_{kl} = w_l - w_k$. The normalized system (system (4) of [AK12], written as system (10) there) is

$$\begin{gathered} z_k = \sum_{l\neq k} m_l\,\delta_{kl}^3\,(z_k - z_l), \qquad w_k = \sum_{l\neq k} m_l\,\delta_{kl}^3\,(w_k - w_l) \qquad (k = 1, \dots, 5), \\ \delta_{kl}^2\, z_{kl}\, w_{kl} = 1 \quad (1 \le k < l \le 5), \qquad z_{12} = w_{12}. \end{gathered} \tag{2.1}$$

Its zero set $A_m \subset \mathbb{C}^{20}$ is the set of *normalized central configurations*. A solution is *real* if $w_k = \bar{z}_k$ for all k and *positive* if moreover $\delta_{kl} > 0$ for all $k < l$ [AK12, Def. 2]. Positive normalized central configurations are in two-to-one correspondence with central configurations modulo rotation, translation and scaling, so Theorem 1.1 asserts that A_m has finitely many positive points. The map $\sigma(z, w, \delta) = (\bar{w}, \bar{z}, \bar{\delta})$ preserves (2.1) and fixes the real solutions. System (2.1) is symmetric under $z \leftrightarrow w$. A permutation of the bodies, with the masses permuted, preserves the equations apart from the choice of gauge. After relabelling, write the coordinates again as (z, w, δ). A rotation $(z, w) \mapsto (\lambda z, \lambda^{-1} w)$ with $\lambda^2 = w_{12}/z_{12}$ restores the gauge $z_{12} = w_{12}$. It preserves U, I, every r_{kl}^2, and the diagram. Along a germ, if $\operatorname{ord}\lambda = a$, it adds a to all z-orders and subtracts a from all w-orders. Thus

$$\operatorname{ord}(\lambda z_{ij}) - \operatorname{ord}(\lambda z_{kl}) = \operatorname{ord} z_{ij} - \operatorname{ord} z_{kl}, \qquad (\lambda z_{ij})(\lambda^{-1} w_{ij}) = r_{ij}^2.$$

We implement these symmetries on finite algebraic covers, with a finite ramification of the germ parameter when required.

More precisely, on a relabelled irreducible curve Γ consider $\widetilde{\Gamma} = \{(q, \lambda) \in \Gamma \times \mathbb{C}^* : \lambda^2 z_{12}(q) = w_{12}(q)\}$. The distance constraints make z_{12} and w_{12} invertible regular functions, so this is a finite unramified cover, and every irreducible component maps onto Γ. Any chosen germ lifts after a finite ramification. Its component maps by the rotation to an irreducible algebraic curve after taking the Zariski closure: some squared distance is nonconstant by Lemma 3.1, and remains nonconstant under the cover and the rotation. If Γ contains a positive arc, each component contains a local lift of that arc; there $|\lambda| = 1$, so its rotated image is again a nonconstant positive arc. The same

argument preserves the existence of a real point. All squared distances, four-products, U and I are unchanged, and a pole of a four-product remains a pole on a lifted germ.

2.2 Potential and inertia

The potential and inertia are

$$U := \sum_{k<l} m_k m_l \delta_{kl}, \qquad I := \sum_k m_k z_k w_k.$$

We show that both have the same constant value on each irreducible curve of solutions. This converts the local limits computed below into restrictions on the entire curve.

Lemma 2.1. *Every solution of (2.1) satisfies $U = I$, and U is constant on every irreducible curve $\Gamma \subset A_m$.*

Proof. Let $B \subset \mathbb{C}^{20}$ be the set where $\delta_{kl}^2 z_{kl} w_{kl} = 1$ for all $k < l$, and let $f = 2U + I$. On B no $z_{kl} w_{kl}$ vanishes, so B is smooth, (z, w) are local coordinates on B, and $d\delta_{kl} = -\frac{1}{2}\delta_{kl}^3 (w_{kl}\, dz_{kl} + z_{kl}\, dw_{kl})$. Hence, on B,

$$\frac{\partial f}{\partial z_k} = m_k \Big(w_k - \sum_{l \neq k} m_l \delta_{kl}^3 (w_k - w_l) \Big), \qquad \frac{\partial f}{\partial w_k} = m_k \Big(z_k - \sum_{l \neq k} m_l \delta_{kl}^3 (z_k - z_l) \Big),$$

so every point of A_m is a critical point of $f|_B$. The scaling $(z, w, \delta) \mapsto (sz, sw, s^{-1}\delta)$ preserves B, and hence

$$0 = \frac{d}{ds}\big(2s^{-1}U + s^2 I\big)\Big|_{s=1} = -2U + 2I$$

at every such point. Therefore $U = I$ on A_m. Let $\Gamma \subset A_m$ be an irreducible curve. At a smooth point of Γ the tangent line lies in the tangent space of B, on which df vanishes; so f is locally constant on the smooth locus of Γ, which is connected, and hence constant on Γ. As $f = 3U$ on A_m, U is constant on Γ. □

The identity $U = I$ also appears in [AK12, p. 541]; in fact, $U(A_m)$ is finite by [AK12, Lemma 2]. We will also use the following consequence of constructibility to produce singular sequences.

Fact AK1 ([AK12, Lemma 1])**.** *If $X \subset \mathbb{C}^N$ is a closed algebraic subset and f a polynomial, then $f(X)$ is finite or the complement of a finite set ("f is dominating").*

2.3 Singular sequences and diagrams

We recall how the dominant coordinate and force terms determine the diagram of a singular sequence. This description will let us read the diagrams from the orders of Puiseux series. Let $\mathcal{Z} = (z_1, \dots, z_5, Z_{12}, \dots, Z_{45})$ and $\mathcal{W} = (w_1, \dots, w_5, W_{12}, \dots, W_{45})$ with $Z_{kl} = z_{kl}\, \delta_{kl}^3$ and $W_{kl} = w_{kl}\, \delta_{kl}^3$; these are, up to mass factors, the terms of the equations (2.1). Let $Z = \max_p |\mathcal{Z}_p|$ and $W = \max_q |\mathcal{W}_q|$. A *singular sequence* [AK12, Def. 3] is a sequence of normalized central configurations with Z or W unbounded, extracted so that the maximal components are fixed and $Z^{-1}\mathcal{Z}$, $W^{-1}\mathcal{W}$ converge. Its *diagram* has a z-circle at k if $\lim z_k/Z \neq 0$, a z-stroke between k and l if $\lim Z_{kl}/Z \neq 0$, and likewise in the second colour [AK12, §3]. After multiplying z by $a > 0$ and w by a^{-1} one may assume $Z = W = \epsilon^{-2}$ with $\epsilon \to 0$ [AK12, p. 547]. We write $a \prec b$, $a \preccurlyeq b$, $a \approx b$, $a \sim b$ as in [AK12, §3].

Fact AK2 ([AK12, §3]). *Rules 1a–1e and 2a–2e and Estimates 1–2 of [AK12, §3] hold for every singular sequence.*

Fact AK3 ([AK12, Prop. 1]). *Proposition 1 of [AK12] (localization of the origin in a cluster) holds.*

Fact AK4 ([AK12, §7]). *Every singular sequence of the five-body problem approaches, after renumbering the bodies and possibly exchanging z and w, one of the sixteen diagrams of [AK12, Fig. 11].*

For pairwise distinct i, j, k, l, m with $\{i, j, k, l, m\} = \{1, \dots, 5\}$ the *four-product* is $p_{ij} = r_{ij}^2 r_{kl}^2 r_{lm}^2 r_{mk}^2$, where $r_{kl}^2 = z_{kl} w_{kl}$.

Fact AK5 ([AK12, Prop. 8]). *Along a singular sequence approaching one of Diagrams 10–16 every four-product is bounded.*

Facts AK4 and AK5 leave Diagrams 1–9 whenever a four-product is unbounded. Lemma 4.2 gives the distance estimates needed for Diagram 9. For Diagrams 1–6 and 8, Propositions 5.10, 6.6, 7.2, 8.1, 9.5 and 11.4 derive the mass conditions from the corresponding limiting equations.

3 Reduction to germs with an unbounded four-product

To apply the diagram classification, we seek an algebraic curve containing a positive arc and a branch at which a four-product has a pole. We first prove that fixed squared distances determine only finitely many normalized solutions. A monomial identity then ensures that every curve has a nonconstant four-product.

Lemma 3.1. *Let $s \in (\mathbb{C}^*)^{10}$. Only finitely many points of A_m satisfy $z_{kl} w_{kl} = s_{kl}$ for all $k < l$.*

Proof. For any such point, $\delta_{kl}^2 = s_{kl}^{-1}$, so δ takes finitely many values; fix one. Multiplying the z-equation of body k by m_k and summing gives $\sum_k m_k z_k = 0$ by antisymmetry, and likewise $\sum_k m_k w_k = 0$.

Put $v_k = (z_k - z_1, w_k - w_1) \in \mathbb{C}^2$ and let B be the symmetric bilinear form with $B((a,b),(a,b)) = ab$. Then $B(v_k - v_l, v_k - v_l) = s_{kl}$, so the Gram matrix $B(v_k, v_l) = \frac{1}{2}(s_{1k} + s_{1l} - s_{kl})$ is fixed. The span V of the v_k is nondegenerate for B: either $V = \mathbb{C}^2$, or $V = \mathbb{C} v_2$ with $B(v_2, v_2) = s_{12} \neq 0$.

If (v_k') is a second family with the same Gram matrix, its span V' is nondegenerate by the same argument, and $\sum_k c_k v_k = 0$ holds if and only if $\sum_k c_k B(v_k, v_l) = 0$ for all l, and likewise for the v_k'; so the v_k and the v_k' satisfy the same linear relations, and $v_k \mapsto v_k'$ is an isometry of nondegenerate subspaces; by Witt's theorem it extends to an element of the orthogonal group of B, which consists of the maps $(a,b) \mapsto (\lambda a, \lambda^{-1} b)$ and $(a,b) \mapsto (\lambda b, \lambda^{-1} a)$, $\lambda \in \mathbb{C}^*$.

The centre-of-mass identities determine (z_1, w_1) from (v_k):

$$(z_1, w_1) = -\frac{1}{\sum_k m_k} \sum_k m_k v_k.$$

For a fixed reference family (v_k^0) with first components a_2^0 and second components b_2^0 of v_2^0 (so $a_2^0 b_2^0 = s_{12} \neq 0$), the gauge $z_{12} = w_{12}$ leaves $\lambda^2 = b_2^0/a_2^0$ or $\lambda^2 = a_2^0/b_2^0$: at most four values. Hence there are finitely many solutions for each δ. □

We now use semialgebraic curve selection to retain positivity while passing from an infinite set of configurations to an algebraic curve.

Lemma 3.2 (Arc lemma)**.** *If A_m has infinitely many positive points, there is an irreducible algebraic curve $\Gamma \subset A_m$ such that*

(i) Γ contains a nonconstant real-analytic arc of positive points;

(ii) $\sigma(\Gamma) = \Gamma$;

(iii) $U = I$ is constant on Γ, with value $c \in (0, \infty)$.

Proof. In the real coordinates $(x, y, \delta) \in \mathbb{R}^{20}$ the positive points form a semialgebraic set. It has finitely many connected components, so one of them is infinite, hence of positive dimension, and the Nash curve selection lemma [BCR98, Prop. 8.1.13] gives a nonconstant Nash arc in it. Its Zariski closure $\Gamma \subset A_m$ (in the complexified coordinates; passing to (z, w) is linear) is an irreducible curve, which proves (i). Since (2.1) has real coefficients, $\sigma(\Gamma)$ is an irreducible curve containing the arc, so $\sigma(\Gamma) = \Gamma$. By Lemma 2.1, $U = I$ is constant on Γ, and on the arc $U = \sum m_k m_l \delta_{kl} > 0$. □

On the projective normalization of this curve, a singular sequence approaches a branch at infinity. The next lemma expresses that branch in a parameter suitable for comparing the orders of the force terms.

Lemma 3.3. *Let $\Gamma \subset A_m$ be an irreducible curve and (q_n) a singular sequence on Γ. There are a Puiseux parametrization γ of a branch of Γ at infinity, $\gamma(t) \to \infty$ as $t \to 0$, and $t_n \to 0$ with $\gamma(t_n)$ a subsequence of (q_n). Every coordinate and every component of $\mathcal{Z}$, $\mathcal{W}$ has a Puiseux expansion $c\,t^\alpha(1+o(1))$ along γ (or vanishes identically), and the diagram of every singular sequence extracted from $(\gamma(t_n))$ is determined by these orders and leading coefficients: the circles and strokes of colour z are the components of $\mathcal{Z}$ of minimal order, and the relations $\prec$, $\preccurlyeq$, $\approx$, $\sim$ of [AK12, §3] become relations between orders and leading coefficients. In particular a four-product unbounded along (q_n) has negative order along γ.*

Proof. The closure of Γ in a projective compactification has finitely many points at infinity, each with finitely many analytic branches; a sequence tending to infinity has a subsequence on one branch, and the normalization of the branch gives the parameter t. Polynomial functions of the coordinates are meromorphic in t, which gives the expansions.

If α is the least order among the components of $\mathcal{Z}$, then $Z = \max_p |\mathcal{Z}_p|$ has order α, and $\mathcal{Z}_p/Z \to 0$ exactly when $\operatorname{ord} \mathcal{Z}_p > \alpha$; this does not depend on the subsequence.

The renormalization of [AK12, p. 547] multiplies all z-quantities by one positive factor $a(t) \approx |t|^{(\beta-\alpha)/2}$ and all w-quantities by a^{-1}, where β is the least order in $\mathcal{W}$. This preserves every relation within one colour and every product $z_{kl}w_{kl}$. Replacing a by the Puiseux series $\lambda(t) = (\mathcal{W}_q(t)/\mathcal{Z}_p(t))^{1/2}$ built from maximal components, with $|\lambda|/a$ bounded above and below, gives an analytic renormalization with the same relations; after reparametrizing $t = s^N$, ϵ is comparable to a power of s. A meromorphic function that is unbounded along $t_n \to 0$ has a pole. □

Definition 3.4. A *Diagram-r germ* is a Puiseux branch γ of a curve in A_m whose diagram, in the sense of Lemma 3.3, is Diagram r of [AK12, Fig. 11] after renumbering the bodies and possibly exchanging z and w. It has an *unbounded four-product* if some p_{ij} has negative order along γ.

The following monomial relations transfer variation of the squared distances to variation of the four-products.

Lemma 3.5. *On every irreducible curve $\Gamma \subset A_m$ some four-product is not constant.*

Proof. Index the squared distances and the four-products by the ten pairs. The pairs disjoint from ij are the sides of the triangle complementary to ij, so the exponent vector of p_{ij} in the variables r_{ab}^2 is the column ij of $E = I + P$, where P is the adjacency matrix of the Petersen graph (two pairs are adjacent if they are disjoint). The eigenvalues of E are 4, 2 (five times) and -1 (four times), so $\det E = 128$, and $N = 128E^{-1}$ is an integer matrix. Since $r_{ab}^2 \neq 0$ on A_m, we obtain

$$(r_{ab}^2)^{128} = \prod_{ij} p_{ij}^{N_{ab,ij}}.$$

If every p_{ij} were constant on Γ, this identity would make every r_{ab}^2 constant on the irreducible curve Γ. Lemma 3.1 would then make Γ finite, a contradiction. □

For the global argument, we also need a relation that couples all four-products to a single meromorphic function.

Lemma 3.6 (The total product)**.** *Put $\Delta = \prod_{k<l} r_{kl}^2$. On A_m, the function Δ is regular and nowhere zero, and*

$$\prod_{i<j} p_{ij} = \Delta^4. \tag{3.1}$$

Proof. The distance constraints give $r_{kl}^2 = z_{kl}w_{kl} \neq 0$ on A_m. Each squared distance occurs in its own four-product and in the three four-products indexed by a pair disjoint from its endpoints. It therefore occurs exactly four times in the product on the left of (3.1). □

Proposition 3.7 (Reduction)**.** *Let $m \in \mathbb{R}^5_{>0}$. If A_m has infinitely many positive points, there are $r \in \{1, \dots, 9\}$, a curve Γ as in Lemma 3.2, and a Diagram-r germ on Γ with an unbounded four-product.*

Proof. Let Γ be given by Lemma 3.2. By Lemma 3.5 some four-product p_{ij} is not constant on Γ, and by Fact AK1 it is dominating there. A sequence on Γ with $p_{ij} \to \infty$ is unbounded, and a singular sequence extracted from it approaches one of Diagrams 1–9 by Facts AK4 and AK5. Lemma 3.3 turns it into a germ. □

Remark 3.8. On a positive arc with $U = I = c > 0$, write $M = \sum_k m_k$. The identities

$$c = \sum_{k<l} \frac{m_k m_l}{r_{kl}} = \frac{1}{M} \sum_{k<l} m_k m_l r_{kl}^2$$

give, for every pair,

$$\frac{m_k m_l}{c} \leq r_{kl} \leq \sqrt{\frac{Mc}{m_k m_l}}.$$

The branches at infinity are therefore reached by continuation on the complex curve. We use identities on that curve to connect their asymptotic behaviour with the positive arc: $U = I = c$ for Diagram 7 and an inverse-distance identity for Diagram 3. For Diagrams 5 and 9, the local conclusion $\operatorname{ord}_\gamma \Delta > 0$ supplies the input to the global argument below.

4 Boundary alternatives and the global argument

We combine the diagram analysis with the meromorphic identity $\Delta^4 = \prod_{i<j} p_{ij}$. On a curve containing a positive arc, the local alternatives force Δ to vanish at every pole of a four-product. A zero and a pole of Δ on the completed curve then yield the contradiction proving finiteness.

Theorem 4.1. *Let $m \in \mathbb{R}^5_{>0}$, $r \in \{1, \dots, 9\}$ and let γ be a Diagram-r germ with an unbounded four-product on an irreducible curve $\Gamma \subset A_m$.*

(a) If $r \in \{1, 2, 4, 6, 8\}$, no such germ exists.

(b) If $r = 7$, then Γ contains no real point.

(c) If $r = 3$, then Γ contains no arc of positive points.

(d) If $r \in \{5, 9\}$, then $\operatorname{ord}_\gamma \Delta > 0$, where $\Delta = \prod_{k<l} r_{kl}^2$.

Lemma 4.2 (Diagram 9)**.** *Let $m \in \mathbb{R}^5_{>0}$ and let γ be a Diagram 9 germ on an irreducible curve $\Gamma \subset A_m$. With the numbering of [AK12, §§7.5.1.2 and 8.9], the zw-triangle is* 123 *and the two w-edges are* 14 *and* 25*. Then*

$$\begin{aligned} r_{12}, r_{13}, r_{23} \approx \epsilon, \qquad r_{14}, r_{25}, r_{45} \approx 1, \\ r_{15}, r_{24}, r_{34}, r_{35} \approx \epsilon^{-1/2}. \end{aligned} \tag{4.1}$$

Consequently $\Delta \approx \epsilon^2 \to 0$ and $\operatorname{ord}_\gamma \Delta > 0$.

Proof. We normalize the germ as in Lemma 3.3 and take $r_{ij} = \delta_{ij}^{-1}$, so that the signs of the square roots are those of the given solution. We first exclude the clustering in [AK12, §8.9.1]. In that case, $r_{14} \approx \epsilon^{1/4}$, and equation (41) of [AK12], together with the estimates following it, gives

$$\frac{m_1 m_2}{r_{12}} + \frac{m_1 m_3}{r_{13}} + \frac{m_2 m_3}{r_{23}} = O(\epsilon^2).$$

Every term of U outside this triangle, apart from $m_1 m_4 / r_{14}$, is bounded. Hence

$$U = \frac{m_1 m_4}{r_{14}} + O(1), \qquad \left| \frac{m_1 m_4}{r_{14}} \right| \longrightarrow \infty,$$

contrary to Lemma 2.1. The same argument applies after the simultaneous exchanges $1 \leftrightarrow 2$ and $4 \leftrightarrow 5$.

We are therefore in the case of [AK12, §8.9.2], where both w-edges are maximal. The estimates for the zw-triangle and the maximal w-edges give

$$z_{12}, z_{13}, z_{23}, w_{12}, w_{13}, w_{23} \approx \epsilon, \qquad z_{14}, z_{25} \approx \epsilon^2, \qquad w_{14}, w_{25} \approx \epsilon^{-2}.$$

Taking differences gives $z_{45} \approx \epsilon$ and, for $ij \in \{15, 24, 34, 35\}$,

$$z_{ij} \approx \epsilon, \qquad w_{ij} \approx \epsilon^{-2}, \qquad r_{ij} \approx \epsilon^{-1/2}.$$

Thus all estimates in (4.1), except that for r_{45}, follow. To obtain the remaining estimate, subtract the z-equations of bodies 4 and 5. The terms from pairs 14 and 25 are $O(\epsilon^2)$, and those from pairs $15, 24, 34, 35$ are $O(\epsilon^{5/2})$. Therefore

$$(z_4 - z_5)\big(1 - (m_4 + m_5)r_{45}^{-3}\big) = O(\epsilon^2).$$

Since $z_4 - z_5 \approx \epsilon$ and $m_4 + m_5 > 0$, we obtain $r_{45}^3 \to m_4 + m_5$, and hence $r_{45} \approx 1$.

Multiplying the ten squared-distance estimates now gives

$$\Delta \approx (\epsilon^2)^3\, 1^3\, (\epsilon^{-1})^4 = \epsilon^2 \longrightarrow 0.$$

The meromorphic function Δ thus has positive order along γ. □

In the numbering of Lemma 4.2, the same estimates give

$$p_{15} = r_{15}^2 r_{23}^2 r_{34}^2 r_{24}^2 \approx \epsilon^{-1}, \qquad p_{24} = r_{24}^2 r_{13}^2 r_{35}^2 r_{15}^2 \approx \epsilon^{-1}.$$

Thus Diagram 9 contributes unbounded four-products together with a vanishing total product. This is the alternative used in the global argument, while Fact AK5 provides boundedness for Diagrams 10–16.

Proof of Theorem 1.1. Suppose that A_m has infinitely many positive points. Proposition 3.7 gives an irreducible curve Γ containing a positive arc and a germ γ_0 on which some four-product has a pole. Theorem 4.1(a)–(c) excludes every diagram except Diagrams 5 and 9. By part (d),

$$\operatorname{ord}_{\gamma_0} \Delta > 0.$$

Let $\overline{\Gamma}$ be the smooth projective completion of the normalization of Γ. All squared distances, four-products and Δ pull back to meromorphic functions on $\overline{\Gamma}$. By Lemma 3.6, Δ is nowhere zero on the affine curve; its zero at γ_0 therefore implies that it is nonconstant. A nonconstant meromorphic function on the compact curve $\overline{\Gamma}$ has a pole, so choose a pole of Δ and let γ_1 be the corresponding germ. Taking orders in (3.1) gives

$$\sum_{i<j} \operatorname{ord}_{\gamma_1} p_{ij} = 4 \operatorname{ord}_{\gamma_1} \Delta < 0,$$

so $\operatorname{ord}_{\gamma_1} p_{ij} < 0$ for at least one pair ij. As in the proof of Proposition 3.7, Facts AK4 and AK5 and Lemma 3.3 put this germ in one of Diagrams 1–9. It lies on the same curve Γ through the positive arc, so Theorem 4.1(a)–(c) again leaves only Diagrams 5 and 9. Part (d) now gives $\operatorname{ord}_{\gamma_1} \Delta > 0$, contradicting the choice of a pole. The finite covers and rotations used to restore the gauge after relabelling preserve Δ, the four-products and the existence of the positive arc, as explained in Section 2. □

The statements of Theorem 4.1 for Diagrams 1–8 are proved in Sections 5–11, and the Diagram 9 statement is Lemma 4.2. Theorem 8.11 and Lemma 4.2 establish $\operatorname{ord}_\gamma \Delta > 0$ for every germ of Diagram 5 or 9. Table 1 records the mass conditions, assumptions and conclusions for each diagram.

Throughout, a statement about a diagram is meant for every renumbering of the bodies (with the masses permuted) and for both colourings; both operations are symmetries of (2.1).

r	mass condition	configurations	method and conclusion
1	(23) on the triangle	complex	triangle virial and Morse–Bott estimate are incompatible
2	K_j, L_j, P_j	complex	isolation of every boundary label at orders 1–3, with folds of corank one and two
3	(29) $m_1 m_3 = m_2 m_4$	positive	orders 3–6; identity $\delta_{15} + 2\delta_{13} = 0$
4	(23)	complex	incompatibility of the force balances at orders one and three
5	(23)	complex	direct force-order estimates give $\operatorname{ord}\Delta > 0$; used in the global pole argument
6	(32) $m_1 = m_2$, $m_3 = m_4$	complex	finite family of coordinate charts
7	none needed	real	inertia identity forces $I \to 0$
8	(40)	complex	order-six and order-nine pairings; elimination on one curve of masses
9	none needed	complex	the distance estimates give $\Delta \approx \epsilon^2 \to 0$

Table 1: The nine boundary arguments. For Diagrams 5 and 9, the product of squared distances vanishes; this conclusion is used in the global argument. Numbers in parentheses refer to the equations of [AK12, §8].

5 Diagram 1

We exclude Diagram 1 by comparing two orders of vanishing of the triangle potential. The limiting triangle has a nondegenerate critical manifold, which gives a quadratic bound in the forces; the virial identity gives an incompatible leading term.

Diagram 1 is one of the two disconnected diagrams remaining in [AK12, §7.1, p. 565]. It consists of the z-stroke 12, with z-circles at 1 and 2, and the w-triangle with strokes $34, 35, 45$ and no circle. There are no other strokes or circles. We use the numbering of [AK12, §8.1]: $P = \{1, 2\}$ is the pair and $T = \{3, 4, 5\}$ is the triangle. Indices i refer to bodies of P, and k, l to bodies of T. Put $M_P = m_1 + m_2$, $M_T = m_3 + m_4 + m_5$, $Z_P = M_P^{-1}(m_1 z_1 + m_2 z_2)$, $Z_T = M_T^{-1} \sum_k m_k z_k$, and define W_P, W_T in the same way. The mass-centre identities $\sum_j m_j z_j = \sum_j m_j w_j = 0$ on A_m were proved in Lemma 3.1; they give $M_P Z_P + M_T Z_T = 0 = M_P W_P + M_T W_T$. Write the coordinate equations of (2.1) as $E_j^z = E_j^w = 0$, where

$$E_j^z = z_j - \sum_{l \neq j} m_l \delta_{jl}^3 (z_j - z_l), \qquad E_j^w = w_j - \sum_{l \neq j} m_l \delta_{jl}^3 (w_j - w_l).$$

Along a germ, $a \prec b$, $a \preccurlyeq b$ and $a \approx b$ in the sense of [AK12, §3] mean $\operatorname{ord} a > \operatorname{ord} b$, $\operatorname{ord} a \geq \operatorname{ord} b$ and $\operatorname{ord} a = \operatorname{ord} b$ (Lemma 3.3).

The first step is to control a triangle body that might approach the w-centre of the triangle more closely than the other two. The torque on the pair rules this out (Lemma 5.5), allowing every germ to be represented in one weighted chart (Proposition 5.6). The limiting triangle then has zero multiplier and its masses satisfy relation (23) of [AK12, §8.1] (Proposition 5.10). Near this limiting triangle, the potential vanishes quadratically in the forces. The virial identity gives a lower vanishing order along the germ, and the two estimates yield the required contradiction.

5.1 The chart

Let γ be a Diagram 1 germ at $m \in \mathbb{R}^5_{>0}$. The symmetries of (2.1) allow us to renumber the bodies and exchange z and w so that its diagram is as described above. Since the coordinate equations are homogeneous of degree one in z and in w separately, and the distance constraints involve only $z_{jl}w_{jl}$, the transformation $(z,w) \mapsto (cz, c^{-1}w)$ preserves both the equations and the diagram [AK12, p. 547]. By Rule 2e [AK12, p. 552], the three z-separations of the triangle have the same order, as do its three w-separations. Choosing c to be a power of the parameter t, we normalize the w-separations of the triangle to have order 0. This choice of rotation fixes the chart normalization. Define $\lambda, \mu, \rho, \sigma$ by

$$\operatorname{ord} z_{12} = -\lambda, \qquad \operatorname{ord} z_{kl} = \mu, \qquad \operatorname{ord} Z_P = \mu + \rho, \qquad \operatorname{ord} W_P = \sigma,$$

with $\rho = +\infty$ (resp. $\sigma = +\infty$) if Z_P (resp. W_P) vanishes identically, in which case the terms carrying $\hat{Z}$ (resp. $\hat{W}$) below are absent. After a finite ramification of t we write

$$\begin{aligned} z_i &= t^{\mu+\rho}\hat{Z} + t^{-\lambda}A_i, \qquad & w_i &= t^{\sigma}\hat{W} + t^{\lambda}B_i \qquad (i \in P), \\ z_k &= -\tfrac{M_P}{M_T}\, t^{\mu+\rho}\hat{Z} + t^{\mu}\zeta_k, \qquad & w_k &= -\tfrac{M_P}{M_T}\, t^{\sigma}\hat{W} + \eta_k \qquad (k \in T), \\ \delta_{12} &= D_{12}, \qquad & \delta_{ik} &= t^{\lambda/2}D_{ik}, \qquad \delta_{kl} = t^{-\mu/2}D_{kl}, \end{aligned} \tag{5.1}$$

where $(A_1, A_2) = (m_2, -m_1)\,\pi$, $(B_1, B_2) = (m_2, -m_1)\,\varpi$ and $\sum_k m_k\zeta_k = \sum_k m_k\eta_k = 0$. The chart coordinates are $\hat{Z}, \hat{W}, \pi, \varpi$, the mass-centred triangle offsets $\zeta = (\zeta_k)$, $\eta = (\eta_k)$, and the ten D_{jl}. To obtain finite limiting equations, we multiply E^z_i, E^w_i, E^z_k and E^w_k by t^{λ}, $t^{-\lambda}$, $t^{\mu/2}$ and $t^{3\mu/2}$, respectively; the ten distance constraints retain weight 1. The exponents will belong to the open cone

$$C = \{\lambda > 0,\ \mu > 0,\ \sigma - \lambda > 0,\ \rho + \tfrac{3}{2}\mu > 0,\ \rho + \lambda + \mu > 0\}. \tag{5.2}$$

For the triangle block put, with sums over T,

$$\phi_k = \sum_{l \neq k} m_l D^3_{kl}(\zeta_k - \zeta_l), \qquad \psi_k = \sum_{l \neq k} m_l D^3_{kl}(\eta_k - \eta_l), \qquad \chi_{kl} = D^2_{kl}(\zeta_l - \zeta_k)(\eta_l - \eta_k) - 1,$$

$U_T = \sum_{k<l} m_k m_l D_{kl}$ and $I_T = \sum_k m_k\zeta_k\eta_k$; note $\sum_k m_k\phi_k = \sum_k m_k\psi_k = 0$ identically. The crossed forces are $X^z_k = \sum_{i\in P} m_i\delta^3_{ik}(z_k - z_i)$ and $X^w_k = \sum_{i\in P} m_i\delta^3_{ik}(w_k - w_i)$.

Lemma 5.1. *After substitution of (5.1) and multiplication by the prescribed weights, each of the ten coordinate equations and ten distance constraints is a finite sum $\sum_e c_e\, t^e$. The exponents e are linear forms in $(\lambda, \mu, \rho, \sigma)$, $e = 0$ occurs, and each c_e is a polynomial in the chart coordinates. Every exponent $e \neq 0$ is a nonnegative rational combination of the five forms in (5.2), with at least one positive coefficient. The constant coefficients c_0 are independent of $\hat{Z}$ and $\hat{W}$. For $k \in T$,*

$$t^{\mu/2}E^z_k = t^{\mu/2}(z_k - X^z_k) - \phi_k, \qquad t^{3\mu/2}E^w_k = t^{3\mu/2}(w_k - X^w_k) - \psi_k,$$

and the exponents occurring in $t^{\mu/2}(z_k - X^z_k)$, respectively $t^{3\mu/2}(w_k - X^w_k)$, are

$$\tfrac{3}{2}\mu,\ \tfrac{1}{2}(\lambda+\mu),\ \tfrac{3}{2}(\lambda+\mu),\ \tfrac{3}{2}\mu+\rho,\ \tfrac{3}{2}(\lambda+\mu)+\rho,$$

respectively

$$\tfrac{3}{2}\mu,\ \tfrac{3}{2}(\lambda+\mu),\ \tfrac{5}{2}\lambda+\tfrac{3}{2}\mu,\ \sigma+\tfrac{3}{2}\mu,\ \tfrac{3}{2}(\lambda+\mu)+\sigma.$$

Proof. Put $h = M_P/M_T$ and, for $i \in P$, $k \in T$, set

$$a_{ik} = A_i - t^{\lambda+\mu}\zeta_k + (1+h)t^{\lambda+\mu+\rho}\hat{Z}, \qquad b_{ik} = \eta_k - t^{\lambda}B_i - (1+h)t^{\sigma}\hat{W}.$$

Then $z_i - z_k = t^{-\lambda}a_{ik}$ and $w_i - w_k = -b_{ik}$. The weighted coordinate equations are

$$t^{\lambda}E_i^z = A_i(1 - M_P D_{12}^3) + t^{\lambda+\mu+\rho}\hat{Z} - t^{3\lambda/2}\sum_k m_k D_{ik}^3 a_{ik},$$

$$t^{-\lambda}E_i^w = B_i(1 - M_P D_{12}^3) + t^{\sigma-\lambda}\hat{W} + t^{\lambda/2}\sum_k m_k D_{ik}^3 b_{ik},$$

$$t^{\mu/2}E_k^z = -\phi_k + t^{3\mu/2}\zeta_k - ht^{3\mu/2+\rho}\hat{Z} + t^{(\lambda+\mu)/2}\sum_i m_i D_{ik}^3 a_{ik},$$

$$t^{3\mu/2}E_k^w = -\psi_k + t^{3\mu/2}\eta_k - ht^{\sigma+3\mu/2}\hat{W} - t^{3(\lambda+\mu)/2}\sum_i m_i D_{ik}^3 b_{ik}.$$

The distance constraints are, respectively, $M_P^2 D_{12}^2 \pi\varpi - 1$, $-D_{ik}^2 a_{ik} b_{ik} - 1$ and χ_{kl}. Every nonconstant power is a nonnegative rational combination of the five forms defining C, with at least one positive coefficient. Indeed, $\sigma = \lambda + (\sigma - \lambda)$, and the terms involving ρ are generated by $\lambda + \mu + \rho$ and $3\mu/2 + \rho$. Expanding a_{ik} and b_{ik} gives the two lists of exponents in the statement. The constant terms give the limiting equations described below. □

The constant terms c_0 are $A_i(1 - M_P D_{12}^3)$ and $B_i(1 - M_P D_{12}^3)$ for the pair equations, $-\phi_k$ and $-\psi_k$ for the triangle equations, and $M_P^2 D_{12}^2 \pi\varpi - 1$, $-D_{ik}^2 A_i \eta_k - 1$ and χ_{kl} for the distance constraints within the pair, between the pair and triangle, and within the triangle. We call their common zero set, in the coordinates other than $\hat{Z}, \hat{W}$, the *boundary fibre* $\mathcal{B}_1$. On $\mathcal{B}_1$, $D_{12}^3 = M_P^{-1}$ and $D_{ik}^2 A_i \eta_k = -1$, while $(\zeta, \eta, D_{34}, D_{35}, D_{45})$ satisfies the *triangle system with zero multiplier* $\phi = \psi = 0$, $\chi = 0$.

5.2 Every germ lies in the chart

We now prove that every Diagram 1 germ has a finite limit in the chart (5.1). We use Rules 1e and 2e and Estimates 1 and 2 of [AK12, §3] (Fact AK2), together with [AK12, Prop. 1] (Fact AK3). The following identities express the forces on each component and the torque on the pair.

Lemma 5.2. *The following polynomial identities hold:*

$$m_1 E_1^w + m_2 E_2^w = m_1 w_1 + m_2 w_2 - \sum_{i\in P,\, k\in T} m_i m_k \delta_{ik}^3 (w_i - w_k),$$

$$\sum_{k\in T} m_k E_k^z = \sum_{k\in T} m_k z_k - \sum_{i\in P,\, k\in T} m_i m_k \delta_{ik}^3 (z_k - z_i),$$

$$T_P := \sum_{i\in P,\, k\in T} m_i m_k \delta_{ik}^3 (z_i w_k - w_i z_k) = -\sum_{i\in P} m_i \big(w_i E_i^z - z_i E_i^w\big).$$

In particular the torque T_P of the triangle on the pair vanishes on A_m.

Proof. The internal force terms cancel in pairs within P and within T. The same cancellation in the weighted coordinate equations gives the torque identity. □

Lemma 5.3. *Let γ be a Diagram 1 germ, normalized as above.*

(i) $\operatorname{ord}\delta_{12} = 0$ *and* $\operatorname{ord} w_{12} = \lambda > 0$*; the least order of the components of* $\mathcal{Z}$ *is* $\operatorname{ord} z_1 = \operatorname{ord} z_2 = \operatorname{ord} Z_{12} = -\lambda$.

(ii) $\mu > 0$, $\operatorname{ord}\delta_{kl} = \operatorname{ord} Z_{kl} = -\mu/2$, *and* $\lambda > \mu/2$.

(iii) $\operatorname{ord} z_{ik} = -\lambda$, $\operatorname{ord} w_{ik} < \lambda$ *and* $\operatorname{ord} w_i \geq \lambda$*; hence* $\operatorname{ord} W_T = \operatorname{ord} W_P \geq \lambda$.

(iv) For $k \in T$ *let* o_k *be the coefficient of* t^0 *in* $w_k - W_T$*. At most one* o_k *vanishes. If* $o_k \neq 0$*, then* $\operatorname{ord} w_{ik} = 0$ *and* $\operatorname{ord}\delta_{ik} = \lambda/2$*. If* $o_k = 0$*, then* $f = \operatorname{ord}(w_k - W_T)$ *satisfies* $0 < f < \lambda$*, and* $\operatorname{ord} w_{ik} = f$, $\operatorname{ord}\delta_{ik} = (\lambda - f)/2$.

Proof. (i) The isolated component $\{1,2\}$ of the z-diagram has circled vertices, so it has a maximal z-stroke by Rule 1e [AK12, p. 546], which can only be 12. Estimate 2 [AK12, p. 548] gives $r_{12} \approx 1$, so $\operatorname{ord}\delta_{12} = 0$ and $\operatorname{ord} w_{12} = -\operatorname{ord} z_{12} = \lambda$. Estimate 1 [AK12, p. 548] with z and w exchanged gives $w_{12} \approx \epsilon^2$, and Estimate 2 with z and w exchanged gives $w_{34} \succcurlyeq \epsilon$ for the w-stroke 34. Hence $w_{12} \prec w_{34}$; a comparison within one colour is not affected by the normalizations, so $\lambda > \operatorname{ord} w_{34} = 0$. Finally $1,2$ are z-circled and 12 is a z-stroke, and $\operatorname{ord} Z_{12} = \operatorname{ord} z_{12} + 3\operatorname{ord}\delta_{12} = -\lambda$.

(ii) Bodies $1,2$ carry no w-stroke, so T is an isolated component of the w-diagram without w-circles. By Rule 1e it has no maximal w-stroke. Estimate 2 with z and w exchanged gives $r_{kl} \preccurlyeq 1$, with $r_{kl} \approx 1$ only for a maximal w-edge, so $r_{kl} \prec 1$; thus $\mu = \operatorname{ord}(z_{kl}w_{kl}) = 2\operatorname{ord} r_{kl} > 0$, $\operatorname{ord}\delta_{kl} = -\mu/2$ and $\operatorname{ord} Z_{kl} = \mu - \frac{3}{2}\mu = -\frac{1}{2}\mu$. Diagram 1 has no z-stroke inside T, so $\operatorname{ord} Z_{kl} > -\lambda$, that is, $\lambda > \mu/2$.

(iii) The bodies of T are not z-circled, so $\operatorname{ord} z_k > -\lambda = \operatorname{ord} z_i$ and $\operatorname{ord} z_{ik} = -\lambda$. Since $Z_{ik}^2 = (z_{ik}w_{ik}^3)^{-1}$, $\operatorname{ord} Z_{ik} = \frac{1}{2}(\lambda - 3\operatorname{ord} w_{ik})$, and the absence of a z-stroke ik gives $\operatorname{ord} Z_{ik} > -\lambda$, i.e. $\operatorname{ord} w_{ik} < \lambda$. By Estimate 1 ($z$ and w exchanged) $w_{2k} \succ \epsilon^2 \approx w_{12}$, because 12 is the only maximal z-edge, and $z_{2k} \approx z_{12}$. So [AK12, Prop. 1, p. 559] with z and w exchanged, applied to bodies $1,2$ as in the proof of [AK12, Prop. 4, p. 564], gives $w_1, w_2 \preccurlyeq w_{12}$, i.e. $\operatorname{ord} w_i \geq \lambda$. This also follows from Lemma 5.2: every term of $M_P W_P = \sum m_i m_k \delta_{ik}^3 (w_i - w_k)$ has order $\frac{1}{2}(3\lambda - \operatorname{ord} w_{ik}) > \lambda$, and $w_i - W_P$ has order $\operatorname{ord} w_{12} = \lambda$.

(iv) Since $\sum_k m_k o_k = 0$, and the o_k do not all vanish because $\operatorname{ord} w_{kl} = 0$, at most one o_k vanishes. Now $w_k - w_i = (w_k - W_T) + (W_T - w_i)$ with $\operatorname{ord}(W_T - w_i) \geq \lambda > 0$ by (iii). If $o_k \neq 0$ this gives $\operatorname{ord} w_{ik} = 0$. If $o_k = 0$ then $f > 0$; $f \geq \lambda$ would give $\operatorname{ord} w_{ik} \geq \lambda$, against (iii), so $f < \lambda$ and $\operatorname{ord} w_{ik} = f$. In both cases $\operatorname{ord}\delta_{ik} = -\frac{1}{2}(\operatorname{ord} z_{ik} + \operatorname{ord} w_{ik})$. □

Lemma 5.4. *Let* $\lambda, \mu, f > 0$ *with* $f < \lambda$ *and let* ρ, σ *be real. Modify* (5.1) *by* $(\eta_3, \eta_4, \eta_5) = (t^f\omega,\ m_5\vartheta,\ -m_3m_5^{-1}t^f\omega - m_4\vartheta)$ *and* $\delta_{i3} = t^{(\lambda-f)/2}d_i$ $(i = 1,2)$*, all other entries unchanged. Then*

$$T_P = t^{(\lambda-f)/2}\Big(m_3 \sum_{i=1,2} m_i d_i \left(d_i^2 A_i \omega\right) + \sum_{g \in G} c_g\, t^g\Big),$$

where the c_g *are polynomials in the chart coordinates and* G *consists of the thirteen exponents*

$$\tfrac{1}{2}f,\ \tfrac{3}{2}f,\ \sigma - f,\ \sigma + \tfrac{1}{2}f,\ 2\lambda + \mu - f,\ 2\lambda + \mu + \tfrac{1}{2}f,\ \lambda + \mu + \rho,\ \lambda + \mu + \sigma - f,\ 2\lambda + \mu + \rho - f,$$
$$\lambda + \mu + \rho + \tfrac{1}{2}f,\ \lambda + \mu + \sigma + \tfrac{1}{2}f,\ 2\lambda + \mu + \rho + \tfrac{1}{2}f,\ \lambda + \mu + \rho + \tfrac{3}{2}f,$$

each a combination of $\lambda,\ \mu,\ \sigma - \lambda,\ \rho + \lambda + \mu,\ f,\ \lambda - f$ *with nonnegative rational coefficients, at least one of them positive. Moreover, if* $d_i^2 A_i \omega = -1$ *for* $i = 1,2$*, then* $(m_1d_1)^2 - (m_2d_2)^2 = -(m_1^3 + m_2^3)/(m_1A_1\omega)$.

Proof. Put $h = M_P/M_T$. The determinant in each summand of the torque is

$$z_i w_k - w_i z_k = t^{-\lambda} A_i \eta_k - h t^{\sigma-\lambda} A_i \hat{W} + t^{\mu+\rho} \hat{Z} \eta_k - t^{\mu+\sigma} \hat{W} \zeta_k + h t^{\lambda+\mu+\rho} B_i \hat{Z} - t^{\lambda+\mu} B_i \zeta_k.$$

For $k = 3$, multiply this expression by $t^{3(\lambda-f)/2} d_i^3$ and use $\eta_3 = t^f \omega$. After removing the common factor $t^{(\lambda-f)/2}$, the first term is $d_i^3 A_i \omega$, and the other five exponents are

$$\sigma - f, \quad \lambda + \mu + \rho, \quad \lambda + \mu + \sigma - f, \quad 2\lambda + \mu + \rho - f, \quad 2\lambda + \mu - f.$$

For $k = 4, 5$, multiply instead by $t^{3\lambda/2} D_{ik}^3$; the constant and t^f parts of η_k give exactly the remaining eight exponents in G. This proves the displayed expansion. Each exponent has the asserted nonnegative representation: the only subtractions needing attention are $\sigma - f = (\sigma - \lambda) + (\lambda - f)$, $2\lambda + \mu - f = \lambda + \mu + (\lambda - f)$ and $2\lambda + \mu + \rho - f = (\rho + \lambda + \mu) + (\lambda - f)$; the remaining cases follow by adding nonnegative multiples of λ, μ, f to these or to $\rho + \lambda + \mu$. Finally $A_2 = -m_1 A_1/m_2$, so the distance constraints give directly

$$(m_1 d_1)^2 - (m_2 d_2)^2 = -\frac{m_1^2}{A_1 \omega} + \frac{m_2^2}{A_2 \omega} = -\frac{m_1^3 + m_2^3}{m_1 A_1 \omega}.$$

□

Lemma 5.5. *Let $m \in \mathbb{R}^5_{>0}$. Along every Diagram 1 germ all three leading triangle w-offsets o_k are nonzero.*

Proof. Suppose not. By Lemma 5.3(iv), after renumbering T, $o_3 = 0 \neq o_4, o_5$, $0 < f < \lambda$, $\operatorname{ord} \delta_{i3} = \frac{1}{2}(\lambda - f)$ and $\operatorname{ord} \delta_{ik} = \frac{1}{2}\lambda$ for $k = 4, 5$. In the first two identities of Lemma 5.2 the crossed terms have orders $\frac{1}{2}(3\lambda - \operatorname{ord} w_{ik})$ and $\frac{1}{2}(\lambda - 3 \operatorname{ord} w_{ik})$, so $\sigma \geq \frac{1}{2}(3\lambda - f)$ and $\mu + \rho \geq \frac{1}{2}(\lambda - 3f)$, that is, $\sigma - \lambda \geq \frac{1}{2}(\lambda - f) > 0$ and $\rho + \lambda + \mu \geq \frac{3}{2}(\lambda - f) > 0$. Hence γ is described by the chart of Lemma 5.4, with chart coordinates of order ≥ 0, of order 0 for d_i, A_i, ω (so these are nonzero at $t = 0$), and every exponent in G is positive. As $\operatorname{ord}(Z_T - Z_P) = \mu + \rho > -\lambda$ and $\sigma > \lambda > f$, we have $z_3 - z_i = -t^{-\lambda} A_i (1 + o(1))$ and $w_3 - w_i = t^f \omega (1 + o(1))$, so the distance constraints $\delta_{i3}^2 (z_3 - z_i)(w_3 - w_i) = 1$ give $d_i^2 A_i \omega = -1$ at $t = 0$. By Lemma 5.4, $t^{-(\lambda-f)/2} T_P$ tends along γ to $-m_3(m_1 d_1 + m_2 d_2)$, evaluated at $t = 0$, and this is nonzero because $(m_1 d_1)^2 - (m_2 d_2)^2 = -(m_1^3 + m_2^3)/(m_1 A_1 \omega) \neq 0$. But $T_P = 0$ along γ by Lemma 5.2. □

Proposition 5.6 (Representation in the chart)**.** *Let $m \in \mathbb{R}^5_{>0}$ and let γ be a Diagram 1 germ, normalized as above.*

(i) $\lambda > \mu/2 > 0$.

(ii) Every leading triangle w-offset o_k is nonzero; hence $\operatorname{ord} w_{ik} = 0$ *and* $\operatorname{ord} \delta_{ik} = \lambda/2$ *for $i \in P$, $k \in T$.*

(iii) $\sigma \geq \frac{3}{2}\lambda$ *and* $\mu + \rho \geq \frac{1}{2}\lambda$. *In particular $(\lambda, \mu, \rho, \sigma) \in C$, with $\sigma - \lambda \geq \frac{1}{2}\lambda$, $\rho + \frac{3}{2}\mu \geq \frac{1}{2}(\lambda + \mu)$ and $\rho + \lambda + \mu \geq \frac{3}{2}\lambda$.*

(iv) Along γ every chart coordinate of (5.1) converges as $t \to 0$; the limits of π, ϖ, of the D_{jl}, of the η_k and of the differences $\zeta_k - \zeta_l$, $\eta_k - \eta_l$ $(k \neq l)$ are nonzero; and the limit point lies in the boundary fibre $\mathcal{B}_1$.

Proof. (i) is Lemma 5.3(i),(ii), and (ii) follows from Lemmas 5.5 and 5.3(iv). (iii) On γ, Lemma 5.2 gives $M_P W_P = \sum m_i m_k \delta_{ik}^3 (w_i - w_k)$ and $M_T Z_T = \sum m_i m_k \delta_{ik}^3 (z_k - z_i)$. By (ii) and Lemma 5.3(iii) the terms of the first sum have order $\frac{3}{2}\lambda$ and those of the second $\frac{3}{2}\lambda - \lambda = \frac{1}{2}\lambda$; as $\operatorname{ord} Z_T = \operatorname{ord} Z_P = \mu + \rho$ the two bounds follow. Substitution gives the three remaining inequalities. (iv) By the definitions of $\lambda, \mu, \rho, \sigma$, by Lemma 5.3 and by (ii), every chart coordinate is a Puiseux series of order ≥ 0, of order exactly 0 for the listed ones; so all converge. By Lemma 5.1 and (iii), each weighted equation equals $c_0 + \sum_{e>0} c_e t^e$ along γ with bounded c_e; it vanishes identically, so c_0 vanishes at the limit point. □

Remark 5.7. Lemma 5.3 and Proposition 5.6 give the clustering scheme $z : 1 \dots 345 \dots 2$, $w : 3 \dots 12 \dots 5 \dots 4$, which [AK12, §8.1, p. 572] derives from Rule 2e, Estimate 2 and [AK12, Prop. 4]. Lemma 5.5 is the torque argument of the proof of [AK12, Prop. 4], carried out in the chart with every competing term controlled.

5.3 The limiting triangle and its masses

The chart reduces the limiting equations to a triangle with zero multiplier. We classify its solutions, identify the mass relation, and compute the rank transverse to the scaling and rotation orbit. This rank calculation will control the potential by the squared forces.

Lemma 5.8. *Let $m_3, m_4, m_5 > 0$ and let $(\zeta, \eta, D_{34}, D_{35}, D_{45})$, with $\sum_k m_k \zeta_k = \sum_k m_k \eta_k = 0$, solve the triangle system with zero multiplier. Put $\alpha_{kl} = \zeta_l - \zeta_k$ and $\beta_{kl} = \eta_l - \eta_k$ for $k < l$ in T. Then*

(a) $\beta_{kl} = \Lambda^{-1} \alpha_{kl}$ for one $\Lambda \in \mathbb{C}^*$, and $(D_{kl}\alpha_{kl})^2 = \Lambda$ for all $k < l$;

(b) $(D_{34}\alpha_{34}, D_{35}\alpha_{35}, D_{45}\alpha_{45}) = s\,(-1, 1, -1)$ with $s^2 = \Lambda$; the other six sign patterns admit no solution;

(c) $m_5 \alpha_{34}^2 = m_3 \alpha_{45}^2 = m_4 \alpha_{35}^2$ and $Q(m_3, m_4, m_5) = 0$, where

$$Q = 2m_3 m_4 m_5 (m_3 + m_4 + m_5) - m_3^2 m_4^2 - m_4^2 m_5^2 - m_5^2 m_3^2.$$

Proof. Let $M = \operatorname{diag}(m_3, m_4, m_5)$ and let B be the symmetric matrix

$$B_{kl} = -m_k m_l D_{kl}^3 \quad (k \neq l), \qquad B_{kk} = \sum_{l \neq k} m_k m_l D_{kl}^3.$$

Then $B\mathbf{1} = 0$, $B\zeta = M\phi = 0$ and $B\eta = M\psi = 0$. All the D_{kl} are nonzero by the distance constraints, so $B \neq 0$. The mass-centred vectors ζ, η cannot be independent. If they were, $\mathbf{1}, \zeta, \eta$ would form a basis of $\mathbb{C}^3$, forcing B to vanish. Thus $\eta = \Lambda^{-1}\zeta$ with $\Lambda \neq 0$, and the distance constraints give (a).

Choose $s^2 = \Lambda$ and write $D_{kl}\alpha_{kl} = s\varepsilon_{kl}$, where $\varepsilon_{kl} \in \{1, -1\}$. Put $u = \alpha_{45}/\alpha_{34}$ and $v = \alpha_{35}/\alpha_{34} = 1 + u$. The equations $\phi_4 = \phi_5 = 0$ give

$$u^2 = \frac{m_5}{m_3}\frac{\varepsilon_{45}}{\varepsilon_{34}}, \qquad v^2 = -\frac{m_5}{m_4}\frac{\varepsilon_{35}}{\varepsilon_{34}}.$$

Both squares are nonzero real numbers. Thus u is real or purely imaginary. The latter possibility is excluded because $(1 + u)^2$ would then have nonzero imaginary part. Hence u, v are nonzero

real numbers. Positivity of the masses now implies $\varepsilon_{45} = \varepsilon_{34} = -\varepsilon_{35}$. Changing the sign of s if necessary gives (b), and the two displayed equations give the equalities in (c).

With $x_k = m_k^{-1/2} > 0$, the equalities just obtained and $\alpha_{35} = \alpha_{34} + \alpha_{45}$ imply that one of x_3, x_4, x_5 is the sum of the other two. The factorization

$$Q = (m_3m_4m_5)^2(x_3 + x_4 + x_5)(-x_3 + x_4 + x_5)(x_3 - x_4 + x_5)(x_3 + x_4 - x_5),$$

therefore gives $Q = 0$. For positive masses, the same factorization shows that $Q = 0$ is equivalent to relation (23). □

The torus $\mathbb{T} = \{(\theta_1, \theta_2, \theta_3) \in (\mathbb{C}^*)^3 : \theta_3^2 = \theta_1\theta_2\}$ acts by $(\zeta, \eta, D) \mapsto (\theta_1\zeta, \theta_2\eta, \theta_3^{-1}D)$. It multiplies ϕ by $\theta_1\theta_3^{-3}$, ψ by $\theta_2\theta_3^{-3}$ and I_T by $\theta_1\theta_2$, and it fixes χ.

Lemma 5.9. *Let $b, c > 0$, $(m_3, m_4, m_5) = (b^2c^2, b^2(b + c)^2, c^2(b + c)^2)$ and $\kappa > 0$. Let y^* be the point with $\eta = \zeta$, $\sum_k m_k\zeta_k = 0$, $\zeta_4 - \zeta_3 = \kappa m_5^{-1/2}$, $\zeta_5 - \zeta_3 = -\kappa m_4^{-1/2}$, $D_{34} = m_5^{1/2}/\kappa$, $D_{35} = m_4^{1/2}/\kappa$ and $D_{45} = -m_3^{1/2}/\kappa$. In the coordinates $y = (\zeta_3, \zeta_4, \eta_3, \eta_4, D_{34}, D_{35}, D_{45})$ let $F = (\phi_3, \phi_4, \psi_3, \psi_4, \chi_{34}, \chi_{35}, \chi_{45})$. Then $F(y^*) = 0$; the Jacobian matrix $F'(y^*)$ has rank 5, and its kernel is spanned by the generators $(\zeta_3, \zeta_4, 0, 0, -\frac{1}{2}D)$ and $(0, 0, \eta_3, \eta_4, -\frac{1}{2}D)$ of the $\mathbb{T}$-action; and*

$$I_T(y^*) = \frac{\kappa^2}{m_3 + m_4 + m_5}\Big(\frac{m_3m_4}{m_5} + \frac{m_3m_5}{m_4} + \frac{m_4m_5}{m_3}\Big) > 0.$$

Proof. Write $q = \zeta = \eta$ at y^*. The mass relation is $m_3^{-1/2} = m_4^{-1/2} + m_5^{-1/2}$, and hence $q_5 - q_4 = -\kappa m_3^{-1/2}$. All three distance constraints hold and $D_{34}(q_4 - q_3) = 1$, $D_{35}(q_5 - q_3) = -1$, $D_{45}(q_5 - q_4) = 1$. Substitution in the two force identities used in the proof of Lemma 5.8 gives $\phi_4 = \phi_5 = 0$; the mass-centre identity gives $\phi_3 = 0$, and $\psi = \phi$. Thus $F(y^*) = 0$.

Use the matrix B from that proof, put $M = \operatorname{diag}(m_3, m_4, m_5)$, and set $L = M^{-1}B$. Since $B\mathbf{1} = Bq = 0$, the vectors $\mathbf{1}, q$ are independent and $B \neq 0$, we have $\operatorname{rank} B = 1$ and $\ker B = \operatorname{span}\{\mathbf{1}, q\}$. On the mass-centred plane $H = \{x : \sum_k m_kx_k = 0\}$, the map L has rank 1 and kernel $\mathbb{C}q$.

The three constraint differentials are independent, since $\partial\chi_{kl}/\partial D_{kl} = 2/D_{kl} \neq 0$ and the other two constraint rows have zero entries in this column. Their zero set is therefore a smooth four-dimensional manifold, with local coordinates $(\zeta, \eta) \in H^2$. On its tangent space at y^*,

$$dD_{kl} = -\frac{D_{kl}}{2(q_l - q_k)}\big(d\zeta_l - d\zeta_k + d\eta_l - d\eta_k\big),$$

and differentiation of the forces gives

$$\begin{pmatrix} d\phi \\ d\psi \end{pmatrix} = -\frac{1}{2}\begin{pmatrix} L & 3L \\ 3L & L \end{pmatrix}\begin{pmatrix} d\zeta \\ d\eta \end{pmatrix}.$$

The scalar 2×2 matrix here is invertible, so this map has rank 2 and kernel $\mathbb{C}(q, 0) \oplus \mathbb{C}(0, q)$. Since either two components determine each mass-centred force vector, the full Jacobian $F'(y^*)$ has rank $3 + 2 = 5$. The formula for dD identifies its two kernel generators with the stated infinitesimal torus actions.

The elementary mass-centre identity

$$\sum_k m_kq_k^2 = \frac{1}{m_3 + m_4 + m_5}\sum_{k<l} m_km_l(q_l - q_k)^2$$

gives the displayed value of $I_T(y^*)$, which is positive because q is a nonzero real vector. Conversely every positive solution of the mass relation has the stated (b, c) parametrization: take $b/c = \sqrt{m_4/m_5}$ and $bc = \sqrt{m_3}$. □

By equivariance, the same rank and kernel statements hold at every point of $O = \mathbb{T} \cdot y^*$, and $I_T \neq 0$ on O.

Proposition 5.10 (The limiting mass condition)**.** *Let $m \in \mathbb{R}^5_{>0}$ and let γ be a Diagram 1 germ. Then $Q(m_3, m_4, m_5) = 0$, that is, the triangle masses satisfy relation (23) of [AK12, §8.1]. After renumbering T so that $m_3^{-1/2} = m_4^{-1/2} + m_5^{-1/2}$, the limit y_0 of $y(t) = (\zeta_3, \zeta_4, \eta_3, \eta_4, D_{34}, D_{35}, D_{45})(t)$ lies on the orbit O of Lemma 5.9.*

Proof. By Proposition 5.6(iv) the limit lies in $\mathcal{B}_1$, so its triangle part solves the triangle system with zero multiplier. In the notation of Lemma 5.8, $\phi_4 = m_3\alpha_{34}D_{34}^3 - m_5\alpha_{45}D_{45}^3$ and $\phi_5 = m_4\alpha_{45}D_{45}^3 + m_3\alpha_{35}D_{35}^3$, and $m_3\phi_3 = -m_4\phi_4 - m_5\phi_5$; likewise for ψ. So $\phi = 0$ and $\psi = 0$ are relations (22) and (21) of [AK12, §8.1] in cross-multiplied form, and Lemma 5.8 gives $Q = 0$. After the renumbering, Lemma 5.8(c) and $\alpha_{35} = \alpha_{34} + \alpha_{45}$ leave only $(\alpha_{34}, \alpha_{45}, \alpha_{35}) = \kappa(x_5, -x_3, -x_4)$ with $\kappa \in \mathbb{C}^*$, and Lemma 5.8(a),(b) then determine η and D from (κ, s). The torus $\mathbb{T}$ acts transitively on these pairs (κ, s), and y^* is the solution with $\kappa > 0$, $s = -1$. Hence the solutions form the single orbit O, and $y_0 \in O$. □

5.4 Excluding Diagram 1

We first express the triangle potential and its differential in terms of the forces. Combined with the preceding rank calculation, these identities yield the quadratic estimate needed to compare orders along a germ.

Lemma 5.11. *For arbitrary $\zeta, \eta \in \mathbb{C}^3$ and $D_{34}, D_{35}, D_{45} \in \mathbb{C}$,*

$$\sum_{k\in T} m_k\zeta_k\psi_k = U_T + \sum_{k<l} m_k m_l D_{kl}\chi_{kl},$$
$$2\,dU_T + \sum_{k\in T} m_k\big(\psi_k\,d\zeta_k + \phi_k\,d\eta_k\big) = \sum_{k<l} m_k m_l\big(D_{kl}\,d\chi_{kl} - 2\chi_{kl}\,dD_{kl}\big).$$

Proof. For a single edge put $\alpha = \zeta_l - \zeta_k$, $\beta = \eta_l - \eta_k$, $D = D_{kl}$ and $\chi = D^2\alpha\beta - 1$. Pairing the k and l terms in the force sum gives $m_k m_l D^3\alpha\beta = m_k m_l D(1 + \chi)$, which proves the first identity after summing over the three edges. For the second use the exact single-edge identity

$$D\,d\chi - 2\chi\,dD = 2\,dD + D^3(\beta\,d\alpha + \alpha\,d\beta).$$

Multiplication by $m_k m_l$ and summation gives the result. □

On the constraint manifold $\{\chi = 0\}$, these identities reduce to $U_T = \sum_k m_k\zeta_k\psi_k$ and $dU_T = -\frac{1}{2}\sum_k m_k(\psi_k\,d\zeta_k + \phi_k\,d\eta_k)$. Thus both the potential and its differential vanish on O. Together with the rank calculation in Lemma 5.9, this gives the quadratic estimate below.

Lemma 5.12. *Every $y_0 \in O$ has a neighbourhood $V \subset \mathbb{C}^7$ with a constant c_V such that $|U_T(y)| \le c_V\,|(\phi_3, \phi_4, \psi_3, \psi_4)(y)|^2$ for every $y \in V$ with $\chi(y) = 0$.*

Proof. By Lemma 5.9 and equivariance, five components F_5 of F have independent differentials at y_0. Their common zero set $F_5^{-1}(0)$ is therefore a 2-dimensional manifold near y_0. The orbit map $\theta \mapsto \theta y_0$ immerses a neighbourhood of $1 \in \mathbb{T}$ onto a 2-dimensional piece of O contained in this manifold. Shrinking V, we obtain $F_5^{-1}(0) \cap V \subset O$. The implicit function theorem then gives $\operatorname{dist}(y, O) \le c\,|F_5(y)| \le c\,|(\phi_3, \phi_4, \psi_3, \psi_4)(y)|$ for $y \in V$ with $\chi(y) = 0$.

Near y_0, the constraint manifold $\{\chi = 0\}$ is the graph of D over $(\zeta_3, \zeta_4, \eta_3, \eta_4)$, because $\partial\chi_{kl}/\partial D_{kl} = 2D_{kl}^{-1}(\chi_{kl} + 1) \neq 0$. In these local coordinates, both U_T and its differential vanish on O (Lemma 5.11). Taylor's formula at a nearest point of O gives $|U_T(y)| \le c' \operatorname{dist}(y, O)^2$. Combining this with the preceding distance estimate proves the assertion. □

Theorem 5.13. *Let $m \in \mathbb{R}^5_{>0}$. No curve in A_m has a Diagram 1 germ (Definition 3.4). In particular there is no Diagram 1 germ with an unbounded four-product, which is the case $r = 1$ of Theorem 4.1 (a).*

Proof. Suppose that γ is a Diagram 1 germ. Normalize it as above and renumber T as in Proposition 5.10. Along γ, the triangle distance constraints give $\chi(y(t)) = 0$ identically, and $y(t) \to y_0 \in O$. We compare the order of the triangle potential obtained from the force estimate with the order given by the virial identity.

Since $E_k^z = E_k^w = 0$ along γ, Lemma 5.1 gives $\phi_k = t^{\mu/2}(z_k - X_k^z)$ and $\psi_k = t^{3\mu/2}(w_k - X_k^w)$, with the exponents listed there and bounded coefficients. By Proposition 5.6(iii), $\frac{3}{2}\mu + \rho \ge \frac{1}{2}(\lambda + \mu)$. Hence $|\phi(y(t))| = O(|t|^r)$ with $r = \min(\frac{3}{2}\mu, \frac{1}{2}(\lambda + \mu)) > 0$, and $|\psi(y(t))| = O(|t|^{3\mu/2})$. Lemma 5.12 therefore gives

$$|U_T(y(t))| = O(|t|^{2r}), \qquad 2r = \min(3\mu, \lambda + \mu).$$

On the other hand, Lemma 5.11 and $\chi = 0$ give $U_T(y(t)) = \sum_k m_k \zeta_k \psi_k = t^{3\mu/2} \sum_k m_k \zeta_k (w_k - X_k^w)$. Because $w_k = W_T + \eta_k$ and $\sum_k m_k \zeta_k = 0$, we have $\sum_k m_k \zeta_k w_k = I_T(y(t))$. Moreover, $\operatorname{ord} X_k^w \ge \frac{3}{2}\lambda$ by Proposition 5.6(ii). Thus

$$U_T(y(t)) = t^{3\mu/2}\big(I_T(y_0) + o(1)\big), \qquad I_T(y_0) \neq 0.$$

Comparison with the quadratic estimate gives

$$\frac{3}{2}\mu \ge \min(3\mu, \lambda + \mu) \quad \Longrightarrow \quad \lambda + \mu \le \frac{3}{2}\mu \quad \Longrightarrow \quad \lambda \le \frac{1}{2}\mu,$$

whereas Proposition 5.6(i) gives $\lambda > \mu/2$. □

Remark 5.14. In the chart family (5.1), the outer bodies are z-circled exactly when $\lambda \ge \mu/2$. At $\lambda = \mu/2$, a rotation gives a Diagram 4 chart; for $\lambda < \mu/2$, the diagram is Diagram 5. These two cases are treated in Section 8.

6 Diagram 2

We exclude Diagram 2 by resolving its pair and triangle in a blow-up chart. The torque identity restricts the boundary points to explicit mass loci; local analytic reduction then turns continuation from the boundary into scalar Taylor conditions. The coefficients of these reduced equations yield the exclusion.

Diagram 2 in [AK12, Fig. 11] consists of the z-segment 12 and the w-triangle 345. Every vertex carries a circle of the same colour as its edges [AK12, §8.2, p. 572]. Since (2.1) is invariant under exchanging z and w and renumbering the bodies, we may assume that a Diagram 2 germ has this form. Throughout this section, A, A' denote the bodies in the pair, $\{A, A'\} = \{1, 2\}$, and B, B' denote bodies in the triangle $\{3, 4, 5\}$. We write $M = m_1 + m_2$, $M_T = m_3 + m_4 + m_5$ and $m' = (m_3, m_4, m_5)$. We reserve t for a chart coordinate and use s for the germ parameter called t in Lemma 3.3; thus ord denotes order in s. The notation x, y, u agrees with [AK12, §8.2].

Proposition 6.4 places every Diagram 2 germ at a finite point of the boundary fibre $t = 0$. The torque identity gives $\Phi = 0$ there and recovers the mass loci L_j, K_j, P_j of [AK12, §8.2] (Proposition 6.6). At a boundary limit with $\Phi = 0$, if twenty selected equations define an analytic graph over t, the remaining equations reduce to

$$tq(t) = 0, \qquad q(0) = \Phi, \qquad q'(0) = c_\sigma^2 Q_T(\tilde{v}).$$

The quadratic form Q_T is determined by the triangle (Lemmas 6.7 and 6.10). At fold points we use one- or two-parameter reductions. Section 6.5 applies these reductions to each mass locus. At a regular boundary point, $q'(0) \neq 0$ gives isolation directly.

6.1 The blow-up chart and its boundary

We use coordinates $(t, Z, W, D) \in \mathbb{C}^{21}$, where $Z = (Z_1, \dots, Z_5)$, $W = (W_1, \dots, W_5)$ and $D = (D_{kl})_{k<l}$. Their relation to the coordinates of (2.1) is

$$\begin{gathered} z_A = t^{-1} Z_A, \quad w_A = t W_A, \quad z_B = t Z_B, \quad w_B = t^{-1} W_B, \\ \delta_{12} = D_{12}, \quad \delta_{BB'} = D_{BB'}, \quad \delta_{AB} = t D_{AB}. \end{gathered} \tag{6.1}$$

Multiplying the coordinate equations of (2.1) for z_A, w_A, z_B, w_B by t, t^{-1}, t^{-1}, t, respectively, gives the polynomial rows

$$\begin{aligned} F_A^Z &= Z_A - m_{A'} D_{12}^3 (Z_A - Z_{A'}) - t^3 \textstyle\sum_B m_B D_{AB}^3 (Z_A - t^2 Z_B), \\ F_A^W &= W_A - m_{A'} D_{12}^3 (W_A - W_{A'}) + t \textstyle\sum_B m_B D_{AB}^3 (W_B - t^2 W_A), \\ F_B^Z &= Z_B - \textstyle\sum_{B' \neq B} m_{B'} D_{BB'}^3 (Z_B - Z_{B'}) + t \textstyle\sum_A m_A D_{AB}^3 (Z_A - t^2 Z_B), \\ F_B^W &= W_B - \textstyle\sum_{B' \neq B} m_{B'} D_{BB'}^3 (W_B - W_{B'}) - t^3 \textstyle\sum_A m_A D_{AB}^3 (W_B - t^2 W_A), \end{aligned} \tag{6.2}$$

The distance equations become $T_{12} = D_{12}^2 (Z_2 - Z_1)(W_2 - W_1) - 1$, $T_{BB'} = D_{BB'}^2 (Z_{B'} - Z_B)(W_{B'} - W_B) - 1$ $(B < B')$ and $T_{AB} = D_{AB}^2 (t^2 Z_B - Z_A)(W_B - t^2 W_A) - 1$. We replace the gauge in (2.1) by $G_1 = Z_2 - Z_1 - 1$ and $G_2 = W_5 - 1$. The common zero set of these 22 polynomials is denoted by $\mathcal{X}_m \subset \mathbb{C}^{21}$, and its boundary fibre by $\mathcal{B}_m = \mathcal{X}_m \cap \{t = 0\}$. A point of $\mathcal{B}_m$ specifies the limiting pair, triangle and crossed inverse distances in this chart.

Definition 6.1 (Shape factors)**.** For $\varsigma \in \{\pm 1\}^3$ let $N_\varsigma(w_3, w_4, w_5)$ be the numerator of $w_3/(\varsigma_3(w_5 - w_4)^2) + w_4/(\varsigma_4(w_3 - w_5)^2) + w_5/(\varsigma_5(w_4 - w_3)^2)$. The quintics S_1, S_2, S_3 are N_ς for $\varsigma = (1, -1, -1), (1, -1, 1), (1, 1, -1)$, and $N_{(1,1,1)}$ is a constant multiple of $S_4 S_5$, where

$$\begin{gathered} S_4 = w_3^3 + w_4^3 + w_5^3 - w_3^2 w_4 - w_3 w_4^2 - w_3^2 w_5 - w_3 w_5^2 - w_4^2 w_5 - w_4 w_5^2 + 3 w_3 w_4 w_5, \\ S_5 = w_3^2 + w_4^2 + w_5^2 - w_3 w_4 - w_3 w_5 - w_4 w_5. \end{gathered}$$

The quartics S_6, S_7, S_8 are the factors of $\mathrm{Res}_p\big((w_4-w_3)^3-p^3(w_5-w_4)^3,\ (w_4-w_3)^3-(-1-p)^3(w_3-w_5)^3\big)$ (which is $S_5^3S_6S_7S_8$ up to a constant) whose coefficients of $w_3^3w_4$ are -2, 1 and -5. These are the factors of the complex three-body central configuration locus of [AK12, Appendix, pp. 585–586].

Lemma 6.2 (computer-assisted)**.** *Let $m \in \mathbb{R}^5_{>0}$. A point $p = (0, Z, W, D)$ is a boundary point if and only if*

(a) $Z_1 = -m_2/M$, $Z_2 = m_1/M$, $D_{12}^3 = 1/M$, $W_1 = -m_2D_{12}$ *and* $W_2 = m_1D_{12}$*;*

(b) $F_B^Z = F_B^W = T_{BB'} = 0$ *at* $t = 0$ *(system (2.1) for the bodies* $3, 4, 5$*, without its gauge) and* $W_5 = 1$*;*

(c) $D_{AB}^2 Z_A W_B = -1$ *for all* A, B*.*

At a boundary point $\sum_B m_B Z_B = \sum_B m_B W_B = 0$*, every* $W_B \neq 0$*, the* W_B *are pairwise distinct, and the* triangle $(Z_B, W_B, D_{BB'})$ *is of one of two kinds.*

(i) Lagrange: $D_{BB'} = M_T^{-1/3}\omega^{a_{BB'}}$, $\omega = e^{2\pi i/3}$, $a_{BB'} \in \{0, 1, 2\}$ *(27 classes), and* $\kappa = (W_4 - W_3)/(W_5 - W_4)$ *is a root of* $\omega^{a_{45}}\kappa^2 + (\omega^{a_{34}} + \omega^{a_{45}} - \omega^{a_{35}})\kappa + \omega^{a_{34}}$*. The root is double exactly for the six classes with* a_{34}, a_{35}, a_{45} *distinct; these are the* fold *triangles.* (W_3, W_4, W_5) *is a zero of* S_5 *if* $a_{34} = a_{35} = a_{45}$ *or the* $a_{BB'}$ *are distinct, of* S_6 *if* $a_{34} = a_{35} \neq a_{45}$*, of* S_7 *if* $a_{34} = a_{45} \neq a_{35}$*, and of* S_8 *if* $a_{35} = a_{45} \neq a_{34}$*.*

(ii) Euler: $Z_B = c^{-2}W_B$ *for some* $c \in \mathbb{C}^*$*,* $D_{BB'} = e_{BB'}c/(W_{B'} - W_B)$ *for* $B < B'$ *with* $e_{BB'} = \pm 1$*, and* (W_3, W_4, W_5) *is a zero of* S_1*,* S_2*,* S_3 *or* S_4S_5 *according to the class of* (e_{34}, e_{35}, e_{45}) *modulo a global sign. An Euler triangle on* S_5 *is a Lagrange fold triangle.*

Proof. At $t = 0$, the crossed terms in (6.2) vanish and $T_{AB} = -D_{AB}^2 Z_A W_B - 1$, giving (b) and (c). Together with G_1, the remaining pair equations are $Z_A = m_{A'}D_{12}^3(Z_A - Z_{A'})$, $W_A = m_{A'}D_{12}^3(W_A - W_{A'})$ and $T_{12} = 0$, whose solutions are those in (a). Summing $m_BF_B^Z$ and $m_BF_B^W$ gives the two centre-of-mass relations. The distance equations also ensure that the triangle positions are nonzero and distinct: if $W_B = 0$, then $T_{1B} = -1$, and if $W_B = W_{B'}$, then $T_{BB'} = -1$.

Suppose first that $(1)_B, (Z_B)_B, (W_B)_B$ are linearly independent. We claim that $D_{BB'}^3 = 1/M_T$ for all three edges. The linear map $X \mapsto (\sum_{B' \neq B} m_{B'}D_{BB'}^3(X_B - X_{B'}))_B$ on $\mathbb{C}^3$ fixes (Z_B) and (W_B) by (b), and annihilates $(1)_B$. By the centre-of-mass relations, $X \mapsto X - M_T^{-1}(\sum_B m_BX_B)(1)_B$ has the same action on this basis. The two maps therefore agree. Comparing their entries at (B, B'), $B \neq B'$, gives $m_{B'}D_{BB'}^3 = m_{B'}/M_T$. The coordinate equations now reduce to the centre-of-mass relations, and the distance equations give $Z_{B'} - Z_B = M_T^{2/3}\omega^{a_{BB'}}/(W_{B'} - W_B)$. Substituting these differences into $Z_5 - Z_3 = (Z_4 - Z_3) + (Z_5 - Z_4)$ gives the quadratic for κ.

If the three vectors are dependent, then $Z = \alpha W + \beta$. The centre-of-mass relation gives $\beta = 0$, and the distance equations give $\alpha \neq 0$. Writing $\alpha = c^{-2}$ yields the asserted formula for $D_{BB'}$. To identify the double roots, write $c_{BB'} = \omega^{a_{BB'}}$. When two of the $c_{BB'}$ agree, the discriminant is $b(b - 4a) \neq 0$, with $|a| = |b| = 1$, where a is the repeated value and b the remaining one. When all three are distinct, $c_{34} + c_{45} = -c_{35}$ and $c_{34}c_{45} = c_{35}^2$, so the discriminant vanishes. The identification of the corresponding shape factors is computer-assisted. □

This identification uses exact arithmetic over $\mathbb{Q}(\omega)$. For each of the 27 Lagrange classes, the corresponding factor S_j, expressed as a polynomial in κ, is reduced modulo the quadratic for κ; for fold classes it is evaluated at the double root. For each of the eight sign vectors (e_{34}, e_{35}, e_{45}), the corresponding N_ς is identified with one of $\pm S_1$, $\pm S_2$, $\pm S_3$, $\pm S_4 S_5$. Finally, for Euler triangles on S_5, the three quantities $D^3_{BB'}$ are checked to coincide modulo S_5 (Appendix A).

Definition 6.3 (Mass polynomials)**.** For positions W_3, W_4, W_5 put $f_B = m_B^2/W_B$, $\Lambda = f_3^2 + f_4^2 + f_5^2 - 2f_3f_4 - 2f_3f_5 - 2f_4f_5$ and $\Theta = 3(f_3^2 + f_4^2 + f_5^2) + 2f_3f_5 + 2f_4f_5 - 6f_3f_4$. Equations (25) and (28) of [AK12, §8.2] are

$$\Lambda = 0, \qquad \Lambda\, y^2 + 2(f_3 - f_4 + f_5)(f_3 - f_4 - f_5)\, y + \tfrac{1}{2}(\Theta - \Lambda) = 0.$$

For a rational expression H in W_3, W_4, W_5 whose denominator is a monomial, let $H^\flat$ be the polynomial in W_4 obtained by clearing the position denominators, substituting $W_5 = 1$ and $W_3 = -(m_4 W_4 + m_5)/m_3$ (centre of mass), and clearing the resulting power of m_3. Define $L_j(m')$ and $P_j(y)$ by

$$\operatorname{Res}_{W_4}(S_j^\flat, \Lambda^\flat) = c_j\, L_j, \qquad \operatorname{Res}_{W_4}\big(S_j^\flat, (28)^\flat\big) = m_3^{2\deg S_j} P_j(y),$$

where c_j has no zero on $\mathbb{R}^3_{>0}$; for $j = 5$ the first resultant is $m_3^4 M_T^2 L_5 L_5'$. Write $P_j(y) = E_j(y^2) + yO_j(y^2)$ and $K_j = P_j(0) = E_j(0)$.

These definitions recover the factors $L_1, \dots, L_8, L_5'$ and the polynomials $P_1, \dots, P_8$ in [AK12, pp. 573–575] from the shape factors in its appendix. They agree with the coefficients and symmetries displayed in [AK12]. In particular, $2L_5 = (m_3 - m_4)^2 + (m_4 - m_5)^2 + (m_5 - m_3)^2$, while L_5' has degree 12 and 91 positive coefficients. Permuting (m_3, m_4, m_5) permutes $\{L_1, L_2, L_3\}$ and $\{L_6, L_7, L_8\}$ up to nonzero constants. Exchanging bodies 3 and 4 sends S_1 to $-S_2$, exchanges $P_1 \leftrightarrow P_2$ and $P_6 \leftrightarrow P_7$, and fixes P_3, P_4, P_5, P_8.

The verification is exact over $\mathbb{Q}$, treating the masses and y as indeterminates. The resultants defining L_j and L_5' are computed and factored. Those defining P_j are determined by evaluations on unisolvent grids modulo three primes, with a priori bounds on the degrees and coefficients. Expansion and substitution then verify the decompositions $P_j = E_j + yO_j$, the leading coefficients and the symmetries (Appendix A).

6.2 Limits of Diagram 2 germs in the chart

We now relate the polynomial chart to the singular sequences of Section 2. The following order estimates give a finite boundary limit for every Diagram 2 germ, so the local analysis of $\mathcal{B}_m$ covers all such germs.

Proposition 6.4 (Existence of a boundary limit)**.** *Let $m \in \mathbb{R}^5_{>0}$ and let γ be a Diagram 2 germ on a curve in A_m. Along γ put $t = (z_{12}w_5)^{-1/2}$ (either branch) and $\lambda = (tz_{12})^{-1}$, and let (Z, W, D) be given by (6.1) applied to the rotated configuration $(\lambda z, \lambda^{-1} w, \delta)$. Then t, Z, W, D are Puiseux series in s with $\operatorname{ord} t > 0$ and nonnegative orders, $(t, Z, W, D) \in \mathcal{X}_m$ for $s \neq 0$, and the limit p as $s \to 0$ is a boundary point at which Z_1, Z_2, W_3, W_4, W_5 and all D_{kl} are nonzero.*

Proof. The equations in (2.1), apart from the gauge, are invariant under $(z, w) \mapsto (\lambda z, \lambda^{-1} w)$. The chart equations are obtained from them by multiplying by powers of t, and the chosen normalization

gives $G_1 = \lambda t z_{12} - 1 = 0$ and $G_2 = \lambda^{-1} t w_5 - 1 = t^2 z_{12} w_5 - 1 = 0$. Thus the chart point belongs to $\mathcal{X}_m$.

To prove convergence, we estimate the order of each coordinate. A rotation of γ before this construction changes neither t nor the resulting chart point. We may therefore use the normalization of Lemma 3.3 and [AK12, p. 547], in which the maximal components of $\mathcal{Z}$ and $\mathcal{W}$ have the same order. We measure orders in units for which this common order is -2. Fact AK2 then gives the following estimates.

1. The pair is an isolated component of the z-diagram with z-circles, so its z-stroke is maximal (Rule 1e): $\operatorname{ord} z_{12} = -2$; by Estimate 2, $r_{12} \approx 1$, so $\operatorname{ord} w_{12} = 2$ and $\operatorname{ord} \delta_{12} = 0$.

2. The triangle is an isolated w-component with w-circles; it has a maximal w-stroke (Rule 1e), and by Rule 2e all three are maximal: $\operatorname{ord} w_{BB'} = -2$, $\operatorname{ord} z_{BB'} = 2$, $\operatorname{ord} \delta_{BB'} = 0$.

3. The circles give $\operatorname{ord} z_A = \operatorname{ord} w_B = -2$. The bodies $3, 4, 5$ carry no z-stroke and $1, 2$ no w-stroke, hence no circle of that colour (Rule 1a): $\operatorname{ord} z_B > -2$, $\operatorname{ord} w_A > -2$.

4. Hence $\operatorname{ord} z_{AB} = \operatorname{ord} w_{AB} = -2$ and $\operatorname{ord} \delta_{AB} = 2$.

5. In $w_A = \sum_{l \neq A} m_l \delta_{Al}^3 (w_A - w_l)$ the pair term has order 2 and the crossed terms order 4, so $\operatorname{ord} w_A \geq 2$; likewise $\operatorname{ord} z_B \geq 2$.

6. So $\operatorname{ord} t = 2$ and $\operatorname{ord} \lambda = 0$; $Z_A = \lambda t z_A$, $W_B = \lambda^{-1} t w_B$, D_{12}, $D_{BB'}$ and $D_{AB} = \delta_{AB}/t$ have order exactly 0, and $W_A = \lambda^{-1} w_A / t$, $Z_B = \lambda z_B / t$ have order ≥ 0.

All chart coordinates consequently converge, with $t \to 0$. Those of order exactly 0 have nonzero limits. Since $\mathcal{X}_m$ is closed, the limit belongs to it and satisfies $t = 0$. □

6.3 The torque identity and the mass conditions

The pair torque gives a scalar equation that must vanish at every boundary limit. We first derive this equation, then eliminate the triangle coordinates to obtain conditions on the masses.

Lemma 6.5 (Syzygies). *Identically in* (t, Z, W, D),

$$\sum_{k=1}^{5} m_k (W_k F_k^Z - Z_k F_k^W) = 0, \qquad \sum_A m_A (W_A F_A^Z - Z_A F_A^W) = t\, Q,$$
$$Q = \sum_{A,B} m_A m_B D_{AB}^3 (t^4 W_A Z_B - Z_A W_B).$$

At $t = 0$ the only rows with a nonzero t-derivative are F_A^W and F_B^Z, with $\partial_t F_A^W = \sum_B m_B D_{AB}^3 W_B$ and $\partial_t F_B^Z = \sum_A m_A D_{AB}^3 Z_A$. At a boundary point, $Q = \Phi := \sum_B m_B \varphi_B$ with $\varphi_B = m_1 D_{1B} + m_2 D_{2B}$, and $\partial_t F_B^Z = -\varphi_B / W_B$.

Proof. Since $W_A F_A^Z = w_A F_A^z$, $Z_A F_A^W = z_A F_A^w$, $W_B F_B^Z = w_B F_B^z$ and $Z_B F_B^W = z_B F_B^w$ for the coordinate equations F_k^z, F_k^w of (2.1), the first identity is the torque identity $\sum_k m_k (w_k F_k^z - z_k F_k^w) = 0$, which holds by antisymmetry. In the second sum the terms with D_{12} cancel by antisymmetry and the crossed terms give tQ. The derivatives are read off from (6.2), and Lemma 6.2(c) gives $D_{AB}^3 Z_A W_B = -D_{AB}$ and $D_{AB}^3 Z_A = -D_{AB}/W_B$. □

To express the torque condition in terms of the triangle, we first describe the crossed inverse distances and their signs. At a boundary point fix v_B with $v_B^2 = 1/W_B$. By Lemma 6.2(a),(c), $D_{1B}^2 = M/(m_2 W_B)$ and $D_{2B}^2 = -M/(m_1 W_B)$, so $D_{1B} = s_B\sqrt{M/m_2}\,v_B$ and $D_{2B} = is'_B\sqrt{M/m_1}\,v_B$ with $s_B, s'_B \in \{\pm 1\}$ (positive square roots); these are the 64 crossed-sign assignments. Put $\sigma_B = s_B s'_B$, $c_\pm = \sqrt{M}\,(m_1/\sqrt{m_2} \pm i\,m_2/\sqrt{m_1})$ and $x = i(m_2/m_1)^{3/2}$. Then $c_+c_- = M(m_1^3 + m_2^3)/(m_1 m_2) > 0$, $\varphi_B = c_{\sigma_B} s_B v_B$, and $m_2 D_{2B}/(m_1 D_{1B}) = \sigma_B x$ with $x^2 = -m_2^3/m_1^3$. There are two relative-sign cases:

$$\sigma_3 = \sigma_4 = \sigma_5 = \sigma, \qquad \sigma_k = -\sigma, \quad \sigma_B = \sigma \quad (B \neq k).$$

We refer to these as the equal-sign case and the opposite-sign case, respectively; they comprise two and six of the eight sign patterns. In the opposite-sign case, k is the body with the opposite sign. In both cases put $\tilde v_B = \varphi_B/c_\sigma$, so that $\tilde v_B = s_B v_B$ if $\sigma_B = \sigma$ and $\tilde v_k = \rho\, s_k v_k$ for the body with the opposite sign k, where $\rho = c_{-\sigma}/c_\sigma$ satisfies $|\rho| = 1$, $\rho \neq \pm 1$. Thus $\Phi = c_\sigma \sum_B m_B \tilde v_B$.

Proposition 6.6 (Necessary mass conditions)**.** *Let $m \in \mathbb{R}^5_{>0}$ and let p be the boundary limit of a Diagram 2 germ (Proposition 6.4). Then $\Phi(p) = 0$, and:*

(a) in the equal-sign case, its triangle satisfies (25), $\Lambda = 0$, and $L_j(m') = 0$ for every $j \neq 5$ with $S_j(W_3, W_4, W_5) = 0$ ($L_5 L'_5 = 0$ if $j = 5$);

(b) in the opposite-sign case with $k = 5$ and common sign $\sigma = \sigma_3 = \sigma_4$, its triangle satisfies (28) with $y = \sigma(x + x^{-1})/2$, and $u := y^2 = -(T-1)^2/(4T)$, $T = (m_1/m_2)^3$. Hence $P_j(y) = 0$ for every j with $S_j(W_3, W_4, W_5) = 0$: if $m_1 = m_2$ then $K_j(m') = 0$, and if $m_1 \neq m_2$ then $u < 0$ and $E_j(u) = O_j(u) = 0$.

Proof. All equations vanish along the germ, and $t \not\equiv 0$. Lemma 6.5 therefore gives $Q \equiv 0$ along it, and continuity implies $\Phi(p) = Q(p) = 0$, including at fold points.

In case (a), $\Phi = c_\sigma \sum_B m_B s_B v_B$ with $c_\sigma \neq 0$. The product of $m_3 v_3 \pm m_4 v_4 \pm m_5 v_5$ over the four sign choices equals Λ, because $(m_B v_B)^2 = f_B$. Hence the vanishing of the torque gives (25). In case (b), put $r_B = m_1 m_B D_{1B}$, so that $r_B^2 = (m_1^2 M/m_2) f_B$, and $x' = \sigma x$. The torque becomes $\Phi = (r_3 + r_4)(1 + x') + r_5(1 - x')$, as in (26) of [AK12, §8.2]. Its product over the four sign changes of r_3, r_4 is $4x'^2(m_1^2 M/m_2)^2$ times the left-hand side of (28), with $y = (x' + 1/x')/2$.

Thus W_4 is a common root of $S_j^\flat$ and $\Lambda^\flat$ in case (a), or of the shape polynomial and $(28)^\flat$ in case (b). The resultants in Definition 6.3 vanish, and their nonvanishing prefactors $c_j \neq 0$, $m_3 \neq 0$ give the asserted mass conditions. Finally, $y = iv$ with $v = \sigma\big((m_2/m_1)^{3/2} - (m_1/m_2)^{3/2}\big)/2 \in \mathbb{R}$, which vanishes if and only if $m_1 = m_2$. Since E_j, O_j have real coefficients, separating the real and imaginary parts of $P_j(iv) = E_j(-v^2) + ivO_j(-v^2) = 0$ proves the last assertion. □

6.4 Local exclusion of boundary limits

We turn the torque equation into a local criterion for excluding a boundary point. An invertible coordinate Jacobian reduces the problem to one analytic function; a corank-two reduction instead yields three equations in two parameters.

Let F_* denote the twenty rows other than F_2^Z and F_5^Z. A boundary point p is *excluded* if no holomorphic curve germ $c\colon (\mathbb{C}, 0) \to (\mathcal{X}_m, p)$ has $t \circ c \not\equiv 0$; an isolated point of $\mathcal{X}_m$ is excluded.

Lemma 6.7 (Local analytic lemma)**.** *Let $m \in \mathbb{R}^5_{>0}$ and $p \in \mathcal{B}_m$.*

(a) *If p is excluded, it is not the boundary limit of a Diagram 2 germ.*

(b) *Suppose $dF_*(p)$ has rank 20, so that near p the set $\{F_* = 0\}$ is a smooth curve C. Then near p the set $\mathcal{X}_m$ consists of the points of C at which $tQ = 0$, and p is excluded if $Q|_C \not\equiv 0$. If moreover $\partial F_*/\partial(Z, W, D)(p)$ is invertible, then C is the graph of an analytic map $t \mapsto \phi(t)$, and p is an isolated point of $\mathcal{X}_m$ as soon as $q(t) = Q(t, \phi(t)) \not\equiv 0$, in particular if $q(0) = \Phi(p) \neq 0$ or $q'(0) \neq 0$.*

(c) *Suppose some twenty of the twenty-two polynomials have a differential of rank 20 at p, so that near p their common zeros form a smooth curve C'. If one of the other two polynomials does not vanish identically on C', then p is an isolated point of $\mathcal{X}_m$.*

Proof. (a) By Proposition 6.4, a Diagram 2 germ with limit p gives, after the substitution $s = \tau^N$ that clears the denominators of the Puiseux exponents, a holomorphic germ $\tau \mapsto (t, Z, W, D)$ in $\mathcal{X}_m$ at p with $t \not\equiv 0$. (b) On C all rows except F_2^Z, F_5^Z vanish and $W_5 = 1$, so Lemma 6.5 gives $m_2 W_2 F_2^Z = tQ = -m_5 F_5^Z$ on C; and $W_2(p) = m_1 D_{12}(p) \neq 0$. This describes $\mathcal{X}_m$ near p. A germ c in $\mathcal{X}_m$ at p lies in C, and if $t \circ c \not\equiv 0$ then $Q \circ c \equiv 0$; when $Q|_C \not\equiv 0$ its zeros on the curve C are isolated, so c is constant, which contradicts $t \circ c \not\equiv 0$. In the graph case $\mathcal{X}_m$ is, near p, the set of points $(t, \phi(t))$ with $tq(t) = 0$, which is $\{p\}$ when $q \not\equiv 0$. (c) Near p, $\mathcal{X}_m$ lies in C' and in the zero set of the other polynomial, whose restriction to the smooth curve C' is analytic and not identically zero, hence has an isolated zero at p. □

At the boundary points where the Jacobian of the twenty-two equations has rank 19, the implicit function theorem leaves two local parameters. The following lemma reduces exclusion to the Taylor coefficients of the three remaining equations. Its proof follows the possible tangent directions of a formal arc by a Newton–Puiseux argument after Lyapunov–Schmidt reduction.

Lemma 6.8 (Corank-two isolation)**.** *Let $m \in \mathbb{R}^5_{>0}$ and $p \in \mathcal{B}_m$. Let F_R be nineteen of the twenty-two chart polynomials, $F_{R'}$ the other three, and $y \neq t$ a chart coordinate such that the derivative of $(F_R, t, y)\colon \mathbb{C}^{21} \to \mathbb{C}^{21}$ at p is invertible. By the implicit function theorem, near p the set $\{F_R = 0\}$ is the image of an injective analytic map ϕ, defined near $0 \in \mathbb{C}^2$, with $\phi(0) = p$ and $(y, t) \circ \phi(a, t) = (y(p) + a, t)$, and near p*

$$\mathcal{X}_m = \phi\big(g^{-1}(0)\big), \qquad g = F_{R'} \circ \phi\colon (\mathbb{C}^2, 0) \to (\mathbb{C}^3, 0).$$

Let $g^{(j)}$ be the homogeneous part of degree j of g, and for $\xi \in \mathbb{C}^2$ let $\Pi_\xi = Dg^{(2)}(\xi)$, a 3×2 matrix. Suppose:

(a) *$g^{(1)} = 0$; equivalently, the Jacobian at p of all twenty-two polynomials has rank 19;*

(b) *the components of $g^{(2)}$ have only finitely many common zeros $[\xi_1], \dots, [\xi_r]$ in $\mathbb{P}^1$, and $\Pi_{\xi_j} \neq 0$ for every j;*

(c) *for every j there is $N \geq 3$ such that the vector $r_N(\xi_j)$ defined below exists and does not lie in $\Pi_{\xi_j}\mathbb{C}^2$.*

Then no nonconstant formal arc $x \in \mathbb{C}[[\tau]]^2$ with $x(0) = 0$ satisfies $g(x) = 0$. Hence p is an isolated point of $\mathcal{X}_m$; in particular p is excluded. The same holds if (a)–(c) *hold for $Kg(L\,\cdot)$, $K \in GL_3(\mathbb{C})$, $L \in GL_2(\mathbb{C})$, in place of g.*

The vectors $r_N(\xi)$ are defined for a common zero ξ of $g^{(2)}$ and a fixed $e \in \mathbb{C}^2 \setminus \mathbb{C}\xi$. As $\Pi_\xi \xi = 2g^{(2)}(\xi) = 0$ and $\Pi_\xi \neq 0$, the vector $\varpi = \Pi_\xi e$ spans the line $\Pi_\xi \mathbb{C}^2$. Put $\beta_1 = 0$. For $n \geq 1$, if β_n is defined, let $r_{n+2}(\xi)$ be the coefficient of λ^{n+2} in $g(\lambda\xi + \beta_n(\lambda)e)$; if moreover $r_{n+2}(\xi) = -b_{n+1}\varpi$ with $b_{n+1} \in \mathbb{C}$, put $\beta_{n+1}(\lambda) = \beta_n(\lambda) + b_{n+1}\lambda^{n+1}$. Thus

$$r_3(\xi) = g^{(3)}(\xi), \qquad r_4(\xi) = g^{(4)}(\xi) + b_2\, Dg^{(3)}(\xi)e + b_2^2\, g^{(2)}(e).$$

Condition (c) with $N = 3$ says that the 3×3 matrix $\big(\Pi_{\xi_j} \,|\, r_3(\xi_j)\big)$ has rank two; with $N = 4$ it says that $\big(\Pi_{\xi_j} \,|\, r_3(\xi_j)\big)$ has rank one and $\big(\Pi_{\xi_j} \,|\, r_4(\xi_j)\big)$ has rank two.

Proof. The points of $\{F_R = 0\}$ near p are the points $\phi(a,t)$, and $\phi(a,t) \in \mathcal{X}_m$ if and only if $g(a,t) = 0$. If p were not isolated in $\mathcal{X}_m$, the curve selection lemma [BCR98, Thm. 2.5.5], applied to the real algebraic set $\mathcal{X}_m \subset \mathbb{C}^{21} = \mathbb{R}^{42}$, would give a nonconstant semialgebraic arc $\gamma\colon [0,\varepsilon) \to \mathcal{X}_m$ with $\gamma(0) = p$. Its coordinates are convergent Puiseux series, so $x = (y \circ \gamma - y(p), t \circ \gamma)$ is, after a substitution $s = \tau^{N'}$, a nonconstant arc in $\mathbb{C}[[\tau]]^2$ with $g(x) = 0$. So the first assertion implies the second. The zero set of $Kg(L\,\cdot)$ is $L^{-1}g^{-1}(0)$, which gives the last assertion.

Let x be a nonconstant formal arc with $x(0) = 0$ and $g(x) = 0$, and write $x = \tau^k\xi' + O(\tau^{k+1})$ with $\xi' \neq 0$. By (a), $g(x) = \tau^{2k}g^{(2)}(\xi') + O(\tau^{2k+1})$, so $\xi' \in \mathbb{C}^*\xi$ for one of the $\xi = \xi_j$ of (b). Write $x = u\xi + \tilde{\theta}e$; then $\operatorname{ord} u = k < \operatorname{ord}\tilde{\theta}$. Take $\lambda = u(\tau)$ as the new parameter; then τ is a power series in $\lambda^{1/k}$ with nonzero linear term, and $x = \lambda\xi + \theta e$ with $\theta \in \mathbb{C}[[\lambda^{1/k}]]$ and $\operatorname{ord}\theta > 1$ ($\theta = 0$ is allowed).

We show by induction on $n \geq 1$ that β_n is defined, that $g(\lambda\xi + \beta_n e) = O(\lambda^{n+2})$ and that $o_n = \operatorname{ord}(\theta - \beta_n) > n$. For $n = N - 2$ this contradicts (c). For $n = 1$ the claim holds by (a), $g^{(2)}(\xi) = 0$ and $\operatorname{ord}\theta > 1$. Taylor's formula in the direction e gives

$$0 = g(\lambda\xi + \theta e) = g(\lambda\xi + \beta_n e) + (\theta - \beta_n)\big(\lambda\varpi + O(\lambda^2)\big) + O(\lambda^{2o_n}),$$

because $Dg(\lambda\xi + \beta_n e)e = \lambda\Pi_\xi e + \beta_n \Pi_e e + O(\lambda^2) = \lambda\varpi + O(\lambda^2)$ by (a) and $\operatorname{ord}\beta_n \geq 2$. If $o_n < n+1$, the term $(\theta - \beta_n)\lambda\varpi$ has order $o_n + 1 < n + 2$, every other term has larger order ($2o_n > o_n + 1$), and its leading coefficient is a nonzero multiple of $\varpi \neq 0$; this is impossible. Hence $o_n \geq n + 1$, and the coefficient of λ^{n+2} gives $r_{n+2}(\xi) + b\varpi = 0$, where b is the coefficient of λ^{n+1} in θ. So β_{n+1} is defined, $b_{n+1} = b$ since $\varpi \neq 0$, and $o_{n+1} > n + 1$. The same expansion with β_{n+1} in place of θ gives $g(\lambda\xi + \beta_{n+1}e) = (r_{n+2}(\xi) + b_{n+1}\varpi)\lambda^{n+2} + O(\lambda^{n+3}) = O(\lambda^{n+3})$. □

Remark 6.9. The ξ_j are the points where the strict transform of $g^{-1}(0)$ could meet the exceptional curve of the blow-up of $0 \in \mathbb{C}^2$, and the induction is the Newton–Puiseux algorithm in the blow-up chart at ξ_j.

We next compute the first derivative of the reduced torque equation using the triangle block and the quadratic form Q_T. For a boundary point p, let $J_T(p)$ be the Jacobian at p of the nine equations $F_3^Z, F_3^W, F_4^Z, F_4^W, F_5^W, T_{34}, T_{35}, T_{45}, G_2$ with respect to the nine triangle coordinates $Z_B, W_B, D_{BB'}$. If $J_T(p)$ is invertible and $\xi \in \mathbb{C}^3$, let (ζ, η, ϑ) be the solution of $J_T(p)(\zeta,\eta,\vartheta) = r(\xi)$. Here $r(\xi)$ has entries ξ_3/W_3 and ξ_4/W_4 in the rows F_3^Z, F_4^Z and 0 elsewhere. Define

$$Q_T(\xi) = -\tfrac{1}{2}\textstyle\sum_B m_B\, \xi_B\, \eta_B / W_B.$$

This quadratic form Q_T depends only on m' and the triangle of p.

Lemma 6.10 (Second-order coefficient)**.** *Let p be a boundary point with $\Phi(p) = 0$ and $J_T(p)$ invertible. Then $\partial F_*/\partial(Z, W, D)(p)$ is invertible and, in the notation of Lemma 6.7 and of the crossed data,*

$$q(0) = 0, \qquad q'(0) = c_\sigma^2\, Q_T(\tilde{v}).$$

Proof. At $t = 0$ the pair rows involve only Z_A, W_A, D_{12}, the triangle rows only the triangle coordinates, and T_{AB} involves no crossed coordinate other than D_{AB}, with $\partial T_{AB}/\partial D_{AB} = 2/D_{AB}$; hence the Jacobian is block lower triangular. To check its pair block directly, write

$$d = D_{12}, \qquad u = Z_2 - Z_1, \qquad h = W_2 - W_1.$$

At the boundary point $u = 1$, $Md^3 = 1$ and $h = Md$. The five selected pair rows are $F_1^Z, F_1^W, F_2^W, T_{12}, G_1$, and at $t = 0$ they satisfy

$$F_1^Z = Z_1 + m_2 d^3 u, \qquad F_2^W - F_1^W = (1 - Md^3)h,$$
$$m_1 F_1^W + m_2 F_2^W = m_1 W_1 + m_2 W_2, \qquad T_{12} = d^2 uh - 1.$$

A vector in their differential kernel first has $\dot{u} = 0$ from the gauge, then $\dot{d} = 0$ from the difference of the two W-rows, whose $\dot{d}$ coefficient is $-3Md^2h \neq 0$. The distance constraint gives $\dot{h} = 0$, the weighted W-sum gives $\dot{W}_1 = \dot{W}_2 = 0$, and the Z-row and gauge give $\dot{Z}_1 = \dot{Z}_2 = 0$. Thus the pair block is invertible. Differentiate $F_*(t, \phi(t)) = 0$ at $t = 0$. By Lemma 6.5, the pair Z-row, distance constraint and gauge have no explicit t-forcing, whereas

$$\partial_t F_A^W = -Z_A^{-1} \sum_B m_B D_{AB}, \qquad \partial_t F_1^W - \partial_t F_2^W = \frac{M\Phi}{m_1 m_2} = 0.$$

The same difference, distance constraint and Z-row calculation therefore gives $D'_{12}(0) = Z'_1(0) = Z'_2(0) = 0$.

The triangle rows give $J_T(p)(Z', W', D')_T(0) = r(\varphi)$, hence $W'_B(0) = c_\sigma \eta_B(\tilde{v})$. On C the distance constraints $T_{AB} = 0$ give $Q = \sum_{A,B} m_A m_B D_{AB} + O(t^2)$ and $D'_{AB} = -\frac{1}{2} D_{AB}(Z'_A/Z_A + W'_B/W_B)$. Therefore $q'(0) = \sum_{A,B} m_A m_B D'_{AB}(0) = -\frac{1}{2} \sum_B m_B \varphi_B W'_B(0)/W_B = c_\sigma^2 Q_T(\tilde{v})$. □

Definition 6.11. For a boundary point p with $J_T(p)$ invertible, put $\Psi_T(p) = Q_T(\tilde{v})$ in the equal-sign case and $\Psi_1(p) = Q_T(\tilde{v})$ in the opposite-sign case. The boundary point is *isolated at second order* if $\Phi(p) = 0$, $J_T(p)$ is invertible and $Q_T(\tilde{v}) \neq 0$; by Lemmas 6.7 and 6.10 it is then an isolated point of $\mathcal{X}_m$, and the omitted rows F_2^Z, F_5^Z vanish to order exactly two along the curve C.

At a boundary point with $\Phi(p) = 0$ the vector $\tilde{v}$ is determined by the triangle and the signs s_B (for the body with the opposite sign k, $m_k \tilde{v}_k = -\sum_{B \neq k} m_B \tilde{v}_B$), so Ψ_T and Ψ_1 do not involve m_1, m_2. For an opposite-sign boundary point with $k = 5$ the condition $\Phi = 0$ requires $|m_3 \tilde{v}_3 + m_4 \tilde{v}_4| = m_5 |v_5|$, i.e. the vanishing of the function $G = |m_3 \tilde{v}_3 + m_4 \tilde{v}_4|^2 - m_5^2$ (recall $W_5 = 1$), which again depends on the triangle only; then $\tilde{v}_5 = \rho s_5 v_5$ fixes ρ up to sign, hence m_1/m_2 up to exchanging m_1 and m_2.

6.5 Exclusion on the mass loci

We apply the local criteria to the mass loci determined by $\Phi = 0$. The relative signs of the crossed inverse distances separate the cases; when one relative sign differs, we also distinguish $m_1 = m_2$ from $m_1 \neq m_2$. Table 2 displays the corresponding scalar obstructions and computational methods. The computer-assisted statements below use exact arithmetic, rational intervals, and IEEE ball arithmetic with proved rounding bounds (Appendix A).

Boundary points	Mass locus	Method	Computation
$\Phi \neq 0$ (any)	all m	$Q(p) = \Phi(p) \neq 0$, $tQ = 0$	exact identities
equal-sign, S_4, or S_5 with $L_5' = 0$	$L_4 = 0$, $L_5' = 0$	no point in $\mathbb{R}^3_{>0}$	exact
equal-sign, S_5	$L_5 = 0$: $m_3 = m_4 = m_5$	order three; corank-two blow-up	exact jets
equal-sign, S_1–S_3, S_6–S_8	rays of L_1–L_3, L_6–L_8	second order, $\Psi_T \neq 0$	Bernstein subdivision, Krawczyk method; interval arithmetic
opposite-sign, $m_1 = m_2$, S_3, S_6–S_8	$K_3, K_6, K_7, K_8 = 0$	no point in $\mathbb{R}^3_{>0}$	Bernstein coefficients
opposite-sign, $m_1 = m_2$, S_1, S_2	$K_1 = 0$, $K_2 = 0$	second order	exact; interval cover
opposite-sign, $m_1 = m_2$, S_4	$K_4 = 0$: cones over four arcs	second order	exact
opposite-sign, $m_1 = m_2$, S_5	$K_5 = 0$: $\mathbb{R}_{>0}(1, 1, \rho_*)$	orders one to four	exact; interval arithmetic
opposite-sign, $m_1 \neq m_2$, not fold	$E_j(u) = O_j(u) = 0$	second order, $\Psi_1 \neq 0$: ball covers, exact lemmas	ball and interval arithmetic; exact
opposite-sign, $m_1 \neq m_2$, fold	$E_5(u) = O_5(u) = 0$	corank one; $Q\|_C$ has a simple zero	exact

Table 2: The Diagram 2 cases. A boundary point with $\Phi = 0$ lies on S_j for the j given in the first column (Lemma 6.2); opposite-sign boundary points are taken with $k = 5$.

Lemma 6.12 (computer-assisted)**.** (a) *The polynomials L_4, L_5', K_3, K_6, K_7 and K_8 have no zero in $\mathbb{R}^3_{>0}$. More precisely, at positive masses every normalized W-shape on S_4 is real (the centre-of-mass section of S_4 is a cubic with positive discriminant) and satisfies $\Lambda \neq 0$. Here the normalized shape is real in the sense $(W_3, W_4, W_5) \in \mathbb{R}^3$.* (b) *If $m_3 = m_4 = m_5$, no S_j with $j \neq 5$ has a root satisfying* (25)*.*

The verification uses exact arithmetic over $\mathbb{Q}$. For L_4, a rational parametrization of S_4 gives the obstruction: at positive masses, a real normalized W-shape has positions W_B of different signs. Then Λ is a sum of nonnegative terms, at least one of which is positive. The coefficients of L_5' are positive. For K_3, K_6, K_7, K_8, nonnegative Bernstein coefficients are checked on twelve projective charts, three for each polynomial. Each chart is split into the two sectors obtained by blowing up a corner of the cone. For (b), the verification shows that $S_j^\flat$ and $\Lambda^\flat$ are coprime when $m_3 = m_4 = m_5$ and $j \neq 5$ (Appendix A).

Lemma 6.13 (computer-assisted)**.** *Let $m \in \mathbb{R}^5_{>0}$ with $m_3 = m_4 = m_5$. Every boundary point at m whose triangle lies on S_5 (an all-equal Lagrange class or a fold triangle) and which satisfies $\Phi = 0$ or is a fold boundary point is an isolated point of $\mathcal{X}_m$. At each such m these are 72 boundary points of the all-equal classes and 1152 fold points, 1224 in all.*

We first normalize the masses. Common mass scaling gives an exact isomorphism of the gauged chart, so we may take $m_3 = m_4 = m_5 = 1$ and hence $\delta_* := M_T^{-1/3} = 3^{-1/3}$. Every positive pair (m_1, m_2) has a unique parametrization by $(\nu, M) \in (0,1) \times \mathbb{R}_{>0}$ of the form

$$m_1 = \frac{4M\nu^2}{(1+\nu^2)^2}, \qquad m_2 = \frac{M(1-\nu^2)^2}{(1+\nu^2)^2},$$

Indeed, $M = m_1 + m_2$ and $\nu = 1/(\chi + \sqrt{1+\chi^2})$, where $\chi = \sqrt{m_2/m_1} = (1-\nu^2)/(2\nu)$.

Consider first the 72 boundary points in the all-equal classes. They satisfy $\Phi = 0$, have invertible $J_T(p)$, and satisfy $q(0) = q'(0) = 0$. The next coefficient is obtained from $F_5^Z = \kappa_3 t^3 + O(t^4)$ along C, which gives $q''(0) = -2\kappa_3$. For the nine boundary points of one sign class with $D_{BB'} = \omega^k \delta_*$, $(W_3, W_4, W_5) = (\omega, \omega^2, 1)$ and $D_{12} = \omega^j M^{-1/3}$, the exact computation gives

$$\kappa_3 = \omega^j M^{5/3} R_1 (1 - i\chi) + 3\omega^k \delta_* M R_2 (1 - i\chi^5), \qquad R_1 = \frac{18\nu^2(1-\nu^2)}{(1+\nu^2)^3}, \quad R_2 = \frac{6\nu^2(1+\nu^2)}{(1-\nu^2)^3}.$$

To obtain these nine coefficient identities from the case $j = k = 0$, first regard ν, b as algebraically independent indeterminates, set $M = b^3$, and work over $\mathbb{Q}(\omega, i, \delta_*)(\nu, b)$, where $\delta_*^3 = 1/3$. The commuting automorphisms

$$\delta_* \mapsto \omega^k \delta_*, \qquad b \mapsto \omega^{-j} b$$

fix ν, ω, i and b^3, hence the masses and the chart polynomials. They carry the representative boundary point and its formal implicit solution to those with $D_{BB'} = \omega^k \delta_*$ and $D_{12} = \omega^j b^{-1}$. In particular, $b^5 \mapsto \omega^j b^5$, giving the displayed identity for κ_3.

We now specialize $0 < \nu < 1$ and $b = M^{1/3} > 0$. The boundary point coordinates are regular throughout this range, and the selected-coordinate Jacobian is invertible: its triangle block is J_T, its pair block is invertible by Lemma 6.10, and its crossed block has diagonal entries $2/D_{AB} \neq 0$. At each Taylor order, the implicit equations determine the next coefficient by applying the inverse of this same Jacobian to expressions in the preceding coefficients. Thus the Taylor coefficients are regular at every such specialization. Clearing the nonvanishing denominators in the formal identities and then specializing proves the formula at all positive masses. The remaining 63 boundary points are obtained by exchanging bodies 3 and 4, conjugating, and reversing t.

It remains to see that the two terms in the displayed coefficient cannot cancel. We have $R_1, R_2 > 0$, and $(1 - i\chi)/(1 - i\chi^5)$ is a positive multiple of $1 + \chi^6 + i(\chi^5 - \chi)$. Moreover, $|\chi^5 - \chi| \leq \chi + \chi^5 \leq 1 + \chi^6$, since $(1 + \chi^6) - (\chi + \chi^5) = (1 - \chi)(1 - \chi^5) \geq 0$. The quotient of the two terms of κ_3 is therefore ω^{j-k} times a number whose argument has absolute value at most $\pi/4$, and cannot equal -1. Thus $\kappa_3 \neq 0$, and Lemma 6.7(b) gives isolatedness.

We treat the fold points of rank 20 first. Among the fold points, 1080 have a Jacobian of rank 20 for the twenty-two polynomials. At each such boundary point, twenty equations together with $h = W_4 - W_4(p)$ have an invertible derivative. For 864 boundary points these equations are F_*; for the remaining 216 they are all equations other than F_2^Z and F_3^Z. Along the resulting local curve,

an omitted equation has a nonzero coefficient of h^3. Lemma 6.7(c) therefore gives isolatedness for these boundary points.

The remaining 72 fold points have rank 19 and form one orbit under the symmetries described below. We choose a representative $p_* = p_*(\nu, M)$ whose pair coordinates are as in Lemma 6.2(a), with $D_{12} = M^{-1/3}$, and whose fold triangle is

$$(W_3, W_4, W_5) = (\omega, \omega^2, 1), \qquad Z_B = -\delta_* W_B, \qquad (D_{34}, D_{35}, D_{45}) = \delta_*(1, \omega, \omega^2).$$

The six crossed inverse distances of this representative are

$$(D_{13}, D_{14}, D_{15}) = -\frac{1+\nu^2}{1-\nu^2}(\omega, \omega^2, 1),$$
$$(D_{23}, D_{24}, D_{25}) = \frac{i(1+\nu^2)}{2\nu}(\omega, \omega^2, 1).$$

This sign choice fixes the coordinate t relative to the Taylor coefficients below; simultaneously reversing all crossed signs would reverse this coordinate. For this representative, we use Lemma 6.8 with $F_{R'} = (F_2^W, F_4^W, F_5^W)$, with F_R consisting of the other nineteen equations, and with $y = Z_3$. The two kernel coordinates are the chart parameter t and $a = Z_3 - Z_3(p_*)$. At $t = 0$, the derivative of (F_R, t, Z_3) is block lower triangular, as in the proof of Lemma 6.10. Its diagonal blocks are (1) for dt; the triangle block (rows $F_3^Z, F_3^W, F_4^Z, F_5^Z, T_{34}, T_{35}, T_{45}, G_2, dZ_3$, columns $Z_B, W_B, D_{BB'}$), with determinant $18\delta_*$; the pair block (rows $F_1^Z, F_1^W, F_2^Z, T_{12}, G_1$, columns Z_A, W_A, D_{12}), with determinant $-3M^{-1/3}$; and the diagonal crossed block $(2/D_{AB})$, with determinant $-512i\,\nu^3(1-\nu^2)^3/(1+\nu^2)^6$. All these determinants are nonzero, so the required invertibility holds for every (ν, M). The differentials of the three omitted equations vanish on the two-dimensional kernel of $dF_R(p_*)$, establishing (a). We write the reduced equations as $g = (\psi_2, \psi_4, \psi_5)$, where $\psi_k = F_k^W \circ \phi$.

At $M = 1$, exact arithmetic over $\mathbb{Q}(\nu)(\sqrt{3}, i, \delta_*)$ gives the jet of g through degree four. Substitution of the corresponding jet of ϕ in the nineteen rows of F_R gives $O((|a|+|t|)^5)$. In the quadratic part, $\psi_2^{(2)} = 0$, there is no a^2 term, and the 3×3 coefficient matrix has rank one. So $g^{(2)} = \psi_4^{(2)}\, n$ for a fixed vector $n = (0, 1, n_5)$, where $\psi_4^{(2)} = t(\gamma_{11} a + \gamma_{02} t)$ and

$$\gamma_{11} = \frac{9i\delta_*(\nu - i)(\nu + i)^3\, \Upsilon(\nu)}{4\nu(\nu^2-1)(\nu^2+1)^2}, \qquad \Upsilon(\nu) = \nu^4 - 2i\nu^3 - 6\nu^2 + 2i\nu + 1 = \tfrac{1}{4}\big(3(\nu - i)^4 + (\nu + i)^4\big).$$

For real ν, $|3(\nu - i)^4| = 3(\nu^2+1)^2 > |(\nu + i)^4|$, so $\Upsilon(\nu) \neq 0$ and $\gamma_{11} \neq 0$ for $0 < \nu < 1$. Hence the common zeros of $g^{(2)}$ are two lines: the *boundary line* $\xi_\partial = (1, 0)$, tangent to $\{t = 0\}$, and the line transverse to $\{t = 0\}$ $\xi_s = (-\gamma_{02}, \gamma_{11})$. Moreover $\Pi_\xi = n\, d\psi_4^{(2)}(\xi)$, with $d\psi_4^{(2)}(\xi_\partial) = (0, \gamma_{11})$ and $d\psi_4^{(2)}(\xi_s) = \gamma_{11}(\gamma_{11}, \gamma_{02})$, both nonzero. So (b) holds. On the transverse line take $N = 3$: the matrix $(\Pi_{\xi_s} \,|\, r_3(\xi_s))$ has the 2×2 minor

$$\mu_3 = -\frac{3^{10}(\nu - i)^4(\nu + i)^{16}\, \Upsilon(\nu)^5}{2^{12}\delta_*\, \nu^6(\nu^2-1)^6(\nu^2+1)^8} \neq 0.$$

On the boundary line take $e = (0, 1)$, so that $\varpi = \gamma_{11} n$, and $N = 4$. Identically in ν, $r_3(\xi_\partial) = g^{(3)}(1, 0) = -b_2\varpi$, and the matrix $(\Pi_{\xi_\partial} \,|\, r_4(\xi_\partial))$ has the minor

$$\mu_4 = -\frac{27i\delta_*(\nu - i)^3(\nu + i)^5\, \Upsilon(\nu)}{64\, \nu^3(\nu^2-1)^3} \neq 0.$$

So (c) holds. Every coefficient used is regular for $0 < \nu < 1$: the jets are polynomials in the coordinates of p_*, the masses and the entries of the inverses of the three diagonal blocks, and the only further division is by γ_{11}. Hence the identities above hold at every $\nu \in (0,1)$.

For general M, $p_*(\nu, M)$ is obtained from $p_*(\nu, 1)$ by multiplying W_1, W_2 by $M^{2/3}$ and D_{12} by $M^{-1/3}$. Exact Laurent arithmetic in $M^{1/3}$ gives the reduced map $\tilde{g}(a, \tilde{t}) = \big(M^{-2/3}\psi_2, \psi_4, \psi_5\big)(a, \tilde{t}/M)$; the selected rows vanish along both lines through the orders used. The quadratic part of $\tilde{g}$ does not depend on M, the compatibility $r_3(\xi_\partial) \in \mathbb{C}\varpi$ persists, and the two minors become $M^{-3}\mu_3(\nu)$ and $M^{-5/3}\mu_4(\nu)$. By Lemma 6.8, applied to $\tilde{g}$, $p_*(\nu, M)$ is an isolated point of $\mathcal{X}_m$ for every $0 < \nu < 1$ and $M > 0$.

To transfer isolatedness from this representative to the other boundary points, we use the symmetries of the chart equations. Because the twenty-two polynomials have real coefficients and $m_3 = m_4$, they are preserved by two commuting involutions: the biholomorphic reversal $(t, D_{AB}) \mapsto (-t, -D_{AB})$ and the antiholomorphic map obtained by conjugating and exchanging bodies 3 and 4. Both fix (ν, M) and together carry p_* to the other three sign choices in its orbit. Multiplying t, Z_B, W_A, D_{12} and $D_{BB'}$ by ω^j multiplies each polynomial by a cube root of unity and preserves the gauges. A permutation of bodies $3, 4, 5$, followed by the chart-torus action restoring the gauges, gives a local biholomorphism of $\mathcal{X}_m$. The implicit function theorem applies because the transversality determinant is $2(Z_2 - Z_1)W_5 = 2$. These $3 \cdot 6$ maps produce eighteen boundary points for each of the four sign choices. The resulting 72 boundary points form precisely one orbit under these discrete symmetries and are disjoint from the 1080 boundary points already treated. Every map used here preserves isolatedness.

The computations in the proof of Lemma 6.13, namely the coefficients κ_3, the ranks and the coefficients of h^3 at the 1080 fold points, the determinants, jets and minors at p_*, and the orbit of p_*, are exact, over number fields containing ω, i and δ_* and over rational function fields over them in ν and $M^{1/3}$ (Appendix A).

Lemma 6.14 (computer-assisted)**.** (a) *The zero set of L_8 in $\mathbb{R}^3_{>0}$ is $\mathbb{R}_{>0}(p_*, q_*, 1) \cup \mathbb{R}_{>0}(q_*, p_*, 1)$ with $p_* = 0.67987758\ldots$ and $q_* = 0.82343785\ldots$.* (b) *The zero set of L_1 in $\mathbb{R}^3_{>0}$ is $\mathbb{R}_{>0}(a_*, b_*, 1) \cup \mathbb{R}_{>0}(a_*, 1, b_*)$ with $a_* = 1.02227400\ldots$ and $b_* = 0.90559635\ldots$; at $(a_*, b_*, 1)$ the zeros of S_1 that satisfy* (25) *form exactly one pair of complex conjugate non-real shapes.* (c) *Consequently the zero sets of $L_1L_2L_3$ and of $L_6L_7L_8$ in $\mathbb{R}^3_{>0}$ are the rays through the six permutations of $(a_*, b_*, 1)$ and of $(1, p_*, q_*)$, respectively.*

For (a), exact Bernstein subdivision of the mass simplex locates the possible zeros, and the Krawczyk method in rational interval arithmetic encloses them. For (b), subresultants eliminate the shape variable. Bernstein subdivision and the signs of resultants locate the mass zeros, which are again enclosed by the Krawczyk method. The quadratic subresultant has negative discriminant in these enclosures, giving the conjugate non-real shapes (Appendix A). Part (c) follows from (a), (b) and the permutation symmetry following Definition 6.3.

Lemma 6.15 (computer-assisted)**.** *Let $m \in \mathbb{R}^5_{>0}$ with m' on one of the rays of Lemma 6.14(c), and let p be a equal-sign boundary point at m with $\Phi(p) = 0$. Then $J_T(p)$ is invertible and $\Psi_T(p) \neq 0$, so p is isolated at second order.*

The enumeration includes all 48 Lagrange triangles at each permutation of $(1, p_*, q_*)$ and the six Euler triangles on the conjugate shape pair at each permutation of $(a_*, b_*, 1)$. At these mass vectors, only the factor defining the ray vanishes among $L_1, \ldots, L_8, L'_5$. Proposition 6.6(a) therefore

excludes any other triangle from carrying an equal-sign boundary point with $\Phi = 0$. For $36 + 36$ triangles, there is a sign vector satisfying $\sum_B m_B s_B v_B = 0$, unique up to a simultaneous sign change. Their J_T are invertible, and complex interval arithmetic gives $|\Psi_T| > 0.36$ (L_6–L_8) and $|\Psi_T| > 0.3$ (L_1–L_3) at the normalized representatives.

Common mass scaling preserves isolatedness and $\Psi_T \neq 0$, extending these conclusions to the entire rays. Moreover, Ψ_T depends on m' and the triangle, so the same conclusion holds for every positive pair (m_1, m_2). The Krawczyk method encloses the shapes and the square and cube roots; the solutions of the linear systems are also enclosed rigorously (Appendix A).

Lemma 6.16 (computer-assisted)**.** *Let $m \in \mathbb{R}^5_{>0}$ with $m_1 = m_2$, let p be an opposite-sign boundary point at m with $k = 5$ and $\Phi(p) = 0$, and let S_j carry its triangle, so that $K_j(m') = 0$. Then p is an isolated point of $\mathcal{X}_m$. More precisely:*

(a) $j \notin \{3, 6, 7, 8\}$, by Lemma 6.12;

(b) if $j \in \{1, 2\}$: at every zero of K_1 in $\mathbb{R}^3_{>0}$ the S_1 shapes compatible with (28) at $y = 0$ reduce to one simple common root, which is real and lies on one of two explicit real branches; $J_T(p)$ is invertible and $q'(0) \neq 0$. The case $j = 2$ is the image of $j = 1$ under the exchange of bodies 3 and 4;

(c) if $j = 4$: m' lies in one of the positive cones over the four arcs $k \mapsto \mathrm{II}(k), \mathrm{III}(k)$, $5^{-1/4} < k < 1$, and their images under $m_3 \leftrightarrow m_4$. Thus $m' = c\,\mathrm{II}(k)$ or $c\,\mathrm{III}(k)$, or an exchanged vector, for some $c > 0$. Here $\alpha = 4k(1 + k^4)$, $\beta = (1 - k^2)(1 + 6k^2 + k^4)$, $\gamma = (1 + k^2)(6k^2 - 1 - k^4)$, $\mathrm{II} = ((\gamma^3 - \beta^3)\alpha, (\alpha^3 - \gamma^3)\beta, (\alpha^3 - \beta^3)\gamma)$ and $\mathrm{III} = ((\gamma^3 - \beta^3)\alpha, (\gamma^3 + \alpha^3)\beta, (\alpha^3 + \beta^3)\gamma)$; $J_T(p)$ is invertible and $q'(0) \neq 0$;

(d) if $j = 5$: $m' \in \mathbb{R}_{>0}(1, 1, \rho_*)$, where $\rho_* \in (1.5736409, 1.573641)$ is the unique positive root of $\rho^8 + 2\rho^7 + 3\rho^6 - 4\rho^5 - 5\rho^4 - 6\rho^3 - 3\rho^2 - 12\rho - 12$, and every boundary point at m with a Lagrange triangle is isolated, for all 27 classes and all 64 crossed-sign assignments.

For (b) and (c), 60 of the 64 crossed-sign assignments satisfy $\Phi \neq 0$. The other four form one orbit under conjugation and $t \mapsto -t$ and have $q'(0) \neq 0$. In (d), the two-equal classes have $\Phi \neq 0$. Among the all-equal classes, 60 assignments have $\Phi \neq 0$ and 4 have $q'(0) \neq 0$. For the fold classes, an augmented system using $h = W_4 - W_4(p)$ gives a nonzero coefficient of order three for 60 assignments and of order four for the remaining 4. The exact grading makes these computations uniform in the common pair mass $m_1 = m_2$.

In (b), Bernstein coefficients establish smoothness of the zero set of K_1 in $\mathbb{R}^3_{>0}$. A linear subresultant over $\mathbb{Q}$ then determines the common shape root. A rational parametrization of S_1, together with Sturm sequences, identifies its two real branches. An exact reduction and an interval cover of the branch parameter give $q'(0) \neq 0$.

For (c), exact computation on the rational parametrizations of the four projective mass arcs classifies the real normalized W-shapes. Lemma 6.12(a) shows that these include every S_4 shape at positive masses. Common mass scaling, followed by restoration of the chart gauges, is a local analytic isomorphism. Isolatedness therefore extends from the arcs to their positive cones.

For (d), exact resultants and a Sturm count determine the zero set of K_5 in $\mathbb{R}^3_{>0}$. Complex rational interval arithmetic classifies the boundary points, using one triangle from each orbit under conjugation and exchange of bodies 3 and 4 (Appendix A). Renumbering the bodies as at the start

of this section gives the same conclusions for every choice of the pair and of the body with the opposite sign.

Lemma 6.17 (computer-assisted)**.** *Let $m \in \mathbb{R}^5_{>0}$ with $m_1 \neq m_2$, and let p be an opposite-sign boundary point at m with $k = 5$, $\Phi(p) = 0$, whose triangle is not a fold triangle. Then $J_T(p)$ is invertible and $\Psi_1(p) \neq 0$, so p is isolated at second order.*

The verification covers the mass simplex $m_3+m_4+m_5 = 1$. For S_1, S_3, S_4, it includes every Euler shape, enclosed by the Krawczyk method or Pellet discs. For S_5, S_6, S_8, it includes every Lagrange shape up to conjugation and the cube-root action. Both relative signs $s_3 s_4$ are considered. On each box, either G has a strict sign or ball arithmetic encloses Ψ_1 away from 0. On boxes meeting $m_3 = 0$ or $m_4 = 0$, where Ψ_1 vanishes, the corresponding transversal derivative is instead enclosed away from zero.

The remaining boundary and degeneracy cases require separate arguments. At the vertices, $G \neq 0$. At the ends $(m_3, m_4) = (0, \frac{1}{2})$ for S_1 and S_4, and $(\frac{1}{2}, 0)$ for S_4, a majority body becomes massless. An exact edge identity gives interval margins 0.787, 0.588, 0.551. At the corner $(\frac{1}{2}, \frac{1}{2}, 0)$ for S_3, an exact weighted blow-up gives weight two in m_5 to the Euler shape parameter. The resulting determinant and response numerator are m_5^{13} and m_5^6 times factors whose moduli are bounded below by $6.96 \cdot 10^8$ and $1.63 \cdot 10^6$, respectively, in interval arithmetic. For the six real points of S_4 where the Euler boundary point degenerates, and for the corner $(\frac{1}{2}, \frac{1}{2}, 0)$ of S_4, an exact formula for Ψ_1 on the Euler shapes cancels the degenerating factor. The remaining factor S_5 is treated by exact algebra (Appendix A).

On S_5, let g be the W_4-component of the solution of the triangle linear system with row F_5^Z omitted and $dW_5 = 0$, when the right-hand side is one in row F_4^Z and zero elsewhere. Put $X = \tilde{v}_4/W_4 - \tilde{v}_3/W_3$. Then $\Psi_1 = -\frac{1}{2} m_4 g X^2$. The numerator of g is a nonzero constant times $m_3(m_3^2 + m_3 m_4 + m_4^2)^2(P_0 + \omega P_1)$, where the real polynomials P_0, P_1 have only one common zero in the simplex, at $m_3 = m_4 = \frac{1}{3}$. Also, $X \neq 0$ except at this point. There $m_3 = m_4 = m_5$, and a boundary point with $G = 0$ would have $\rho = -1$, which is impossible. The cases S_2 and S_7 follow from S_1 and S_6 by exchanging bodies 3 and 4.

Lemma 6.18 (computer-assisted)**.** *Let $m \in \mathbb{R}^5_{>0}$ and let p be an opposite-sign boundary point at m with $k = 5$, $\Phi(p) = 0$ and a fold triangle. Then $dF_*(p)$ has rank 20 and $Q|_C$ has a simple zero at p; in particular p is excluded.*

The triangle Jacobian $J_T(p)$ has corank exactly one. Its kernel vector and left null vector ℓ are computed explicitly. The t-column has entries $-\varphi_B/W_B$ in the triangle Z-rows. Applying ℓ to this column gives $c_\sigma X$, with X as in Lemma 6.17. Here $X \neq 0$ at every boundary point under consideration: the exceptional point $m_3 = m_4 = m_5$ has no opposite-sign boundary point with $\Phi = 0$ at positive masses. Thus the additional parameter resolves the corank, and the curve C exists with local parameter $h = W_4 - W_4(p)$. At p, it satisfies $dt/dh = 0$ and $dQ/dh = -\frac{1}{2} c_\sigma m_4 X \neq 0$, which gives the required simple zero.

These statements are verified exactly over $\mathbb{Q}(m_3, m_4)(\omega)$, treating m_3, m_4 as indeterminates. The computation gives the kernel and left null vectors, an order-eight minor of $J_T(p)$ that is nonzero throughout the simplex, and the zero set of X (Appendix A).

Theorem 6.19. *For $m \in \mathbb{R}^5_{>0}$, no curve in A_m has a Diagram 2 germ. In particular, Theorem 4.1*(a) *holds for $r = 2$.*

Proof. Suppose that γ is a Diagram 2 germ on a curve in A_m, and let p be its boundary limit given by Proposition 6.4. For each B, the quantity $\sigma_B x$ is the limit of $m_2\delta_{2B}/(m_1\delta_{1B})$ along γ. This limit is independent of the chart, so the crossed-sign pattern of p is determined by γ. Renumbering bodies $3, 4, 5$ before applying Proposition 6.4 permutes the pattern. We may therefore take $k = 5$ in the opposite-sign case.

By Lemma 6.7(a), it suffices to exclude p. Proposition 6.6 gives $\Phi(p) = 0$. Choose a shape factor S_j vanishing on the triangle of p, as in Lemma 6.2.

Suppose first that the relative signs are equal. Proposition 6.6(a) gives $L_j(m') = 0$, or $L_5 L_5' = 0$ if $j = 5$. Lemma 6.12 rules out $j = 4$ and $L_5' = 0$. If $L_5 = 0$, then $m_3 = m_4 = m_5$, and Lemma 6.12(b) gives $j = 5$; Lemma 6.13 excludes these boundary points. In all other cases, $j \in \{1, 2, 3, 6, 7, 8\}$ and m' lies on one of the rays in Lemma 6.14(c), so Lemma 6.15 applies.

For an opposite-sign boundary point with $m_1 = m_2$, exclusion follows from Lemma 6.16. When $m_1 \neq m_2$, Lemma 6.18 applies to fold triangles and Lemma 6.17 to all other triangles.

Thus every possible p is excluded; for those isolated at second order this follows from Definition 6.11. This contradicts Lemma 6.7(a). □

7 Diagram 3

We classify Diagram 3 germs by reducing the chart equations to two analytic functions. Their Taylor coefficients determine the masses and the possible germs. The surviving analytic families satisfy an inverse-distance identity on the whole algebraic curve, which excludes every curve containing a positive point.

Diagram 3 of [AK12, Fig. 11] occurs at infinity on Roberts' continuum [AK12, §7.2]. We use the numbering of [AK12, §5.2 and §8.3], in which the clusters are $z : 12\ldots34$, $w : 14\ldots23$. The diagram is the quadrilateral 1234, with z-edges 14, 23 and w-edges 12, 34; neither the diagonals nor body 5 carry a stroke. Exchanging z and w has the same effect as exchanging bodies 2 and 4. Thus, after renumbering, every Diagram-3 germ has the form considered below.

7.1 The crossed-cluster chart

Apart from the gauge, the equations in (2.1) are invariant under $(z, w, \delta) \mapsto (\mu z, \mu^{-1} w, \delta)$, $\mu \in \mathbb{C}^*$. We therefore impose $z_{13} = w_{13}$, following [AK12, p. 543] for Roberts' continuum. To resolve the two pairs of clusters, introduce a parameter t and coordinates $A, B, x, y, \hat{z}_5, C, D, u, v, \hat{w}_5$ and D_{kl} by the following substitutions:

$$z_1 = \tfrac{A}{t}, \quad z_2 = \tfrac{A}{t} + tx, \quad z_3 = \tfrac{B}{t}, \quad z_4 = \tfrac{B}{t} + ty, \quad z_5 = t^2\hat{z}_5,$$
$$w_1 = \tfrac{C}{t}, \quad w_2 = \tfrac{D}{t}, \quad w_3 = \tfrac{D}{t} + tv, \quad w_4 = \tfrac{C}{t} + tu, \quad w_5 = t^2\hat{w}_5,$$
$$\delta_{kl} = D_{kl} \ \ (kl = 12, 14, 23, 34), \qquad \delta_{kl} = t\, D_{kl} \ \ (kl = 13, 24, 15, 25, 35, 45).$$

Multiply the z- and w-equations for bodies $1, \ldots, 4$, together with the gauge equation, by t, and those for body 5 by t^{-2}. The resulting 21 equations are polynomial: $F = F^0 + t^2F^2 + \cdots + t^6F^6 = 0$.

Setting $t = 0$ gives the *boundary equations* $F^0 = 0$, whose left-hand sides are

$$\begin{gathered}
R^z_1 = A - m_4 D^3_{14}(A-B), \qquad R^z_4 = B - m_1 D^3_{14}(B-A),\\
R^z_2 = A - m_3 D^3_{23}(A-B), \qquad R^z_3 = B - m_2 D^3_{23}(B-A),\\
R^w_1 = C - m_2 D^3_{12}(C-D), \qquad R^w_2 = D - m_1 D^3_{12}(D-C),\\
R^w_3 = D - m_4 D^3_{34}(D-C), \qquad R^w_4 = C - m_3 D^3_{34}(C-D),\\
\hat{z}_5 + (m_1 D^3_{15} + m_2 D^3_{25})A + (m_3 D^3_{35} + m_4 D^3_{45})B,\\
\hat{w}_5 + (m_1 D^3_{15} + m_4 D^3_{45})C + (m_2 D^3_{25} + m_3 D^3_{35})D,\\
D^2_{12}x(D-C) - 1, \quad D^2_{34}y(C-D) - 1, \quad D^2_{14}u(B-A) - 1,\\
D^2_{23}v(B-A) - 1, \quad D^2_{13}(B-A)(D-C) - 1, \quad D^2_{24}(B-A)(C-D) - 1,\\
D^2_{15}AC - 1, \quad D^2_{25}AD - 1, \quad D^2_{35}BD - 1, \quad D^2_{45}BC - 1, \quad (B-A) - (D-C);
\end{gathered} \tag{7.1}$$

Here R^z_k, R^w_k are the boundary equations for body k. We restrict to the slice $B = A + 1$. Proposition 7.1 justifies this choice by showing that every germ can be represented with $t = 1/z_{13}$.

7.2 Representation of germs in the chart

We first verify that this chart contains every germ with the prescribed diagram. We write s for the parameter of a germ.

Proposition 7.1 (Representation of germs)**.** *Let γ be a Diagram-3 germ, renormalized as in Lemma 3.3.*

(i) Bodies $1, \dots, 4$ are circled in both colours, body 5 in neither.

(ii) The sides are maximal edges: $r_{12}, r_{14}, r_{23}, r_{34} \approx 1$, $z_{14}, z_{23}, w_{12}, w_{34} \approx \epsilon^{-2}$ and $w_{14}, w_{23}, z_{12}, z_{34} \approx \epsilon^2$.

(iii) $z_{13}, w_{13}, z_{24}, w_{24}, z_{k5}, w_{k5} \approx \epsilon^{-2}$ for $k \le 4$, and $z_5, w_5 \preccurlyeq \epsilon^4$.

(iv) Renormalize instead by $\lambda = (w_{13}/z_{13})^{1/2}$, so that $z_{13} = w_{13}$, and put $t = 1/z_{13} \approx \epsilon^2$. Then $B - A = 1$ and every chart coordinate converges as $s \to 0$; the limit X_0 satisfies (7.1), and A, B, C, D, x, y, u, v and all D_{kl} are nonzero at X_0.

Proof. Lemma 3.3 allows us to apply the order relations of [AK12, §3] along γ, and hence the rules in Fact AK2.

For (i), each side is the only stroke of its colour incident to either endpoint. Rule 1a [AK12, pp. 545–546] therefore circles $1, \dots, 4$ in both colours. Since every circle requires a stroke of the same colour, 5 has no circle.

For (ii), the isolated component $\{1, 4\}$ of the z-diagram has z-circles. Rule 1e [AK12, p. 546] makes its stroke 14 maximal. The same argument applies to 23, and to 12, 34 in the w-diagram. The right equality case of Estimate 2 [AK12, p. 548] gives the asserted orders.

For (iii), (ii) gives $z_{13} = z_{14} - z_{34}$, $w_{13} = w_{12} + w_{23}$, $z_{24} = z_{23} + z_{34}$ and $w_{24} = w_{14} - w_{12}$, all of order $\approx \epsilon^{-2}$. Part (i) also gives $z_k \approx Z = \epsilon^{-2} \succ z_5$ for $k \le 4$, and the corresponding relations for w. In the equation $z_5(1 - \sum_{k\le4} m_k \delta^3_{k5}) = -\sum_{k\le4} m_k \delta^3_{k5} z_k$ for body 5, we have $\delta_{k5} \approx \epsilon^2$. It follows that $z_5 \preccurlyeq \epsilon^4$. The same argument applies to w_5.

For (iv), (iii) shows that λ differs from the normalization factor in Lemma 3.3 by a Puiseux series of order 0. Thus the orders in (i)–(iii) remain unchanged. We now have $B - A = tz_{13} = 1$, and the coordinates $A = z_1/z_{13}$, $B = z_3/z_{13}$, $C = w_1/w_{13}$, $D = w_2/w_{13}$, $x = z_{12}z_{13}$, $y = z_{34}z_{13}$, $u = w_{14}w_{13}$, $v = w_{23}w_{13}$ and D_{kl} ($= \delta_{kl}$ for the sides, $\delta_{kl}z_{13}$ otherwise) are ≈ 1. The remaining coordinates satisfy $\hat{z}_5 = z_5 z_{13}^2$, $\hat{w}_5 = w_5 w_{13}^2 \preccurlyeq 1$. Every chart coordinate therefore has a finite limit, and the limiting point satisfies $F = 0$. □

By Proposition 7.1(ii) and (iii), every Diagram-3 germ has an unbounded four-product: $p_{12} = r_{12}^2 r_{34}^2 r_{35}^2 r_{45}^2 \approx \epsilon^{-8}$.

7.3 The first mass relation

A weighted sum of the boundary equations gives the mass relation $m_1 m_3 = m_2 m_4$. This relation will also make the two omitted boundary equations vanish in the analytic reduction.

Proposition 7.2 (First condition)**.** *At every point of $\{t = 0\}$ satisfying (7.1), $m_1 m_3 = m_2 m_4$. Hence every Diagram-3 germ satisfies (29) of [AK12, §8.3].*

Proof. Since $m_1 R_1^z + m_4 R_4^z = m_1 A + m_4 B$ and $m_2 R_2^z + m_3 R_3^z = m_2 A + m_3 B$ identically,

$$(B - A)(m_1 m_3 - m_2 m_4) = (m_1 + m_4)(m_2 R_2^z + m_3 R_3^z) - (m_2 + m_3)(m_1 R_1^z + m_4 R_4^z), \tag{7.2}$$

and likewise $(D - C)(m_1 m_3 - m_2 m_4) = (m_1 + m_2)(m_3 R_3^w + m_4 R_4^w) - (m_3 + m_4)(m_1 R_1^w + m_2 R_2^w)$. With $T_{13} = D_{13}^2(B - A)(D - C) - 1$ (the distance equation of 13) and $g = (B - A) - (D - C)$ (the gauge), $D_{13}^2(B - A)^2 - 1 = T_{13} + D_{13}^2(B - A)g$. At a boundary point $T_{13} = g = 0$, so $D_{13}^2(B - A)^2 = 1$, $B - A \neq 0$, and (7.2) gives $m_1 m_3 = m_2 m_4$. The second statement follows from Proposition 7.1(iv). □

These identities express in chart coordinates the argument of [AK12, §5.2]: Rule 1c makes $m_1 z_1 + m_4 z_4$ and $m_2 z_2 + m_3 z_3$ non-maximal, whereas $z_1 \sim z_2$ and $z_3 \sim z_4$.

7.4 Boundary solutions and analytic continuation

Assume henceforth that $m \in \mathbb{R}^5_{>0}$ satisfies (29), and set $M = m_1 + m_2 + m_3 + m_4$. After eliminating $B = A + 1$, denote the remaining 19 chart coordinates by X. We will solve nineteen independent equations by the implicit function theorem, leaving two equations that must vanish along any germ.

Let F_{z_k}, F_{w_k} denote the rescaled equations for body k. Define Λ to be the 19×21 matrix selecting the combinations $m_1 F_{z_1} + m_4 F_{z_4}$, $F_{z_4} - F_{z_1}$, $F_{z_3} - F_{z_2}$, $m_1 F_{w_1} + m_2 F_{w_2}$, $F_{w_2} - F_{w_1}$, $F_{w_3} - F_{w_4}$, together with the other 13 equations. The two remaining linear combinations are $\ell_z F = F_{z_1} - F_{z_2} - \frac{m_3}{m_2} F_{z_3} + \frac{m_4}{m_1} F_{z_4}$ and $\ell_w F = F_{w_1} + \frac{m_2}{m_1} F_{w_2} - \frac{m_2}{m_1} F_{w_3} - F_{w_4}$.

Lemma 7.3. *(a) On the slice, (7.1) has exactly $3^4 \cdot 2^6 = 5184$ boundary points. They are obtained as follows: $D_{12}, D_{34}, D_{14}, D_{23}$ are arbitrary cube roots of $\frac{1}{m_1+m_2}$, $\frac{1}{m_3+m_4}$, $\frac{1}{m_1+m_4}$, $\frac{1}{m_2+m_3}$; $D_{13}, D_{24}, D_{15}, D_{25}, D_{35}, D_{45}$ are arbitrary square roots of 1, -1, $\frac{M}{m_3}$, $-\frac{M}{m_4}$, $\frac{M}{m_1}$, $-\frac{M}{m_2}$; $A = -\frac{m_4}{m_1+m_4}$, $C = -\frac{m_2}{m_1+m_2}$, $D = \frac{m_1}{m_1+m_2}$, $x = (m_1 + m_2)D_{12}$, $y = -(m_3 + m_4)D_{34}$, $u = (m_1 + m_4)D_{14}$, $v = (m_2 + m_3)D_{23}$, and $\hat{z}_5, \hat{w}_5$ are determined by the equations for body 5. These boundary solutions are independent of the mass m_5.*

(b) At every boundary point the matrix $\partial_X(\Lambda F^0)$ is invertible.

Proof. On the slice $B = A + 1$, the three selected combinations of the z-equations are

$$m_1 R_1^z + m_4 R_4^z = (m_1 + m_4)A + m_4, \qquad R_4^z - R_1^z = 1 - (m_1 + m_4)D_{14}^3,$$

$$R_3^z - R_2^z = 1 - (m_2 + m_3)D_{23}^3.$$

The gauge gives $D - C = 1$, while the weighted combination of the w-equations gives $m_1 C + m_2 D = 0$. The other two selected combinations are

$$R_2^w - R_1^w = (D - C)\big(1 - (m_1 + m_2)D_{12}^3\big), \qquad R_3^w - R_4^w = (D - C)\big(1 - (m_3 + m_4)D_{34}^3\big).$$

These equations determine A, C, D and the four cubes listed in (a). The four distance equations for the sides then determine x, y, u, v. The other six distance equations determine the six squares, and the equations for body 5 determine $\hat{z}_5, \hat{w}_5$. The relation $m_1 m_3 = m_2 m_4$ ensures that the omitted boundary coordinate rows also vanish. Since all roots are nonzero and simple, each choice of the four cube roots and six square roots gives exactly one boundary point. This accounts for all $3^4 2^6$ boundary points.

The same elimination proves (b). Let a vector belong to the kernel of the differential. The linear equations force its variations of A, C, D to vanish. The four cube equations then force the variations of the side coordinates D_{kl} to vanish, because their coefficients are nonzero multiples of D_{kl}^2. The four side distance equations next force the variations of x, y, u, v to vanish. Each remaining distance equation does the same for its own D_{kl}, since its derivative in that coordinate is $2/D_{kl} \neq 0$. Finally, the two equations for body 5 force the variations of $\hat{z}_5, \hat{w}_5$ to vanish. The square matrix $\partial_X(\Lambda F^0)$ therefore has trivial kernel. □

For all X, we have $\ell_z F^0 = -B(m_1 m_3 - m_2 m_4)/(m_1 m_2)$ and $\ell_w F^0 = -(D - C)D_{34}^3(m_1 m_3 - m_2 m_4)/m_1$. Thus ℓ_z, ℓ_w annihilate $J_0 = \partial_X F^0(X_0)$. Since ΛJ_0 is invertible, J_0 has rank 19, ℓ_z, ℓ_w span its left kernel, and the 21 rows of Λ, ℓ_z, ℓ_w are independent.

Lemma 7.4. *Let X_0 be a boundary point. There is a unique analytic $t \mapsto X(t)$ near 0 with $X(0) = X_0$ and $\Lambda F(t, X(t)) = 0$; put $G(t) = (\ell_z F(t, X(t)), \ell_w F(t, X(t))) = \sum_{k \geq 1} G_k t^k$. If the chart coordinates of a Diagram-3 germ converge to X_0, then along the germ they are $X(t)$, and $G \equiv 0$. Every analytic $Y(t)$ with $Y(0) = X_0$ and $F(t, Y(t)) \equiv 0$ equals $X(t)$.*

Proof. The analytic implicit function theorem applies by Lemma 7.3(b) and gives the unique analytic graph. After a reparametrization $s \mapsto s^N$, the chart coordinates $Y(s)$ and $t(s) = 1/z_{13}(s)$ of the germ are convergent power series (Lemma 3.3 and Proposition 7.1), with $Y(0) = X_0$, $t(0) = 0$ and $t \not\equiv 0$. The identity $\Lambda F(t(s), Y(s)) = 0$ and uniqueness imply $Y(s) = X(t(s))$. Consequently, $F(t, X(t))$ vanishes on the image of $t(s)$, which contains a neighbourhood of 0. Thus $F(t, X(t)) \equiv 0$ and $G \equiv 0$. The last assertion follows from the same uniqueness. □

Lemma 7.4 reduces the existence of a germ to $G_k = 0$ for all k. Since $F^1 = 0$ and ΛJ_0 is invertible, $X_1 = 0$ and

$$X(t) = X_0 + X_2 t^2 + X_3 t^3 + \cdots .$$

Put $\ell = (\ell_z, \ell_w)$. The identity $\ell J_0 = 0$ removes X_k from the coefficient G_k, giving, with all derivatives evaluated at X_0,

$$\begin{aligned} G_2 &= \ell F^2, \qquad G_3 = \ell F^3, \\ G_4 &= \ell\big(F^4 + \partial F^2 \cdot X_2 + \tfrac{1}{2}\partial^2 F^0[X_2, X_2]\big), \\ G_5 &= \ell\big(F^5 + \partial F^3 \cdot X_2 + \partial F^2 \cdot X_3 + \partial^2 F^0[X_2, X_3]\big). \end{aligned}$$

Thus each successive obstruction is determined by the coefficients already obtained.

7.5 Orders three and four

The third-order coefficient gives two polynomial mass equations. The fourth-order coefficient then imposes equality of two masses, reducing the candidates to four explicit one-parameter families up to scaling.

Lemma 7.5 (computer-assisted)**.** *At every boundary point $G_2 = 0$ and $G_3 = \big(\frac{m_1+m_2}{m_1^2 m_2}\Omega, \frac{m_1+m_4}{m_1^2 m_4}\Omega\big)$, where $\Omega = \Omega_1 + \Omega_2$,*

$$\begin{aligned} \Omega_1 &= m_1 m_2 m_4 D_{13} + m_5 m_1^2 D_{15} + m_5 m_2 m_4 D_{35}, \\ \Omega_2 &= m_1 m_2 m_4 D_{24} + m_5 m_1 m_2 D_{25} + m_5 m_1 m_4 D_{45}. \end{aligned}$$

This is an exact symbolic computation over $\mathbb{Q}(m_1, m_2, m_4, m_5)$, with $m_3 = m_2 m_4/m_1$. It evaluates ℓF^2 and ℓF^3 at the boundary points and reduces the results modulo the cubic and quadratic equations for D_{kl}. The verification is described in Appendix A.

Corollary 7.6. *If $G_3 = 0$ at some boundary point, then $N_1(m) = N_2(m) = 0$, where $Q = m_5^2 M$ and*

$$\begin{aligned} N_1 &= \big((m_1 m_2 m_4)^3 - Q(m_1^6 + m_2^3 m_4^3)\big)^2 - 4Q^2 m_1^6 m_2^3 m_4^3, \\ N_2 &= \big((m_1 m_2 m_4)^3 - Q m_1^3(m_2^3 + m_4^3)\big)^2 - 4Q^2 m_1^6 m_2^3 m_4^3. \end{aligned}$$

As $N_1 - N_2 + QS = 0$ with $S = (m_1^3 - m_2^3)(m_1^3 - m_4^3)\big(2(m_1 m_2 m_4)^3 - Q(m_1^3 + m_2^3)(m_1^3 + m_4^3)\big)$, also $S(m) = 0$.

Proof. At positive masses D_{13}, D_{15}, D_{35} are real and D_{24}, D_{25}, D_{45} purely imaginary (Lemma 7.3), so $\Omega_1 = \Omega_2 = 0$. If $\alpha+\beta+\gamma = 0$ then $(\alpha^2-\beta^2-\gamma^2)^2 - 4\beta^2\gamma^2 = \prod_{\pm}(\alpha+\beta\pm\gamma)\prod_{\pm}(\alpha-\beta\pm\gamma) = 0$. Inserting the squares $(m_1 m_2 m_4)^2$, $m_5^2 m_1^4 M/m_3$, $m_5^2 m_2^2 m_4^2 M/m_1$ of the terms of Ω_1, using $m_3 = m_2 m_4/m_1$ and multiplying by $(m_1 m_2 m_4)^2$ gives $N_1 = 0$; the terms of Ω_2 are i times reals with squares $(m_1 m_2 m_4)^2$, $m_5^2 m_1^2 m_2^2 M/m_4$, $m_5^2 m_1^2 m_4^2 M/m_2$, which gives $N_2 = 0$. Finally $Q > 0$. □

Lemma 7.7 (computer-assisted)**.** *At every boundary point $G_4 = -\frac{\Pi}{m_1^2}\big(m_4(m_1+m_2), m_2(m_1+m_4)\big)$ with $\Pi = (m_1 D_{12} - m_4 D_{34})(m_1 D_{14} - m_2 D_{23})$.*

As in Lemma 7.5, the computation uses exact arithmetic and reduction modulo the equations for D_{kl}. After solving for X_2, it evaluates the preceding formula for G_4 (Appendix A). At every boundary point, $(m_1 D_{12})^3 - (m_4 D_{34})^3 = m_1(m_1+m_4)(m_1-m_4)/(m_1+m_2)$ and $(m_1 D_{14})^3 - (m_2 D_{23})^3 = m_1(m_1+m_2)(m_1-m_2)/(m_1+m_4)$, so $\Pi = 0$ forces $m_1 = m_4$ or $m_1 = m_2$; with (29), $m = (p, q, q, p, m_5)$ or (p, p, q, q, m_5).

Lemma 7.8. *Let $m \in \mathbb{R}^5_{>0}$ satisfy (29), $N_1(m) = N_2(m) = 0$ and $m_1 = m_2$ (resp. $m_1 = m_4$). Write $m_1 = a^2$, $m_3 = b^2$ (resp. $m_2 = b^2$) with $a, b > 0$, and $c = (2(a^2 + b^2))^{1/2}$. Then*

$$m_5 = \frac{a^3 b^3}{c\,|a^3 + b^3|} \qquad \text{or} \qquad m_5 = \frac{a^3 b^3}{c\,|a^3 - b^3|} \quad (a \neq b).$$

Conversely N_1, N_2 and S vanish on these four families, called $E^{\pm}_{12}$ (if $m_1 = m_2$) and $E^{\pm}_{14}$ (if $m_1 = m_4$). They meet only at $a = b$, i.e. at m proportional to $(1, 1, 1, 1, \frac{1}{4})$, which lies on $E^{+}_{12} \cap E^{+}_{14}$.

Proof. For either mass pattern, the equations $N_1 = 0$ and $N_2 = 0$ coincide. Their common polynomial factors as

$$N_1 = N_2 = a^{12}\big(a^6 b^6 - m_5^2 c^2 (a^3 + b^3)^2\big)\big(a^6 b^6 - m_5^2 c^2 (a^3 - b^3)^2\big).$$

Positivity of a, b, c, m_5 shows that the first factor gives exactly the plus formula, and the second gives the minus formula, with $a \neq b$. One of the first two factors of S vanishes, so $S = 0$ as well. The two mass patterns coincide only when all four outer masses are equal. In that case the minus formula is impossible, while the plus formula gives $m_5 = a^2/4$. For each mass pattern, the plus and minus formulas are distinct when $a, b > 0$. □

Lemma 7.9. *Let m lie on one of the four families.*

(a) If $a \neq b$, exactly 108 boundary points have $G_3 = G_4 = 0$: two of the eight sign choices for (D_{13}, D_{15}, D_{35}) give $\Omega_1 = 0$, two of those for (D_{24}, D_{25}, D_{45}) give $\Omega_2 = 0$, and $\Pi = 0$ exactly when $D_{14} = D_{23}$ (on $E^{\pm}_{12}$) or $D_{12} = D_{34}$ (on $E^{\pm}_{14}$), for 27 of the 81 choices of cube roots.

(b) If $a = b$, exactly 180 boundary points have $G_3 = G_4 = 0$: those with $D_{15} = D_{35} = -2D_{13}$, $D_{25} = D_{45} = -2D_{24}$, and $D_{12} = D_{34}$ or $D_{14} = D_{23}$ (45 choices of cube roots).

Proof. Exchanging 2 and 4 together with z and w reduces the proof to E^{ϵ}_{12}, $\epsilon \in \{1, -1\}$. Write $A_0 = a^3$, $B_0 = b^3$, $\nu = \operatorname{sgn}(A_0 + \epsilon B_0)$, and set $D_{13} = \sigma$, $D_{15} = s_1 c/b$, $D_{35} = s_2 c/a$. After division by a nonzero real factor, $\Omega_1 = 0$ becomes

$$(\sigma\nu + s_1)A_0 + (\epsilon\sigma\nu + s_2)B_0 = 0.$$

For $\epsilon = 1$, any nonzero coefficients have the same sign, so the equation forces both coefficients to vanish. For $\epsilon = -1$, cancellation between two nonzero coefficients would require $A_0 = B_0$, which is excluded on the minus family. There are therefore exactly two real sign triples. Applying the same argument to $D_{24}/i, D_{25}/i, D_{45}/i$ gives exactly two imaginary sign triples. If $a \neq b$, the first factor of Π cannot vanish because the cubes of $a^2 D_{12}$ and $b^2 D_{34}$ are $a^4/2$ and $b^4/2$. Consequently $\Pi = 0$ exactly when $D_{14} = D_{23}$, leaving $3^3 = 27$ cube-root choices and hence $4 \cdot 27 = 108$ boundary points.

At $a = b$, the sign conditions reduce to $D_{15} = D_{35} = -2D_{13}$ and $D_{25} = D_{45} = -2D_{24}$, again giving four choices. Each equality $D_{12} = D_{34}$ or $D_{14} = D_{23}$ admits 27 cube-root choices, and their intersection admits 9. Thus their union gives $4(27 + 27 - 9) = 180$ boundary points; the intersection gives $4 \cdot 9 = 36$. □

7.6 Orders five and six

The fifth-order coefficient excludes the unequal-mass points of the four families. At their intersection, the sixth-order coefficient leaves 36 boundary points. We construct the corresponding analytic germs explicitly and record the identity that will exclude positive points on their algebraic curves.

Lemma 7.10 (computer-assisted)**.** *Let m lie on one of the four families with $a \neq b$. Then $G_5 \neq 0$ at each of the* 108 *boundary points of Lemma 7.9(a). At $a = b$, $G_5 = 0$ at all* 180 *boundary points of Lemma 7.9(b).*

The fifth-order calculation gives the following identities. On E_{12}^{+}, where $D_{14} = D_{23}$, the z-component of G_5 is a nonzero factor times $(a-b)L_{+}(\xi_1, \xi_2)$, where $\xi_1 = D_{12}/D_{14}$, $\xi_2 = D_{34}/D_{14}$ and

$$L_\epsilon(\xi_1, \xi_2) = a^2 F_{4,\epsilon}\xi_1 - a^{-2}b^4 F_{4,\epsilon}\xi_2 - b^2(a^2 - b^2)F_{2,\epsilon},$$

$$F_{4,\epsilon} = a^4 + \epsilon a^3 b + a^2 b^2 + \epsilon a b^3 + b^4, \qquad F_{2,\epsilon} = a^2 + \epsilon ab + b^2, \qquad \epsilon \in \{1, -1\},$$

with $\xi_1^3 = (a^2 + b^2)/(2a^2)$ and $\xi_2^3 = (a^2 + b^2)/(2b^2)$. The product of the nine values of L_{+} is, up to sign and a nonzero monomial factor, $(a-b)^3(a+b)^3 P_{66}(a, b)$. On E_{12}^{-}, the calculation gives $G_5 = 0 \Rightarrow P_{66}(a, -b) = 0$ at each of the 108 boundary points. All factors removed in obtaining this implication are regular and nonzero for $a, b > 0$, $a \neq b$. These identities are the computer-assisted part of the argument. Their required nonvanishing follows from the elementary estimates below.

We treat both signs together. Interchanging a, b and ξ_1, ξ_2 multiplies L_ϵ by the nonzero factor $-a^2/b^2$. We may therefore assume $a > b > 0$ and write

$$r = b/a \in (0, 1), \quad c_0 = \left(\frac{1 + r^2}{2}\right)^{1/3}, \quad f_{4,\epsilon} = 1 + \epsilon r + r^2 + \epsilon r^3 + r^4, \quad f_{2,\epsilon} = 1 + \epsilon r + r^2.$$

Writing $\xi_1 = c_0\omega^j$ and $\xi_2 = c_0 r^{-2/3}\omega^k$, $\omega = e^{2\pi i/3}$, gives

$$a^{-6}L_\epsilon = A_\epsilon \omega^j - B_\epsilon \omega^k - C_\epsilon, \quad A_\epsilon = c_0 f_{4,\epsilon}, \quad B_\epsilon = A_\epsilon r^{10/3}, \quad C_\epsilon = r^2(1 - r^2)f_{2,\epsilon}.$$

All three coefficients are positive, with $A_\epsilon > B_\epsilon$. Since

$$\operatorname{Im}(A_\epsilon\omega^j - B_\epsilon\omega^k - C_\epsilon) = A_\epsilon \operatorname{Im}\omega^j - B_\epsilon \operatorname{Im}\omega^k, \qquad \operatorname{Im}\omega^j \in \{0, \sqrt{3}/2, -\sqrt{3}/2\},$$

vanishing of the imaginary part forces $j = k = 0$. We therefore compare the real coefficients. For the plus sign, $c_0 > r^2$ and $f_{4,+} > f_{2,+}$. For the minus sign, we have

$$\frac{f_{4,-}}{f_{2,-}} = \frac{1 + r^5}{1 + r^3} > r^{4/3}, \qquad c_0 > r^{2/3},$$

since

$$1 + r^5 - r^{4/3} - r^{13/3} = (1 - r^{4/3})(1 - r^{13/3}) + r^5 - r^{17/3} > 0.$$

In both cases, therefore, $c_0 f_{4,\epsilon} > r^2 f_{2,\epsilon}$, and

$$A_\epsilon - B_\epsilon = c_0 f_{4,\epsilon}(1 - r^{10/3}) > r^2 f_{2,\epsilon}(1 - r^2) = C_\epsilon.$$

This excludes the remaining real-root case, so all nine values of L_ϵ are nonzero. The product identity for L_+ gives $P_{66}(a,b)\neq 0$. Replacing b by $-b$ in the same identity gives $P_{66}(a,-b)\neq 0$ for $a\neq b$. The fifth-order identities thus exclude both families when $a\neq b$.

Exchanging bodies 2 and 4 together with z and w maps the chart at m onto the chart at (m_1,m_4,m_3,m_2,m_5), $E_{12}^{\pm}$ onto $E_{14}^{\pm}$ and the 108 boundary points onto the 108 boundary points; the relabelling (13)(24) exchanges a and b, accounting for the minus families with $a<b$. At $a=b$, the calculation gives zero for both components of G_5 at the boundary points of E_{12}^{+}. The exchange then gives vanishing at all 180 boundary points.

Lemma 7.11 (computer-assisted)**.** *Let $m=(\frac12,\frac12,\frac12,\frac12,\frac18)$. Of the 180 boundary points of Lemma 7.9(b), the 144 boundary points with $D_{12}\neq D_{34}$ or $D_{14}\neq D_{23}$ have $G_6\neq 0$.*

The lemma is proved by computing the jets modulo two primes (Appendix A). All data lie in $R=\mathbb{Z}[\frac12,i,\omega]$, where $\omega=e^{2\pi i/3}$. Choose a prime $\mathfrak{p}$ of R above $p\in\{13,37\}$. Since $p\equiv 1 \pmod{12}$, we have $R/\mathfrak{p}=\mathbb{F}_p$.

If a germ passed through a boundary point, Lemma 7.4 would give $F(t,X(t))\equiv 0$. At each order, X_k would then be uniquely determined by the order-k equations, using any 19 rows with an invertible minor, and all 21 residuals would vanish. We use 19 rows whose minor is a unit modulo $\mathfrak{p}$. The calculation produces a nonzero order-6 residual at each of the 144 boundary points. Changing $\mathfrak{p}$ acts on the boundary points through $\mathrm{Gal}(\mathbb{Q}(i,\omega)/\mathbb{Q})$, with $\omega\mapsto\omega^2$ and $i\mapsto -i$. This action preserves the set of 144 boundary points, and the two primes give the same excluded set.

Lemma 7.12. *Let $m=(\frac12,\frac12,\frac12,\frac12,\frac18)$, $\alpha,\beta\in\{1,\omega,\omega^2\}$ and $\varepsilon,\eta\in\{\pm1\}$. Let (P,Q,Y) be the analytic solution near $\tau=0$ of*

$$(1-P)(1+Q)=\alpha\tau^2,\qquad (1+P)(1-Q)=\beta\tau^2,\qquad 4PQY^2+1=0$$

with $(P,Q,Y)(0)=(1,1,\eta i/2)$; the Jacobian there is $\pm 16i$. Then

$$z=\tfrac{1}{\tau}(1,P,-1,-P,0),\qquad w=\tfrac{1}{\tau}(1,-Q,-1,Q,0),$$

$$\delta_{12}=\delta_{34}=\alpha,\quad \delta_{14}=\delta_{23}=\beta,\quad \delta_{13}=\tfrac{\varepsilon\tau}{2},\quad \delta_{15}=\delta_{35}=-\varepsilon\tau,\quad \delta_{24}=\tau Y,\quad \delta_{25}=\delta_{45}=-2\tau Y$$

solves (2.1) with the gauge $z_{13}=w_{13}$. In the chart, with $t=-\tau/2$, it is an analytic solution of $F=0$ through the boundary point with $D_{12}=D_{34}=\alpha$, $D_{14}=D_{23}=\beta$, $D_{13}=-\varepsilon$ and $D_{24}=-\eta i$. These 36 boundary points are the boundary points of Lemma 7.9(b) with $D_{12}=D_{34}$ and $D_{14}=D_{23}$. Along these germs

$$\delta_{15}+2\delta_{13}=\delta_{35}+2\delta_{13}=\delta_{25}+2\delta_{24}=\delta_{45}+2\delta_{24}=0. \tag{7.3}$$

Proof. The derivative of the three defining equations with respect to (P,Q,Y) at $(\tau,P,Q,Y)=(0,1,1,\eta i/2)$ is

$$\begin{pmatrix} -2 & 0 & 0\\ 0 & -2 & 0\\ -1 & -1 & 4\eta i\end{pmatrix},$$

with determinant $16\eta i\neq 0$. The implicit function theorem gives the claimed analytic solution. Since the equations depend on τ only through τ^2, uniqueness makes P,Q,Y even, and $P-1,Q-1=O(\tau^2)$.

We verify the body equations first. For either coordinate $q = z$ or $q = w$, the two adjacent side forces at outer body k sum to q_k: the cubes of their inverse distances are 1, the adjacent masses are $1/2$, and the adjacent bodies have opposite positions. At body 1 or 3, the diagonal force and the force from body 5 cancel, since

$$\tfrac{1}{2}(\varepsilon\tau/2)^3(2q_k) + \tfrac{1}{8}(-\varepsilon\tau)^3 q_k = 0.$$

At body 2 or 4 the same cancellation is

$$\tfrac{1}{2}(\tau Y)^3(2q_k) + \tfrac{1}{8}(-2\tau Y)^3 q_k = 0.$$

Pairing opposite bodies also proves the equation for body 5. For the distance equations, the pairs $12, 34$ reduce to $\alpha^2(1-P)(1+Q)/\tau^2 = 1$, the pairs $14, 23$ to $\beta^2(1+P)(1-Q)/\tau^2 = 1$, and $24, 25, 45$ to $-4PQY^2 = 1$; the remaining distance equations hold identically. The gauge follows from $z_{13} = w_{13} = -2/\tau$. Finally $t = -\tau/2$ makes the chart coordinates analytic, since the only apparent quotients include $(P-1)/\tau^2$ and $(Q-1)/\tau^2$. The displayed inverse distances give (7.3) identically and yield precisely the $9 \cdot 4 = 36$ boundary points stated in the lemma. □

7.7 Exclusion of curves containing a positive point

We now pass from the local analytic classification to the original algebraic curve. The inverse-distance identities hold along its germ and hence, by irreducibility, on the whole curve. Their signs at a positive point give the required contradiction.

Theorem 7.13. *Let $m \in \mathbb{R}^5_{>0}$, let $\Gamma \subset A_m$ be an irreducible curve, and let γ be a Diagram-3 germ on Γ, with the bodies numbered as in Diagram 3. Then $m_1 = m_2 = m_3 = m_4 = 4m_5$, and (7.3) holds identically on Γ. In particular Γ contains no positive point, and Theorem 4.1(c) holds.*

Proof. Exchanging z and w and renumbering if necessary puts γ in the form described at the start of this section. These operations preserve the conclusions. Proposition 7.1 gives a limiting chart point X_0 satisfying (7.1) and $B - A = 1$. Relation (29) holds by Proposition 7.2, so Lemma 7.3 identifies X_0 as a boundary point. Lemma 7.4 then gives $G \equiv 0$ at X_0.

The third-order condition, from Lemma 7.5 and Corollary 7.6, gives $N_1(m) = N_2(m) = 0$. The fourth-order condition in Lemma 7.7 gives $m_1 = m_4$ or $m_1 = m_2$. Thus m belongs to one of the four families in Lemma 7.8. Lemmas 7.9(a) and 7.10 exclude every point with $a \neq b$, leaving $a = b$, or equivalently $m_1 = m_2 = m_3 = m_4 = 4m_5$.

The covariance of (2.1) under $(m, z, w, \delta) \mapsto (\kappa^3 m, \kappa z, \kappa w, \kappa^{-1}\delta)$, $\kappa > 0$, preserves both Diagram-3 germs and (7.3). We may therefore normalize to $m = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{8})$. Lemmas 7.9(b) and 7.11 now imply $D_{12} = D_{34}$ and $D_{14} = D_{23}$ at X_0. By Lemmas 7.12 and 7.4, the germ γ coincides in the chart with the analytic germ of Lemma 7.12 through X_0. Since renormalization leaves δ_{kl} unchanged, (7.3) holds along γ.

Each expression in (7.3) is a regular function on the irreducible curve Γ. It vanishes at infinitely many points of γ, and hence vanishes identically on Γ. A positive point would have $\delta_{15} + 2\delta_{13} > 0$, a contradiction. □

Theorem 7.13 classifies all Diagram 3 germs: the mass vector is proportional to $(4, 4, 4, 4, 1)$ and, up to the stated symmetries, the germ is one of the 36 analytic families in Lemma 7.12. The modified Roberts continuum provides an explicit member of this classification.

Remark 7.14 (The modified Roberts continuum). The family of Roberts [Rob99] has the following signed-distance variant in [AK12]. Let $m = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{8})$, proportional to $(4, 4, 4, 4, 1)$ [AK12, Rem. 8, p. 583]. The curve

$$z = (a, b, -a, -b, 0), \quad w = (a, -b, -a, b, 0), \quad b^2 = a^2 - 1,$$

$$\delta_{12} = \delta_{23} = \delta_{34} = \delta_{14} = 1, \quad \delta_{13} = \tfrac{1}{2a}, \quad \delta_{24} = \tfrac{i}{2b}, \quad \delta_{15} = \delta_{35} = -\tfrac{1}{a}, \quad \delta_{25} = \delta_{45} = -\tfrac{i}{b}$$

is the modified Roberts continuum [AK12, p. 543]. It solves (2.1) with the gauge $z_{13} = w_{13}$, $U = I = 1$ on it, and (7.3) holds, so it has no positive point. Since $w_{12}/z_{12} = (a + b)^2$ on it, the rotation by $\mu = a + b$ gives an irreducible curve $\Gamma_R \subset A_m$. For $0 < a < 1$ and $b \in i\mathbb{R}$, $|\mu| = 1$ and the points are real, with $\delta_{13} > 0 > \delta_{15} = \delta_{35}$. Before this rotation, the end $a \to \infty$ with $b \sim a$ is precisely the analytic germ of Lemma 7.12 with $(\alpha, \beta, \varepsilon, \eta) = (1, 1, 1, 1)$, $\tau = 1/a$ and $P = Q = b/a$. In this balanced $z_{13} = w_{13}$ representative its strokes are z_{14}, z_{23}, w_{12}, w_{34}, so the corresponding end of Γ_R is a Diagram-3 germ (Roberts' continuum at infinity [AK12, §7.2]); the sides keep length 1, $r_{13}, r_{24}, r_{k5} \to \infty$, $p_{12} = a^2(1 - a^2)$ is unbounded, and (29) holds. Thus Γ_R gives a real arc with signed inverse distances, $U = I = 1$, and an unbounded four-product at its Diagram 3 end. The identity $\delta_{15} + 2\delta_{13} = 0$ describes its sign constraint directly.

8 The triangle in Diagrams 4 and 5

Diagrams 4 and 5 contain the same isolated zw-triangle 345 with uncircled vertices. We first identify its limiting shape and mass relation. We then use the forces between this triangle and the two remaining bodies to exclude Diagram 4 and to prove $\Delta \to 0$ in Diagram 5. These are the two local conclusions required by Theorem 4.1.

Proposition 8.1 (Relation (23)). *Let $m \in \mathbb{R}^5_{>0}$ and let γ be a Diagram 4 or Diagram 5 germ, renormalized as in Lemma 3.3 so that $\mathcal{Z}$ and $\mathcal{W}$ have the same least order $-2\nu < 0$. Let z_c and w_c be the centres of mass of $T = \{3, 4, 5\}$. Then the limits*

$$\zeta_k = \lim_{t\to 0} t^{-\nu}(z_k - z_c), \qquad \eta_k = \lim_{t\to 0} t^{-\nu}(w_k - w_c), \qquad D_{kl} = \lim_{t\to 0} t^{\nu}\delta_{kl} \qquad (k, l \in T)$$

exist and solve the zero-multiplier triangle system of Section 5, and $\sum_k m_k\zeta_k = \sum_k m_k\eta_k = 0$. Hence (ζ, η, D) is as in Lemma 5.8, and $Q(m_3, m_4, m_5) = 0$, that is, the masses satisfy (23).

Proof. The sides of T are zw-edges. Estimate 1 and the left equality case of Estimate 2, as stated in Fact AK2 and justified for germs by Lemma 3.3, give $\operatorname{ord} z_{kl} = \operatorname{ord} w_{kl} = \nu$ and $\operatorname{ord}\delta_{kl} = -\nu$ for $k, l \in T$. Since $z_k - z_c$ and $w_k - w_c$ are linear combinations of these differences, the limits exist. The choice of the centres of mass makes both sums $\sum_k m_k\zeta_k$, $\sum_k m_k\eta_k$ vanish.

For $k \in T$, consider the equation $z_k = \sum_{l\neq k} m_l\delta_{kl}^3(z_k - z_l)$. The terms with $l \in T$ have order -2ν. Every other term has larger order: z_k because k is uncircled, and the terms with $l \notin T$ because T is isolated and no z-stroke leaves it. Multiplication by $t^{2\nu}$ followed by the limit $t \to 0$ therefore gives $\phi_k = 0$. The same argument gives $\psi_k = 0$. The distance equations within T give $\chi_{kl} = 0$. Lemma 5.8 applies, and for positive masses the relation $Q = 0$ is equivalent to (23). □

8.1 Diagram 4

Diagram 4 consists of the isolated zw-triangle 345, whose vertices are not circled, and the isolated z-stroke 12 with z-circles at 1 and 2; there are no other strokes or circles [AK12, §7.4, p. 567, and Fig. 11]. The common triangle gives relation (23), as in [AK12, §8.4]. We shall combine it with two force balances to exclude the diagram. We use the notation of Section 5: $P = \{1, 2\}$, $T = \{3, 4, 5\}$, $i \in P$, $k, l \in T$, M_P, M_T, Z_P, Z_T, W_P, W_T, the equations E^z_j, E^w_j, the triangle functions $\phi_k, \psi_k, \chi_{kl}$, the torque T_P of Lemma 5.2 and the orbit O of Lemma 5.9. Renumbering inside P or inside T does not change the diagram.

Theorem 8.2. *Let $m \in \mathbb{R}^5_{>0}$. No curve in A_m carries a Diagram 4 germ.*

To prove part (a) of Theorem 4.1 for $r = 4$, we proceed in two steps.

First, we show that the chart contains every germ. A torque identity rules out a triangle body at the triangle's w-centre (Lemma 8.5). Every germ then has a limit in the boundary fibre $\mathcal{B}_4$, which is smooth near that limit (Proposition 8.6 and Lemma 8.7).

Second, we derive two necessary balances of the leading forces. The Lyapunov–Schmidt reduction in Lemma 8.8 expresses them as successive obstructions to the existence of a germ. The first obstruction requires equal leading w-forces on the two outer bodies. The second vanishes identically, and the third requires equal leading z-forces (Lemma 8.9). Positivity makes these two balances incompatible (Lemma 8.10).

8.1.1 The chart

The invariance of (2.1), apart from the gauge, under $(z, w, \delta) \mapsto (cz, c^{-1}w, \delta)$, $c \in \mathbb{C}^*$ allows us to use $z_{34} = w_{34}$ in place of $z_{12} = w_{12}$. With parameter t, write

$$(\zeta_3, \zeta_4, \zeta_5) = (m_5 u,\ m_5 v,\ -m_3 u - m_4 v), \qquad (\eta_3, \eta_4, \eta_5) = (m_5 p,\ m_5 q,\ -m_3 p - m_4 q),$$

so that $\sum_k m_k \zeta_k = \sum_k m_k \eta_k = 0$, and

$$\begin{aligned} z_i &= t^{-4} A_i, \qquad w_i = t^4 B_i \qquad (i \in P), \\ z_k &= t^{-1}\hat{Z} + t^2 \zeta_k, \qquad w_k = t^5 \hat{W} + t^2 \eta_k \qquad (k \in T), \\ \delta_{12} &= D_{12}, \qquad \delta_{ik} = t\, D_{ik}, \qquad \delta_{kl} = t^{-2} D_{kl}. \end{aligned} \tag{8.1}$$

The chart has twenty coordinates x: $A_1, A_2, B_1, B_2, \hat{Z}, \hat{W}, u, v, p, q$ and the ten D_{jl}. The substitution is invertible for $t \neq 0$. For convenience, set

$$\begin{aligned} \Sigma_i &= \sum_k m_k D^3_{ik}, \qquad f_i = \sum_k m_k D^3_{ik} \eta_k, \\ R^z_i &= A_i - m_{i'} D^3_{12}(A_i - A_{i'}), \qquad R^w_i = B_i - m_{i'} D^3_{12}(B_i - B_{i'}), \end{aligned}$$

where $\{i, i'\} = P$. Substituting (8.1) gives

$$\begin{aligned} t^4 E^z_i &= R^z_i - \textstyle\sum_k m_k D^3_{ik}\,(t^3 A_i - t^6 \hat{Z} - t^9 \zeta_k), \\ t^{-4} E^w_i &= R^w_i + t\, f_i - \textstyle\sum_k m_k D^3_{ik}\,(t^3 B_i - t^4 \hat{W}), \\ t^4 E^z_k &= -\phi_k + t^3 \hat{Z} + t^6 \zeta_k - \textstyle\sum_i m_i D^3_{ik}\,(t^6 \hat{Z} + t^9 \zeta_k - t^3 A_i), \\ t^4 E^w_k &= -\psi_k + t^6 \eta_k + t^9 \hat{W} - \textstyle\sum_i m_i D^3_{ik}\,(t^9 \eta_k + t^{12} \hat{W} - t^{11} B_i), \end{aligned} \tag{8.2}$$

The distance equations have left-hand sides $D_{12}^2(A_1 - A_2)(B_1 - B_2) - 1$, the crossed expressions $D_{ik}^2(t^3\hat{Z} + t^6\zeta_k - A_i)(\eta_k - t^2B_i + t^3\hat{W}) - 1$, and χ_{kl}. The gauge becomes $t^{-2}(z_{34} - w_{34}) = (\zeta_4 - \zeta_3) - (\eta_4 - \eta_3)$. Together these give 21 polynomials, denoted by $F(t, x) = \sum_{j\geq 0} t^j F_j(x)$. Their zero set at the boundary is the *boundary fibre* $\mathcal{B}_4 = \mathcal{B}_4(m)$, the zero set of F_0. Equation (8.2) shows that F_1 vanishes except in the two rows $t^{-4}E_i^w$, where it equals f_i. Likewise, F_2 vanishes except in the crossed distance equations, where it equals $D_{ik}^2 A_i B_i$.

8.1.2 Representation of germs in the chart

Let γ be a Diagram 4 germ at $m \in \mathbb{R}^5_{>0}$. Normalize it as in Lemma 3.3, so that $Z \approx W \approx \epsilon^{-2}$. After a base change of the branch parameter τ, all coordinates are Laurent series in τ, with $\operatorname{ord}_\tau Z = \operatorname{ord}_\tau W = -4q$ for an integer $q \geq 1$. Set $t = \tau^q$ and measure orders in units of $\operatorname{ord} t$. The maximal components of $\mathcal{Z}$ and $\mathcal{W}$ then have order -4, and $\operatorname{ord}\epsilon = 2$.

Lemma 8.3. *Along γ, for $i \in P$ and $k \neq l$ in T:*

(i) $\operatorname{ord} z_k > -4$ *and* $\operatorname{ord} z_{ik} = -4$*;*

(ii) $\operatorname{ord} z_1 = \operatorname{ord} z_2 = \operatorname{ord} z_{12} = -4$, $\operatorname{ord}\delta_{12} = 0$, $\operatorname{ord} w_{12} = 4$, $\operatorname{ord} w_1, \operatorname{ord} w_2 \geq 4$, *and hence* $\operatorname{ord} W_T = \operatorname{ord} W_P \geq 4$*;*

(iii) $\operatorname{ord} z_{kl} = \operatorname{ord} w_{kl} = 2$ *and* $\operatorname{ord}\delta_{kl} = -2$.

Proof. Part (i) follows because the bodies in T have no z-circles, whereas 1 and 2 do.

For (ii), the z-circles at bodies $1, 2$ give $\operatorname{ord} z_1 = \operatorname{ord} z_2 = -4$. The isolated component $\{1, 2\}$ of the z-diagram has circled vertices, so Rule 1e [AK12, p. 546] makes its stroke 12 maximal. The right equality case of Estimate 2 [AK12, p. 548] gives $r_{12} \approx 1$, $z_{12} \approx \epsilon^{-2}$ and $w_{12} \approx \epsilon^2$. The triangle edges are not maximal: part (iii), which is independent of (ii), gives $z_{kl} \approx \epsilon$ on them. Thus 12 is the only maximal z-edge. Estimate 1 [AK12, p. 548], with z and w exchanged, now gives $w_{12} \prec w_{ik}$ for $i \in P$, $k \in T$.

Apply [AK12, Prop. 1, p. 559], with z and w exchanged, to bodies 1 and 2, as in the proof of [AK12, Prop. 4, p. 564]. It places the origin in the w-cluster formed by w_1 and w_2, whose size is w_{12}. Hence $w_1, w_2 \preccurlyeq w_{12}$. The relation $M_P W_P + M_T W_T = 0$ gives the assertion about the two centres.

For (iii), the triangle edges are zw-edges. Estimate 1 and the left equality case of Estimate 2 give $r_{kl} \approx z_{kl} \approx w_{kl} \approx \epsilon$. □

Lemma 8.4. *Let $0 < e < 2$. In (8.1) replace $\hat{Z}$ by $t^{-3e/2}\hat{Z}$, $\hat{W}$ by $t^{-e/2}\hat{W}$, (η_3, η_4, η_5) by $(t^e\omega,\ m_5 q,\ -m_3 m_5^{-1} t^e\omega - m_4 q)$ and D_{i3} by $t^{-e/2}d_i$ $(i \in P)$. Then*

$$T_P = t^{1-e/2}\Big(m_3 \sum_{i\in P} m_i d_i \,(d_i^2 A_i \omega) + \sum_{g\in G} c_g\, t^g\Big),$$

where the c_g are polynomials in the chart coordinates and $G = \{\frac{1}{2}e,\ \frac{3}{2}e,\ 3,\ 3 - \frac{3}{2}e,\ 5 - \frac{5}{2}e,\ 5 - e,\ 8 - e,\ 8 + \frac{1}{2}e\}$; all elements of G are positive.

Proof. After the stated substitutions,

$$z_i w_k - w_i z_k = t^{-2} A_i \eta_k + t^{1-e/2} A_i \hat{W} - t^{3-3e/2} B_i \hat{Z} - t^6 B_i \zeta_k.$$

For $k=3$, we have $\eta_3 = t^e\omega$ and $\delta^3_{i3} = t^{3-3e/2}d_i^3$. The four terms, divided by $t^{1-e/2}$, have exponents 0, $3-3e/2$, $5-5e/2$ and $8-e$, and the first coefficient is $d_i^3A_i\omega$. For $k=4,5$, multiply by $t^3D^3_{ik}$ instead. The constant and t^e parts of η_k give the exponents $e/2$ and $3e/2$; the last three terms give 3, $5-e$ and $8+e/2$. These exponents give the eight elements of G, all of which are positive because $0<e<2$. □

Lemma 8.5. *For $k\in T$ let o_k be the coefficient of t^2 in $w_k - W_T$ along γ. Then $o_3, o_4, o_5 \neq 0$.*

Proof. Lemma 8.3(iii) identifies $o_l - o_k$ as the leading coefficient of w_{kl}. Thus at most one o_k vanishes. Renumbering T, suppose that $o_3 = 0$. Write $\operatorname{ord}(w_3 - W_T) = 2+e$ with $e>0$ when $w_3 - W_T \not\equiv 0$, and put $e=+\infty$ otherwise. By Lemma 8.3(ii), $\operatorname{ord}(W_T - w_i)\geq 4$.

There are two cases. If $e\geq 2$, then $\operatorname{ord} w_{i3}\geq 4$, so $\operatorname{ord}\delta_{i3} = -\frac12(\operatorname{ord} z_{i3} + \operatorname{ord} w_{i3}) \leq 0$ and $\operatorname{ord} Z_{i3} = \operatorname{ord} z_{i3} + 3\operatorname{ord}\delta_{i3} \leq -4$. This would make Z_{i3} a maximal component of $\mathcal{Z}$, producing a z-stroke between i and 3, contrary to Diagram 4.

Suppose instead that $0<e<2$. We have $\operatorname{ord} w_{i3} = 2+e$ and $\operatorname{ord}\delta_{i3} = 1-\frac{e}{2}$, whereas $\operatorname{ord} w_{ik} = 2$ and $\operatorname{ord}\delta_{ik} = 1$ for $k=4,5$. The first two identities of Lemma 5.2 express M_PW_P and M_TZ_T as sums of crossed terms $m_im_k\delta^3_{ik}(w_i-w_k)$ and $m_im_k\delta^3_{ik}(z_k-z_i)$. These have orders at least $5-\frac{e}{2}$ and $-1-\frac32 e$, respectively. Together with $M_PW_P + M_TW_T = 0$, this gives $\operatorname{ord} W_T \geq 5-\frac{e}{2}$ and $\operatorname{ord} Z_T \geq -1-\frac32 e$.

The modified chart in Lemma 8.4 therefore represents γ with all coordinates of nonnegative order, and with A_i, ω, d_i of order 0. At $t=0$, the distance equations $\delta^2_{i3}z_{i3}w_{i3}=1$ give $d_i^2A_i\omega = -1$, while $m_1A_1 + m_2A_2 = t^4(m_1z_1+m_2z_2) = -t^4M_TZ_T$ vanishes. Lemma 8.4 now shows that $t^{e/2-1}T_P$ tends to $-m_3(m_1d_1+m_2d_2)$. This limit is nonzero, since

$$(m_1d_1)^2 - (m_2d_2)^2 = -\frac{m_1^2}{A_1\omega} + \frac{m_2^2}{A_2\omega} = -\frac{m_1^3+m_2^3}{m_1A_1\omega} \neq 0.$$

This contradicts $T_P = 0$ on A_m, proved in Lemma 5.2. □

Proposition 8.6 (Coverage)**.** *Let $m\in\mathbb{R}^5_{>0}$ and let γ be a Diagram 4 germ. Then m satisfies (23), and $\operatorname{ord} Z_T\geq -1$, $\operatorname{ord} W_T\geq 5$ along γ. Renormalize by $(z,w)\mapsto(cz, c^{-1}w)$ with $c=(w_{34}/z_{34})^{1/2}$, so that $z_{34}=w_{34}$. After a base change of τ, every chart coordinate of (8.1) is a power series in τ; the value y of x at $\tau=0$ lies in $\mathcal{B}_4(m)$, and $\eta_k(y)\neq 0$ for $k\in T$.*

Proof. Relation (23) is Proposition 8.1. By Lemmas 8.3 and 8.5, $\operatorname{ord} w_{ik}=2$ for all $i\in P$, $k\in T$, hence $\operatorname{ord}\delta_{ik}=1$. The first two identities of Lemma 5.2 then give $\operatorname{ord} Z_T \geq 3-4 = -1$ and $\operatorname{ord} W_P \geq 3+2 = 5$, and $\operatorname{ord} W_T = \operatorname{ord} W_P$. The factor c has order 0 by Lemma 8.3(iii), so all orders persist after the renormalization. Inverting (8.1), $A_i = t^4z_i$, $B_i = t^{-4}w_i$, $\hat{Z} = tZ_T$, $\hat{W} = t^{-5}W_T$, $\zeta_k = t^{-2}(z_k - Z_T)$, $\eta_k = t^{-2}(w_k - W_T)$, $D_{12}=\delta_{12}$, $D_{ik} = t^{-1}\delta_{ik}$ and $D_{kl} = t^2\delta_{kl}$; by Lemma 8.3 all have nonnegative order, and $\eta_k \to o_k \neq 0$. The rows of F are monomial multiples of the equations, so they vanish along γ, and $F_0(y)=0$. □

8.1.3 The boundary fibre

On $\mathbb{C}^{20}$ consider the scalings

$$\alpha\cdot x:\ (A,B,D_{ik}) \mapsto (\alpha^2A,\ \alpha^{-2}B,\ \alpha^{-1}D_{ik}),$$
$$\beta\cdot x:\ (\zeta,\eta,D_{kl},D_{ik}) \mapsto (\beta^2\zeta,\ \beta^2\eta,\ \beta^{-2}D_{kl},\ \beta^{-1}D_{ik}),$$

Here $\alpha, \beta \in \mathbb{C}^*$, and coordinates not displayed remain fixed. Together with translations of $(\hat{Z}, \hat{W})$, these transformations preserve the zero set of F_0: each row changes only by a nonzero factor. They therefore act on $\mathcal{B}_4(m)$. We use these boundary symmetries to identify its tangent spaces and the cokernels of dF_0.

Lemma 8.7. *Let $m \in \mathbb{R}^5_{>0}$ satisfy (23) and let $y \in \mathcal{B}_4(m)$.*

(a) $D_{12}^3 = M_P^{-1}$, $A_2 = \rho A_1$ and $B_2 = \rho B_1$ with $\rho = -m_1/m_2$, $D_{12}^2(A_1 - A_2)(B_1 - B_2) = 1$ and $D_{ik}^2 A_i \eta_k = -1$; in particular $A_i, B_i, \eta_k \neq 0$. The point $(\zeta_3, \zeta_4, \eta_3, \eta_4, D_{34}, D_{35}, D_{45})$ solves the zero-multiplier triangle system, so, after renumbering T, it lies on the orbit O of Lemma 5.9.

(b) $dF_0(y)$ has rank 16. Its kernel is spanned by $\partial_{\hat{Z}}$, $\partial_{\hat{W}}$ and the generators of the two scalings. The annihilator of its image, equivalently the dual of its cokernel, is spanned by the linear form

$$v_P(r) = (B_1 - B_2)\,(r_1^z - r_2^z) - (A_1 - A_2)\,(r_1^w - r_2^w),$$

where r_i^z, r_i^w are the components of $r \in \mathbb{C}^{21}$ in the rows $t^4 E_i^z$, $t^{-4} E_i^w$, and by four linear forms that involve only the rows $t^4 E_k^z$, $t^4 E_k^w$, χ_{kl} and the gauge.

(c) Near y, $\mathcal{B}_4(m)$ coincides with the orbit of y under the scalings and the translations; it is a complex submanifold of dimension 4 with tangent space $\ker dF_0(y)$ at y.

Proof. For (a), the identities $R_1^z - R_2^z = (A_1 - A_2)(1 - M_P D_{12}^3)$ and $m_1 R_1^z + m_2 R_2^z = m_1 A_1 + m_2 A_2$ hold everywhere, as do the corresponding identities for R^w. The distance equation for 12 gives $A_1 \neq A_2$ and $B_1 \neq B_2$. Thus $F_0(y) = 0$ implies $D_{12}^3 = M_P^{-1}$, $m_1 A_1 + m_2 A_2 = 0$ and $m_1 B_1 + m_2 B_2 = 0$. The crossed distance equations are $D_{ik}^2 A_i \eta_k = -1$. The triangle equations in F_0 are $\phi_k, \psi_k, \chi_{kl}$ together with the gauge. The proof of Proposition 5.10 gives the last assertion.

For (b), order the coordinates as $(D_{ik};\, A, B, D_{12};\, u, v, p, q, D_{kl};\, \hat{Z}, \hat{W})$. Order the rows as the crossed distance equations, the pair equations R_i^z, R_i^w and the distance equation for 12, followed by the triangle equations. The pair and triangle equations involve only their respective coordinates. Each D_{ik} occurs only in its own crossed distance equation, where $\partial/\partial D_{ik} = -2 D_{ik} A_i \eta_k \neq 0$. So $dF_0(y)$ is block upper triangular, and its rank is $6 + \operatorname{rank} J_P + \operatorname{rank} J_T$, where J_P, J_T are the pair and triangle blocks. We compute their ranks separately.

At y, the differentials of $R_1^z - R_2^z$ and $R_1^w - R_2^w$ are $-3 M_P D_{12}^2$ times $(A_1 - A_2)\, dD_{12}$ and $(B_1 - B_2)\, dD_{12}$, those of $m_1 R_1^z + m_2 R_2^z$ and $m_1 R_1^w + m_2 R_2^w$ are $m_1 dA_1 + m_2 dA_2$ and $m_1 dB_1 + m_2 dB_2$, and the distance equation for 12 adds D_{12}^2 times $(B_1 - B_2)(dA_1 - dA_2) + (A_1 - A_2)(dB_1 - dB_2)$ modulo dD_{12}. These relations give $\operatorname{rank} J_P = 4$, with kernel spanned by $(A, -B, 0)$. The linear form v_P, obtained by differentiating at y the identity $(B_1 - B_2)(R_1^z - R_2^z) - (A_1 - A_2)(R_1^w - R_2^w) = 0$, spans the annihilator of its image.

By Lemma 5.9, the rows $\phi_3, \phi_4, \psi_3, \psi_4, \chi_{kl}$ have rank 5 at y, and their kernel is spanned by the generators $(\zeta, 0, -\frac{1}{2}D)$ and $(0, \eta, -\frac{1}{2}D)$ of $\mathbb{T}$. The rows ϕ_5 and ψ_5 are redundant because $\sum_k m_k \phi_k = \sum_k m_k \psi_k = 0$. On the first generator, the gauge row takes the value $\zeta_4 - \zeta_3$, which is nonzero by $\chi_{34} = 0$; it takes the value 0 on $(\zeta, \eta, -D)$. It follows that $\operatorname{rank} J_T = 6$, with kernel spanned by $(\zeta, \eta, -D)$.

The total rank is therefore 16. Completing the four stated vectors in their D_{ik} components by the crossed distance equations gives a basis of the kernel. Any linear form annihilating the image has zero coefficients on the crossed distance rows, as is seen from the columns D_{ik}. It is

consequently the sum of a form annihilating the image of J_P and one annihilating the image of J_T; the latter annihilator has dimension $10 - 6 = 4$. For (c), the differential of the orbit map is onto $\ker dF_0(y)$. The orbit is thus a submanifold of dimension 4 near y, contained in $\mathcal{B}_4(m)$. Conversely, (b) and the rank theorem place $\mathcal{B}_4(m)$ near y inside a submanifold of dimension $20 - 16 = 4$. The two submanifolds therefore coincide locally. □

8.1.4 A Lyapunov–Schmidt lemma

We now reduce continuation from the smooth boundary fibre to the successive Taylor coefficients in the cokernel of dF_0. A nonzero coefficient forces $t = 0$ near the boundary point and therefore excludes a germ entering $t \neq 0$. For a polynomial curve $x(t)$, denote by $[F]_j$ the coefficient of t^j in $F(t, x(t))$.

Lemma 8.8 (Lyapunov–Schmidt)**.** *Let $F(t, x) = \sum_{j\geq 0} t^j F_j(x)$ be a polynomial map $\mathbb{C} \times \mathbb{C}^n \to \mathbb{C}^p$, let $Z = F_0^{-1}(0)$, and let $y \in Z$ be a point near which Z is a complex submanifold with $T_{y'}Z = \ker dF_0(y')$ for all $y' \in Z$. For $y' \in Z$ let $Q_{y'}$ be the cokernel of $dF_0(y')$. Let $N \geq 1$ and suppose that every $y' \in Z$ near y is the value at $t = 0$ of a polynomial curve $x(t)$ with $[F]_j = 0$ for $j < N$.*

(a) The class $\Omega_N(y') \in Q_{y'}$ of $[F]_N$ does not depend on the choice of such a curve.

(b) If $\Omega_N(y) \neq 0$, no holomorphic germ $c\colon (\mathbb{C}, 0) \to (F^{-1}(0), (0, y))$ has $t \circ c \not\equiv 0$.

For $N = 1$ the hypothesis is empty and $\Omega_1(y')$ is the class of $F_1(y')$.

Proof. Let $K = \ker dF_0(y)$ and $R = dF_0(y)(\mathbb{C}^n)$. Choose complements $\mathbb{C}^n = K \oplus K'$ and $\mathbb{C}^p = R \oplus R'$, and let π project onto R along R'. The map $\pi\, dF_0(y)$ sends K' isomorphically onto R. The implicit function theorem therefore gives a holomorphic function $h(t, k) \in K'$, defined near $(0, 0) \in \mathbb{C} \times K$, with $h(0, 0) = 0$. Near $(0, y)$, the equation $\pi F(t, y + k + h') = 0$, for small $h' \in K'$, holds precisely when $h' = h(t, k)$. This uniqueness also holds for formal power series in t.

The remaining equations are $G(t, k) = F(t, y + k + h(t, k)) \in R'$. Thus, near $(0, y)$, we have $F^{-1}(0) = \{(t, y + k + h(t, k)) : G(t, k) = 0\}$. At the boundary, the map $k \mapsto y + k + h(0, k)$ parametrizes $\{\pi F_0 = 0\}$. This submanifold contains Z and has the same dimension $\dim K = \dim Z$; it therefore coincides locally with Z. In particular, $G(0, k) = 0$. Write $G = \sum_{j\geq 1} t^j G_j(k)$. For every $y' \in Z$ near y, the space R' remains a complement of $dF_0(y')(\mathbb{C}^n)$, and hence $R' \to Q_{y'}$ is an isomorphism.

We now compare coefficients along a polynomial curve. Fix $j \leq N$ and $y' = y + k' + h(0, k')$, and suppose that $G_1, \ldots, G_{j-1}$ vanish near k'. Let $x(t) = y + \kappa(t) + \eta(t)$, with $\kappa(t) \in K$ and $\eta(t) \in K'$, be a polynomial curve satisfying $x(0) = y'$ and $[F]_i = 0$ for $i < j$. Since $\pi F(t, x(t)) = O(t^j)$, formal uniqueness gives $\eta(t) - h(t, \kappa(t)) = e\, t^j + O(t^{j+1})$ for some e. Taylor's formula yields $F(t, x(t)) = F(t, y + \kappa + h(t, \kappa)) + dF_0(y')(e)\, t^j + O(t^{j+1})$, while $F(t, y + \kappa + h(t, \kappa)) = G(t, \kappa(t)) = t^j G_j(k') + O(t^{j+1})$. Consequently, $[F]_j \equiv G_j(k')$ modulo $dF_0(y')(\mathbb{C}^n)$.

For $j < N$, the curves in the hypothesis give $G_j(k') \in R' \cap dF_0(y')(\mathbb{C}^n) = 0$ for all k' near 0. Induction shows that $G_1, \ldots, G_{N-1}$ vanish near 0. The case $j = N$ proves (a): $\Omega_N(y')$ is the image of $G_N(k')$.

If $\Omega_N(y) \neq 0$, then $G = t^N H$ for a holomorphic H with $H(0, 0) = G_N(0) \neq 0$. Thus, near $(0, 0)$, the equation $G = 0$ forces $t = 0$. Every germ c in (b) has the form $(t, y + k + h(t, k))$ and satisfies $G(t, k) = 0$, so $t \circ c \equiv 0$. □

8.1.5 The first three obstructions

We apply the reduction to the chart equations. The first and third coefficients will impose the two force balances used to exclude the germ. For $y \in \mathcal{B}_4(m)$, define $S_i = \sum_k m_k D_{ik}$ and

$$\Theta := \sum_k m_k \frac{D_{1k} - D_{2k}}{\eta_k} = A_2\Sigma_2 - A_1\Sigma_1.$$

The second expression, as well as $f_i = -S_i/A_i$, follows from $1/\eta_k = -A_i D_{ik}^2$. Under the boundary scalings, S_i/A_i acquires the factors α^{-3} and β^{-1}, while Θ acquires α^{-1} and β^{-3}. Neither expression involves $\hat{Z}$, $\hat{W}$, D_{12} or m_1, m_2.

These quantities describe the leading forces from the triangle. On body i, the crossed forces are $\sum_k m_k \delta_{ik}^3 (z_i - z_k) = t^{-1} A_i \Sigma_i + O(t^2)$ and $\sum_k m_k \delta_{ik}^3 (w_i - w_k) = -t^5 f_i + O(t^7)$. Thus $f_1 = f_2$ and $\Theta = 0$ express equality of the leading forces on the two outer bodies in the w- and z-coordinates, respectively.

Lemma 8.9. *Let $m \in \mathbb{R}^5_{>0}$ satisfy (23), let $y \in \mathcal{B}_4(m)$, and let Ω_N be the classes of Lemma 8.8 for F and $Z = \mathcal{B}_4(m)$. We regard v_P as a linear functional on the cokernel Q_y.*

(a) $\Omega_1(y) = 0$ if and only if $f_1 = f_2$, that is, $S_1/A_1 = S_2/A_2$.

(b) If $f_1 = f_2$ at y, then the hypothesis of Lemma 8.8 holds at y for $N = 3$, and

$$v_P\big(\Omega_3(y)\big) = \tfrac{3}{2}\,(A_1 - A_2)\,\frac{B_1}{A_1}\,\Theta.$$

Proof. For (a), Lemma 8.7(b) shows that $\Omega_1(y)$, the class of $F_1(y)$, vanishes precisely when $F_1(y)$ is annihilated by v_P and the four triangle forms. The triangle components of F_1 vanish, while $v_P(F_1(y)) = -(A_1 - A_2)(f_1 - f_2)$.

For (b), homogeneity implies $f_1 = f_2$ throughout the orbit of y. By Lemma 8.7(c), this orbit is a neighbourhood of y in $\mathcal{B}_4(m)$. Choose a point y' in it and $N \in \{2, 3\}$. Consider a polynomial curve $x(t)$ with $x(0) = y'$ and $F(t, x(t)) = O(t^N)$; part (a) supplies such a curve for $N = 2$. Write $A_i(t), \dots$ for its components and $A_i, \dots$ for their values at y', and set $\Phi(t) = 1 - M_P D_{12}(t)^3$.

We first examine the pair equations. Equation (8.2) gives $t^4 E_i^z = R_i^z + O(t^3)$. Since $R_1^z - R_2^z = (A_1(t) - A_2(t))\Phi(t)$ and $m_1 R_1^z + m_2 R_2^z = m_1 A_1(t) + m_2 A_2(t)$, we obtain $\Phi = O(t^N)$ and $A_i(t) = A_i(1 + a(t)) + O(t^N)$ for a common series $a(t) = O(t)$.

Next, the triangle rows of F agree with those of F_0 up to $O(t^3)$. The triangle components of $x(t)$ therefore solve the triangle block of F_0 modulo t^N. The proof of Lemma 8.7(b),(c) identifies the zero set of that block near y' with the curve $\beta \mapsto \beta \cdot y'$, whose dimension is $\dim \ker J_T$. Consequently, the triangle components of $x(t)$ agree modulo t^N with a curve on this orbit. In particular, $\eta_k(t) = \theta(t)\eta_k + O(t^N)$, where $\theta(0) = 1$. For $N = 2$, the triangle components of $[F]_2$ therefore belong to the image of J_T.

The crossed distance equations give $D_{ik}(t)^2 = -1/\big(A_i(t)(\eta_k(t) - t^2 B_i)\big) + O(t^N)$, and hence

$$D_{ik}(t)^3 \eta_k(t) = -\frac{D_{ik}(t)}{A_i(t)}\Big(1 + t^2 \frac{B_i}{\eta_k}\Big) + O(t^N), \qquad D_{ik}(t) = \frac{D_{ik}}{\sqrt{(1+a)\theta}}\Big(1 + \frac{t^2 B_i}{2\eta_k}\Big) + O(t^N),$$

and, with $f_i(t) = \sum_k m_k D_{ik}(t)^3 \eta_k(t)$,

$$f_i(t) = -\frac{S_i/A_i}{(1+a)^{3/2}\theta^{1/2}} - \frac{3}{2}\,t^2\,\frac{B_i}{A_i}\sum_k m_k \frac{D_{ik}}{\eta_k} + O(t^N).$$

Since $S_1/A_1 = S_2/A_2$ and $B_1/A_1 = B_2/A_2$, $f_1(t) - f_2(t) = -\frac{3}{2}\,t^2\,\frac{B_1}{A_1}\,\Theta + O(t^N)$.

We can now evaluate the obstruction. Applying v_P to $F(t, x(t))$, with its coefficients fixed at y', gives, by (8.2),

$$\Phi(t)\big[(B_1-B_2)(A_1(t)-A_2(t))-(A_1-A_2)(B_1(t)-B_2(t))\big]-(A_1-A_2)\,t\,\big(f_1(t)-f_2(t)\big)-t^3\,\Delta_4+O(t^4),$$

with $\Delta_4 = (B_1 - B_2)(A_1\Sigma_1 - A_2\Sigma_2) - (A_1 - A_2)(B_1\Sigma_1 - B_2\Sigma_2)$, which vanishes because $A_2 = \rho A_1$ and $B_2 = \rho B_1$. The bracket vanishes at $t = 0$ and $\Phi = O(t^N)$.

For $N = 2$, this gives $v_P([F]_2) = 0$. Every linear form annihilating the image of $dF_0(y')$ has zero coefficients on the crossed distance rows. Together with the preceding result for the triangle part, this gives $\Omega_2(y') = 0$. We can therefore correct the curve at order 2 to obtain a curve with $F(t, x(t)) = O(t^3)$. This proves the hypothesis of the Lyapunov–Schmidt lemma for $N = 3$. For $N = 3$, the displayed expression gives $v_P([F]_3) = -(A_1 - A_2)\cdot\big(-\frac{3}{2}\frac{B_1}{A_1}\Theta\big)$, as claimed. □

8.1.6 Incompatibility of the two force balances

The triangle classification gives a real parametrization of its limiting offsets. We use it to express the two force balances as scalar equations and then apply positivity and strict convexity. After renumbering T, write $(\sqrt{m_3}, \sqrt{m_4}, \sqrt{m_5}) = \lambda(bc, b(b+c), c(b+c))$ with $\lambda, b, c > 0$. Exchanging 4 and 5 if necessary allows us to assume $b > c$. The case $b = c$ would give a zero triangle offset, contrary to Proposition 8.6. The triangle offsets are proportional to

$$(o_3, o_4, o_5) = \Big(-\frac{b^2 - c^2}{bc}, \frac{c(2b+c)}{b(b+c)}, -\frac{b(b+2c)}{c(b+c)}\Big).$$

Put $a = \sqrt{-o_3/o_4} > 0$, $d = \sqrt{o_3/o_5} > 0$, and $(g_3, g_4, g_5) = (1, ia, d)$. If $x = b/c > 1$, then

$$a^2 = \frac{(x-1)(x+1)^2}{2x+1}, \qquad d^2 = \frac{(x-1)(x+1)^2}{x^2(x+2)}.$$

In particular $0 < d < a$ and $d < 1$: the differences of the relevant denominators are $x^2(x+2) - (2x+1) = (x-1)(x^2+3x+1) > 0$ and $x^2(x+2) - (x-1)(x+1)^2 = x^2 + x + 1 > 0$. The mass-centre relation $\sum_k m_k o_k = 0$ becomes

$$\frac{m_4}{a^2} = m_3 + \frac{m_5}{d^2}. \tag{8.3}$$

For $s = (s_{13}, s_{14}, s_{15}, s_{23}, s_{24}, s_{25}) \in \{1, -1\}^6$ and $j = 1, 3$, define

$$X_j = \sum_{k\in T} s_{1k} m_k g_k^j, \qquad Y_j = \sum_{k\in T} s_{2k} m_k g_k^j.$$

Lemma 8.10. *For every sign vector s and every $\nu > 0$, the two equations*

$$\nu^3 X_1 + iY_1 = 0, \qquad \nu X_3 = iY_3$$

cannot hold simultaneously.

Proof. Put $r = m_3$, $u = m_4 a$, $v = m_5 d$ and $\varepsilon = s_{13}s_{15}$, $\delta = s_{23}s_{25}$. By (8.3) and $a > d$,

$$u = a^3 r + \frac{m_5 a^3}{d^2} > m_5 d = v.$$

The real and imaginary parts of the first equation give

$$\nu^3 s_{13}(r + \varepsilon v) = s_{24} u, \qquad \nu^3 s_{14} u = -s_{23}(r + \delta v).$$

Thus $|(r+\varepsilon v)(r+\delta v)| = u^2$. Both factors $r+\varepsilon v$ and $r+\delta v$ must be positive. Indeed, if a factor is nonpositive then $v \geq r$ and at least one sign is negative; the absolute product is then either $(v-r)^2$ or $v^2 - r^2$, in both cases strictly less than $v^2 < u^2$. It follows that $s_{24} = s_{13}$ and $s_{14} = -s_{23}$, and hence

$$\nu^3(r + \varepsilon v) = u, \qquad \nu^3 u = r + \delta v. \tag{8.4}$$

Since $g_4^3 = -ia^3$, the real and imaginary parts of the second equation now give

$$\nu(r + \varepsilon v d^2) = u a^2, \qquad \nu u a^2 = r + \delta v d^2. \tag{8.5}$$

If $\varepsilon = \delta = 1$, then (8.4) gives $\nu = 1$ and $u = r + v$. Put $w_0 = r/(r + m_5/d^2)$ and $w_1 = (m_5/d^2)/(r + m_5/d^2)$. These weights are positive and sum to 1. Equations (8.3), (8.4) and (8.5) give, respectively,

$$a^3 = w_0 + w_1 d^3, \qquad a^5 = w_0 + w_1 d^5.$$

But strict convexity of $t \mapsto t^{5/3}$ on $(0, \infty)$, applied to 1 and $d^3 \neq 1$, gives $w_0 + w_1 d^5 > (w_0 + w_1 d^3)^{5/3} = a^5$, a contradiction.

In the other cases (8.4) gives either $u = r - v < r$ or $u^2 = r^2 - v^2 < r^2$, so $u < r$. This forces $a < 1$: otherwise (8.3) would give $r < m_4 \leq m_4 a = u$. Thus $d < a < 1$. If $\varepsilon = \delta = -1$, then $\nu = 1$ and the two balance equations give $r = u + v = ua^2 + vd^2 < u + v$, again a contradiction. Finally, if $\varepsilon = -\delta$, adding the two relations in each of (8.4) and (8.5) yields

$$2r = u(\nu^3 + \nu^{-3}) = ua^2(\nu + \nu^{-1}).$$

Consequently $a^2 = (\nu^3 + \nu^{-3})/(\nu + \nu^{-1}) = \nu^2 + \nu^{-2} - 1 \geq 1$, contradicting $a < 1$. □

8.1.7 Proof of Theorem 8.2

Suppose that γ is a Diagram 4 germ at $m \in \mathbb{R}^5_{>0}$. Proposition 8.6 gives relation (23) and a holomorphic chart representation $\tau \mapsto (t(\tau), x(\tau))$, with $t = \tau^q$, in $F^{-1}(0)$. It satisfies $t \not\equiv 0$ and $x(0) = y \in \mathcal{B}_4(m)$. Lemma 8.7(c) permits the application of Lemma 8.8 at y, with $Z = \mathcal{B}_4(m)$. Taking $N = 1$ gives $\Omega_1(y) = 0$, or $f_1 = f_2$ by Lemma 8.9(a). Lemma 8.9(b) then permits the same argument for $N = 3$, giving $\Omega_3(y) = 0$. Since $(A_1 - A_2)B_1/A_1 \neq 0$, it follows that $\Theta = 0$ at y.

Both conditions are symmetric in the bodies of T and homogeneous under the boundary scalings. We may therefore renumber T as in §8.1.6 with $b > c$. Indeed, $b = c$ would contradict $\eta_k(y) \neq 0$ in Proposition 8.6, because $o_3 = 0$ when $b = c$. The boundary scalings allow us to normalize $A_1 = 1$ and $\eta_3 = -1$. Then $\eta_k = -o_k/o_3$ and $D_{1k}^2 = -1/\eta_k = g_k^2$. With $\nu = \sqrt{m_1/m_2} > 0$ we have $A_2 = -\nu^2$ and $D_{2k}^2 = -D_{1k}^2/\nu^2$, so $D_{1k} = s_{1k} g_k$ and $D_{2k} = s_{2k} i g_k/\nu$ for a sign vector s. With the original triangle masses in the definitions of X_j, Y_j, we obtain

$$S_1 = X_1, \qquad S_2 = iY_1/\nu, \qquad \Sigma_1 = X_3, \qquad \Sigma_2 = -iY_3/\nu^3.$$

Thus $S_1/A_1 = S_2/A_2$ and $A_2\Sigma_2 - A_1\Sigma_1 = 0$ become $\nu^3 X_1 + iY_1 = 0$ and $\nu X_3 = iY_3$. Lemma 8.10 gives the contradiction. □

8.2 Diagram 5

Diagram 5 consists of the isolated zw-triangle 345, with no circled vertex and no other stroke [AK12, §7.4 and Fig. 11]. We write $T = \{3, 4, 5\}$ and call $1, 2$ the outer bodies. We compare the shrinking triangle with its distances to the outer bodies by adding the orders of all ten squared distances. Our aim is $\operatorname{ord} \Delta > 0$, which gives the vanishing required in the global argument.

Theorem 8.11. *Let $m \in \mathbb{R}^5_{>0}$ and let γ be a Diagram 5 germ on an irreducible curve $\Gamma \subset A_m$. Then*

$$\Delta := \prod_{1 \le i < j \le 5} r_{ij}^2 \longrightarrow 0 \quad \text{along } \gamma.$$

8.2.1 Coordinates relative to the triangle

We normalize as in Lemma 3.3 and measure orders in a Puiseux scale t for which $\epsilon \approx |t|$ and the least orders of both $\mathcal{Z}$ and $\mathcal{W}$ are -2. If initially $\epsilon \approx |s|^\nu$, we take $t = cs^\nu$, $c \neq 0$, after a finite ramification as necessary. This choice only multiplies all orders by a positive factor. As usual, $\operatorname{ord} 0 = +\infty$.

Let $M_T = \sum_{k \in T} m_k$, $M = \sum_{k=1}^5 m_k$, and let z_c, w_c be the mass centres of the triangle. We distinguish the internal coordinates of the triangle from the positions of the outer bodies relative to its centre:

$$\xi_k = z_k - z_c, \quad \eta_k = w_k - w_c \quad (k \in T),$$
$$X_j = z_j - z_c, \quad Y_j = w_j - w_c \quad (j = 1, 2).$$

Proposition 8.1 and Lemma 5.8 give relation (23) and show that the limiting triangle has zero multiplier. After a constant change of t and a constant rotation $(z, w) \mapsto (az, a^{-1}w)$, its internal coordinates have the form

$$\xi_k = t(s_k + o(1)), \qquad \eta_k = t(s_k + o(1)), \qquad s_k \in \mathbb{R}, \quad \sum_{k \in T} m_k s_k = 0, \tag{8.6}$$

where the three s_k are pairwise distinct. Every squared distance, as well as U and I, is preserved by these changes.

The rules and estimates in Fact AK2 now give the orders that we shall use. Each coordinate difference within the triangle has order 1 in both colours, so each of its three squared distances has order 2. Since there is no circled vertex, all body coordinates, and therefore all differences from the triangle centre, have order strictly greater than -2. Every crossed force term $\delta^3_{jk}(z_j - z_k)$ and $\delta^3_{jk}(w_j - w_k)$, $j \in \{1, 2\}$, $k \in T$, has order strictly greater than -2, since no stroke joins an outer body to the triangle. In particular,

$$\operatorname{ord} r_{kl}^2 = 2 \quad (k < l,\ k, l \in T), \qquad \operatorname{ord} X_j, \operatorname{ord} Y_j > -2 \quad (j = 1, 2), \qquad \operatorname{ord} r_{ij} \le 1,$$

with equality in the last bound exactly on zw-edges.

The scale of the triangle is thus fixed. Relative to this scale, each outer body belongs to exactly one of the following three types:

- *far*: $\operatorname{ord} X_j < 1$ and $\operatorname{ord} Y_j < 1$;
- *near in z*: $\operatorname{ord} X_j \ge 1$ and $\operatorname{ord} Y_j = \beta_j < 1$;

- *near in w*: the preceding type with the colours exchanged.

Indeed, $\operatorname{ord} X_j, \operatorname{ord} Y_j \geq 1$ would give $\operatorname{ord} r_{jk} \geq 1$ for every $k \in T$, hence additional zw-edges. In a near type, the coordinate of order at least 1 may vanish identically.

For a body near in z, the limit $\theta_j = \lim t^{-1} X_j$ exists. Equation (8.6) shows that at least two of the differences $z_j - z_k$ have order 1. The third can have a higher order $h_j > 1$ only when its leading coefficient vanishes at one triangle vertex. We record this distinction by the excess order

$$e_j := \begin{cases} 0, & \theta_j \neq s_k \text{ for every } k \in T, \\ h_j - 1, & \theta_j = s_k \text{ for some } k \in T. \end{cases}$$

In the second case, we call the unique such k the contact vertex. The distance constraints give $z_j - z_k \neq 0$, so $h_j < \infty$. Summing the three squared-distance orders gives

$$\sum_{k \in T} \operatorname{ord} r_{jk}^2 = 3(1 + \beta_j) + e_j \qquad \text{for a body near in } z.$$

We now separate the forces exerted by the triangle on each outer body from the mutual force of the two outer bodies. These are, respectively,

$$F_j = \sum_{k \in T} m_k \delta_{jk}^3 (z_j - z_k), \qquad G_j = \sum_{k \in T} m_k \delta_{jk}^3 (w_j - w_k),$$
$$P = \delta_{12}^3 (X_1 - X_2), \qquad H = \delta_{12}^3 (Y_1 - Y_2).$$

In these relative coordinates, the original equations and the total mass-centre identity become

$$\begin{aligned} X_1 - X_2 &= (m_1 + m_2) P + F_1 - F_2, \\ X_1 &= m_2 P + (1 + m_1/M_T) F_1 + (m_2/M_T) F_2, \\ X_2 &= -m_1 P + (m_1/M_T) F_1 + (1 + m_2/M_T) F_2. \end{aligned} \tag{8.7}$$

The same identities hold with (X, P, F) replaced by (Y, H, G). Their weighted sums are

$$m_1 X_1 + m_2 X_2 = \frac{M}{M_T}(m_1 F_1 + m_2 F_2), \qquad m_1 Y_1 + m_2 Y_2 = \frac{M}{M_T}(m_1 G_1 + m_2 G_2). \tag{8.8}$$

For example, multiplying the two original outer z-equations by m_1, m_2 and adding cancels their mutual force. The mass-centre identity gives $m_1 z_1 + m_2 z_2 = (M_T/M)(m_1 X_1 + m_2 X_2)$, which proves the first identity. The coefficients of (F_1, F_2) in the last two equations of (8.7) form the matrix

$$C = \begin{pmatrix} 1 + m_1/M_T & m_2/M_T \\ m_1/M_T & 1 + m_2/M_T \end{pmatrix}, \qquad C^{-1} = \frac{1}{M} \begin{pmatrix} M_T + m_2 & -m_2 \\ -m_1 & M_T + m_1 \end{pmatrix}. \tag{8.9}$$

Its determinant is $M/M_T > 0$. In particular, leading terms of the two triangle forces cannot cancel in both outer-body equations simultaneously.

Throughout this subsection put

$$a = \operatorname{ord}(X_1 - X_2), \qquad b = \operatorname{ord}(Y_1 - Y_2), \qquad S = a + b, \qquad d = \operatorname{ord} \Delta.$$

Because 12 is an actual edge, a, b are finite. The distance constraints give

$$\operatorname{ord} P = a - \frac{3}{2}S = -\frac{a+3b}{2}, \qquad \operatorname{ord} H = b - \frac{3}{2}S = -\frac{3a+b}{2}. \tag{8.10}$$

If the coordinate differences on a crossed edge have orders h, v, its contributions to F_j, G_j have orders $-(h+3v)/2$ and $-(3h+v)/2$, respectively. Cancellation in the sum can only increase the order of a force. When $e_j > 0$ and body j is near in z, the edge to its contact vertex gives the unique term of lowest order in each force. Hence

$$\operatorname{ord} F_j = -\frac{h_j + 3\beta_j}{2}, \qquad \operatorname{ord} G_j = -\frac{3h_j + \beta_j}{2}. \tag{8.11}$$

When both bodies are far, we shall also need the inertia. Lemma 2.1 shows that I is constant on every irreducible complex solution curve. Its decomposition relative to the triangle centre is

$$\begin{gathered} I = I_T + J, \qquad I_T = \sum_{k\in T} m_k \xi_k \eta_k = O(t^2), \\ J = \frac{m_1 m_2 (X_1 - X_2)(Y_1 - Y_2) + m_1 M_T X_1 Y_1 + m_2 M_T X_2 Y_2}{M}. \end{gathered} \tag{8.12}$$

Since the internal inertia tends to zero, J is bounded.

8.2.2 The leading external force on the triangle

When both outer bodies are near in the same colour, the triangle equations resolve the possible cancellation of their forces. We shall prove that the leading external force is equal at the three vertices. Subtracting the mass-weighted mean then removes this force from the internal equations.

To study the internal forces, put $\hat{x} = \xi/t$, $\hat{y} = \eta/t$, and let

$$\widehat{U}(\hat{x}, \hat{y}) = \sum_{k<l,\ k,l\in T} m_k m_l \big((\hat{x}_k - \hat{x}_l)(\hat{y}_k - \hat{y}_l)\big)^{-1/2},$$

where the inverse-distance branches are those selected by the germ. Let ϕ, ψ denote the reduced triangle forces in the z, w directions. Differentiation and homogeneity give

$$d\widehat{U} = -\frac{1}{2}\sum_k m_k(\psi_k\, d\hat{x}_k + \phi_k\, d\hat{y}_k), \qquad \widehat{U} = \sum_k m_k \hat{x}_k \psi_k = \sum_k m_k \hat{y}_k \phi_k. \tag{8.13}$$

In particular, both forces tend to zero at $(\hat{x}, \hat{y}) = (s, s)$.

The next estimate compares the orders of the two reduced forces. It will force the leading external force to vanish after subtraction of its mean.

Lemma 8.12. *For a germ of reduced triangles tending to (s, s), put $\kappa_\psi = \operatorname{ord}\psi$ and $\kappa_\phi = \operatorname{ord}\phi$, where the order of a vector is the least order of its components. If $\kappa_\psi < \kappa_\phi$, then $\kappa_\phi \le 2\kappa_\psi$. The assertion also holds with the two forces exchanged.*

Proof. Translations and independent scalings in the two colours preserve the zero-multiplier equations. The resulting orbit is

$$O = \{(As + u\mathbf{1}, Bs + v\mathbf{1}) : A, B \neq 0\}.$$

On this orbit $d\widehat{U} = 0$ and $\widehat{U} = 0$ by (8.13). Let d_{kl} be the inverse distances at (s, s) and define the symmetric matrix

$$L_{kl} = -m_k m_l d_{kl}^3 \quad (k \neq l), \qquad L_{kk} = \sum_{l \neq k} m_k m_l d_{kl}^3.$$

The relations $L\mathbf{1} = Ls = 0$ express the translation and scaling directions. Since its off-diagonal entries are nonzero, the matrix has rank one and kernel $\operatorname{span}\{\mathbf{1}, s\}$. Explicitly, $L = c\, nn^{\mathsf{T}}$, $c \neq 0$, for $n = (s_4 - s_5, s_5 - s_3, s_3 - s_4)^{\mathsf{T}}$, so $\ell := L_{33} \neq 0$. At (As, Bs), choose $\varrho^2 = AB$ consistently with the branches. Differentiating each potential term twice gives

$$\nabla^2 \widehat{U} = \frac{1}{4\varrho} \begin{pmatrix} 3A^{-2}L & (AB)^{-1}L \\ (AB)^{-1}L & 3B^{-2}L \end{pmatrix}.$$

Its kernel is precisely the tangent space of O. The remaining two directions describe changes of shape. We may choose the $\hat{x}_3, \hat{y}_3$ directions as a complement because $s_4 \neq s_5$. On this complement, the Hessian K satisfies

$$\det K = \frac{\ell^2}{2\varrho^6} \neq 0, \qquad (K^{-1})_{11} = \frac{3A^2\varrho}{2\ell} \neq 0, \qquad (K^{-1})_{22} = \frac{3B^2\varrho}{2\ell} \neq 0.$$

A nearby triangle can therefore be written as $o(u) + v_1 e_3^x + v_2 e_3^y$, with $o(u) \in O$. Expanding in the two transverse directions gives $\widehat{U} = \frac{1}{2} v^{\mathsf{T}} K(u) v + O(|v|^3)$ and $\partial_v \widehat{U} = K(u)v + O(|v|^2) = -\frac{m_3}{2}(\psi_3, \phi_3)$. Here $v = (v_1, v_2)$ measures the departure from the zero-multiplier orbit; u may vary along the germ. Since $K(u(t))$ tends to an invertible matrix, $v(t)$ and $\partial_v \widehat{U}$ have the same order. The Taylor estimates are uniform for nearby orbit coordinates. Consequently

$$\operatorname{ord} \widehat{U} \geq 2 \operatorname{ord} v = 2 \operatorname{ord} \partial_v \widehat{U} \geq 2 \min(\kappa_\psi, \kappa_\phi).$$

Suppose $\kappa_\psi < \kappa_\phi$. Then $0 < \kappa_\psi < \infty$. The leading coefficient of ψ satisfies both $\sum_k m_k \psi_{k,0} = 0$ and $\sum_k m_k s_k \psi_{k,0} = 0$, the latter by (8.13) and $\operatorname{ord} \widehat{U} > \kappa_\psi$. These two relations imply that its third component is nonzero: otherwise $s_4 \neq s_5$ would force the other two components to vanish as well. Thus the transverse gradient has leading term $t^{\kappa_\psi}(\chi, 0)$, $\chi \neq 0$. Substituting it into

$$\widehat{U} = \frac{1}{2} (\partial_v \widehat{U})^{\mathsf{T}} K(u)^{-1} (\partial_v \widehat{U}) + O(|v|^3)$$

and using $(K^{-1})_{11} \neq 0$ gives $\operatorname{ord} \widehat{U} = 2\kappa_\psi$. The last identity in (8.13) also gives $\operatorname{ord} \widehat{U} \geq \kappa_\phi$, which proves the assertion. The same argument applies if one force vanishes identically. □

Lemma 8.13 (Equality of the leading external forces)**.** *Suppose both outer bodies are near in z. Let κ_* be the least order of the six individual terms $t^2 m_j \delta_{jk}^3 (w_k - w_j)$, $j \in \{1, 2\}$, $k \in T$. After summing over j, the coefficient of t^{κ_*} is the same at all three triangle vertices. The same assertion holds with the colours exchanged.*

Proof. Write $\operatorname{ord}(z_j - z_k) = 1 + e_{jk}$, $e_{jk} \geq 0$, and $\operatorname{ord}(w_j - w_k) = \beta_j \in (-2, 1)$. The orders of the reduced forcing terms in the two colours are

$$\kappa_{jk} = \frac{1 - \beta_j - 3e_{jk}}{2}, \qquad \frac{3 - 3\beta_j - e_{jk}}{2} = 2\kappa_{jk} + \frac{1 - \beta_j + 5e_{jk}}{2} > 2\kappa_{jk}.$$

The absence of crossed strokes gives $\kappa_* > 0$. At least two edges from each outer body satisfy $e_{jk} = 0$, and $\beta_j > -2$, so also $\kappa_* < 3/2$. Let $g_k = t^2 \sum_j m_j \delta_{jk}^3 (w_k - w_j)$ and let g^z be the analogous z-forcing. A hat on either forcing denotes subtraction of its mass-weighted mean over T. The exact centred triangle equations are

$$\psi = t^3 \hat{y} - \widehat{g}, \qquad \phi = t^3 \hat{x} - \widehat{g^z}.$$

If the coefficient of t^{κ_*} in g were not constant across the vertices, it would survive subtraction of the mean. Then $\kappa_\psi = \kappa_*$, while every term contributing to ϕ has order strictly greater than $2\kappa_*$, including $t^3\hat{x}$. This contradicts Lemma 8.12. □

Lemma 8.14 (Support of the leading external force)**.** *For a single body near in z with $e_j = 0$, its three lowest w-forcing coefficients are not all equal. If $e_j > 0$, its lowest forcing is supported at its contact vertex. If both bodies are near in z and $e_1, e_2 > 0$, their contact vertices coincide, their lowest forcings cancel there, and*

$$3h_1 + \beta_1 = 3h_2 + \beta_2. \tag{8.14}$$

Proof. When $e_j = 0$, $z_j - z_k = t(\theta_j - s_k + o(1))$ with all $\theta_j - s_k \neq 0$. The squares of the three forcing coefficients equal a common nonzero constant times $(\theta_j - s_k)^{-3}$. If the coefficients were equal, the three distinct numbers $\theta_j - s_k$ would have equal nonzero cubes and hence would be all three cube roots of one number. Their sum would be zero, making $\theta_j = (s_3 + s_4 + s_5)/3$ real. Injectivity of the real cube would then force the three numbers to coincide, a contradiction.

When $e_j > 0$, the edge to the contact vertex gives the unique lowest-order force term. If $e_1, e_2 > 0$, unequal lowest orders would leave the lowest coefficient supported at one vertex. Equal orders with different contact vertices would leave it supported at two vertices. Both are forbidden by Lemma 8.13: a coefficient supported at one or two vertices cannot be constant at all three. Therefore their vertices and orders agree, and their leading coefficients cancel at the common vertex. The unscaled order is $-(3h_j + \beta_j)/2$, which proves (8.14). □

8.2.3 Two far bodies

We first consider the case in which both outer bodies are far. Let $u_j = \operatorname{ord} X_j < 1$, $v_j = \operatorname{ord} Y_j < 1$, and $A = u_1 + v_1$, $B = u_2 + v_2$. All squared distances from body 1 to the triangle have order A, and those from body 2 have order B, so

$$d = 6 + 3A + 3B + S. \tag{8.15}$$

Individual force terms give

$$\operatorname{ord} F_1 \geq u_1 - \frac{3}{2}A, \quad \operatorname{ord} G_1 \geq v_1 - \frac{3}{2}A, \qquad \operatorname{ord} F_2 \geq u_2 - \frac{3}{2}B, \quad \operatorname{ord} G_2 \geq v_2 - \frac{3}{2}B.$$

Thus $A < 0$ makes the first two force orders strictly greater than u_1, v_1, and $B < 0$ gives the analogous conclusion for body 2.

To show that the product has positive order, we exclude negative orders among these three groups of distances. The three terms in J in (8.12) have orders S, A, B and nonzero mass coefficients. Since J is bounded, a negative minimum must occur at least twice. Thus, if any order is negative,

either $A, B < 0$, or, after exchanging the outer bodies, $A \geq 0 > B$ and $S = B < 0$. Indeed $S < 0$ alone would be a unique negative minimum.

Suppose first that $A, B < 0$. In the first equation of (8.8) the right side has order greater than $\min(u_1, u_2)$, so

$$u_1 = u_2 = u, \qquad m_1 \widehat{X}_1 + m_2 \widehat{X}_2 = 0.$$

Here the hats denote nonzero leading coefficients. Since $m_1 + m_2 > 0$, they are different and $a = u$. The w equation similarly gives $v_1 = v_2 = v$ and $b = v$. Hence $A = B = S = u + v < 0$. In the first equation of (8.7), the left side has order u, whereas P, F_1, F_2 all have orders greater than u, a contradiction.

Suppose instead that $A \geq 0 > B$ and $S < 0$. The original equation $z_2 = -m_1 P + F_2$ and the total mass centre give

$$(m_1 + M_T) X_2 - m_1 X_1 = M(-m_1 P + F_2).$$

Its right side has order greater than $\min(a, u_2) = \min(u_1, u_2)$. The equality follows directly from $X_1 - X_2$: if $u_1 \neq u_2$, then $a = \min(u_1, u_2)$; otherwise $a \geq u_2$. Consequently $u_1 = u_2 = u$ and $(m_1 + M_T)\widehat{X}_2 = m_1 \widehat{X}_1$. Since $M_T > 0$, the leading coefficients differ and $a = u$. The same argument in w gives $v_1 = v_2 = v$ and $b = v$. This forces $A = B = S$, contradicting $A \geq 0 > B$. Exchanging A, B treats the remaining possibility. Thus $A, B, S \geq 0$, and (8.15) gives $d \geq 6 > 0$.

8.2.4 One far body and one near body

By exchanging the outer bodies and the colours if necessary, we may suppose that body 1 is far and body 2 is near in z. Write

$$u = \operatorname{ord} X_1 < 1, \quad v = \operatorname{ord} Y_1 < 1, \quad \alpha = \operatorname{ord} X_2 \geq 1, \quad \beta = \operatorname{ord} Y_2 < 1.$$

The orders u, v, β are finite and greater than -2; α may be infinite. Here $a = u$. Put $c = \operatorname{ord}(Y_1 - Y_2) \geq \min(v, \beta) > -2$, $x = u + v$ and $y = 1 + \beta$, and let $e \geq 0$ be the excess order of body 2. Then $S = u + c$ and

$$d = 6 + S + 3x + 3y + e. \tag{8.16}$$

If $x \geq 0$, then $u > -1$, $S > -3$ and $y > -1$, so $d > 0$.

It remains to consider $x < 0$. In this case the triangle forces on body 1 have higher order than its coordinates:

$$\operatorname{ord} F_1 \geq u - \frac{3}{2}x > u, \qquad \operatorname{ord} G_1 \geq v - \frac{3}{2}x > v.$$

The first equation of (8.8) has left side of order u, hence $\operatorname{ord} F_2 = u$. The equation for X_2 in (8.7) then forces $\operatorname{ord} P = u$, since its left side and its F_1 term have higher orders. But $\operatorname{ord} P = -(u+3c)/2$, so $c = -u$ and $S = 0$. As $v < c$, cancellation in $Y_1 - Y_2$ requires $\beta = v$ and equal nonzero leading coefficients in Y_1, Y_2. Positive masses therefore make $m_1 Y_1 + m_2 Y_2$ have order v. The second equation of (8.8) now implies $\operatorname{ord} G_2 = v$.

If $e > 0$, (8.11) gives

$$u = -\frac{h + 3v}{2}, \qquad v = -\frac{3h + v}{2}.$$

These identities give $v = -h$ and $u = h > 1$, a contradiction. Thus $e = 0$ and $u = \operatorname{ord} F_2 \geq -(1 + 3v)/2$. Since $u < 1$, this also gives $v > -1$. Using $S = 0$, $\beta = v$ in (8.16), we obtain

$$d = 9 + 3u + 6v \geq \frac{15 + 3v}{2} > 6 > 0.$$

8.2.5 Two near bodies of opposite colours

Let body 1 be near in z and body 2 near in w, and write

$$\operatorname{ord} X_1 \geq 1, \quad v = \operatorname{ord} Y_1 < 1, \qquad u = \operatorname{ord} X_2 < 1, \quad \operatorname{ord} Y_2 \geq 1.$$

Here $u, v > -2$, $a = u$, $b = v$ and $S = u + v$. If $e_1, e_2 \geq 0$ are the two excess orders, summing the ten squared-distance orders gives

$$d = 12 + 4S + e_1 + e_2.$$

Suppose $d \leq 0$. Then $S \leq -3$, and $u, v > -2$ imply $u, v < -1$. Equation (8.10) gives $\operatorname{ord} P > u$ and $\operatorname{ord} H > v$. The mutual forces therefore do not contribute at the leading coordinate orders. In matrix form, the outer-body equations read

$$C\begin{pmatrix} F_1 \\ F_2 \end{pmatrix} = \begin{pmatrix} X_1 - m_2 P \\ X_2 + m_1 P \end{pmatrix}, \qquad C\begin{pmatrix} G_1 \\ G_2 \end{pmatrix} = \begin{pmatrix} Y_1 - m_2 H \\ Y_2 + m_1 H \end{pmatrix}.$$

The right sides have leading terms $(0, X_2)^{\mathsf{T}}$ at order u and $(Y_1, 0)^{\mathsf{T}}$ at order v, respectively. Using (8.9) yields

$$F_1 \sim -\frac{m_2}{M} X_2, \quad F_2 \sim \frac{M_T + m_1}{M} X_2, \qquad G_1 \sim \frac{M_T + m_2}{M} Y_1, \quad G_2 \sim -\frac{m_1}{M} Y_1.$$

All four coefficients are nonzero, so $\operatorname{ord} F_1 = \operatorname{ord} F_2 = u$ and $\operatorname{ord} G_1 = \operatorname{ord} G_2 = v$. If $e_1 = 0$, then $\operatorname{ord} G_1 \geq -(3+v)/2 > v$, a contradiction. If $e_1 > 0$, then $v = -(3h_1 + v)/2$ gives $v = -h_1$, whence $\operatorname{ord} F_1 = -(h_1 + 3v)/2 = h_1 > 1$, also a contradiction. Thus $d > 0$.

8.2.6 Two near bodies of the same colour

It remains to consider two bodies near in z; the other colour is symmetric. Put $\beta_j = \operatorname{ord} Y_j \in (-2, 1)$ and retain $e_j \geq 0$ for their excess orders. Now $a \geq 1$, $b \geq \min(\beta_1, \beta_2)$, so $S > -1$, and

$$d = 12 + 3(\beta_1 + \beta_2) + S + e_1 + e_2.$$

Suppose $d \leq 0$. If $S \geq 0$, the right side is positive since $\beta_j > -2$. If, for example, $\beta_1 \geq -1$, then $\beta_2 > -2$ and $S > -1$ make it greater than 2. Therefore

$$S < 0, \qquad \beta_1 < -1, \qquad \beta_2 < -1. \tag{8.17}$$

If $e_1 = e_2 = 0$, then $\operatorname{ord} G_j \geq -(3+\beta_j)/2 > \beta_j$. The w equation in (8.8) rules out unequal β_1, β_2, since its left side would have a unique lowest term. Thus $\beta_1 = \beta_2 = \beta$; writing $Y_j = v_j t^{\beta}(1 + o(1))$, it gives $m_1 v_1 + m_2 v_2 = 0$. Positivity makes $v_1 \neq v_2$, hence $b = \beta$. By $S < 0$, $\operatorname{ord} H = b - 3S/2 > \beta$. Both outer w-equations in (8.7) now have right sides of order greater than β, whereas their left sides have order β, a contradiction.

Suppose next that $e_1 > 0$ and $e_2 = 0$, interchanging the bodies if needed, and put $h = 1 + e_1 > 1$. If the lowest individual w-forcing orders differed, only one body would contribute at the lowest order. By Lemma 8.14, its coefficient vector could not be constant across the triangle vertices in either case $e_j = 0$ or $e_j > 0$. Lemma 8.13 therefore requires the two lowest orders to agree:

$$3h + \beta_1 = 3 + \beta_2, \qquad \beta_1 < \beta_2.$$

Their common unscaled order is $q = -(3h + \beta_1)/2 = -(3 + \beta_2)/2$. Equation (8.17) implies

$$h + \beta_1 = \frac{3 + \beta_2 + 2\beta_1}{3} < 0, \qquad q - \beta_1 = -\frac{3}{2}(h + \beta_1) > 0.$$

Both G_j have order at least q. But the w equation of (8.8) has a unique lowest term of order β_1 on the left, and right side of order at least $q > \beta_1$, a contradiction.

Finally suppose $e_1, e_2 > 0$. Lemma 8.14 gives a common contact vertex and $3h_1 + \beta_1 = 3h_2 + \beta_2$. If $h_1 = h_2 = h$, then $\beta_1 = \beta_2 = \beta$, $a \geq h$ and $b \geq \beta$. Thus $h + \beta \leq S < 0$ and

$$\operatorname{ord} G_j = -\frac{3h + \beta}{2} = \beta - \frac{3}{2}(h + \beta) > \beta, \qquad \operatorname{ord} H = b - \frac{3}{2}S > b \geq \beta.$$

The outer w-equations again give a contradiction. If $h_1 \neq h_2$, interchange the bodies to arrange $h_1 < h_2$, and put $\eta = h_2 - h_1 > 0$. Then

$$\beta_1 = \beta_2 + 3\eta, \qquad a = h_1 = h_2 - \eta, \qquad b = \beta_2.$$

Here $a = h_1$ follows by subtracting the two actual differences to their common contact vertex. Both G_j have the same exact order $q = -(3h_2 + \beta_2)/2$, while $\operatorname{ord} H = \beta_2 - 3S/2 > \beta_2$. If $q > \beta_2$, the second outer w-equation has no term on the right of order β_2. If $q < \beta_2$, both left sides and H have orders greater than q, so the two nonzero leading coefficients of G_1, G_2 would be annihilated by the invertible matrix C, also impossible. Thus $q = \beta_2$, giving $\beta_2 = -h_2$ and $S = -\eta$. The exact force orders in (8.11) and (8.10) now give

$$\operatorname{ord} F_1 = h_2 - 4\eta, \qquad \operatorname{ord} F_2 = h_2, \qquad \operatorname{ord} P = h_2 + \frac{\eta}{2}.$$

In the first equation of (8.7), F_1 is the unique lowest term, forcing $a = h_2 - 4\eta$, contrary to $a = h_2 - \eta$. Hence $d > 0$ also when both bodies are near in the same colour.

8.2.7 Completion of the proof

Up to exchanging the outer bodies and the colours, the four cases cover all possible positions of the outer bodies. They show that every Diagram 5 germ on a complex solution curve has $\operatorname{ord} \Delta > 0$, proving Theorem 8.11.

□

9 Diagram 6

We first represent every Diagram 6 germ in a chart with finite boundary limits. A torque identity restricts the masses and the signs of the crossed inverse distances. We then classify the corresponding boundary points and use exact computer-assisted Taylor calculations to prove that they are isolated. This excludes Diagram 6 and establishes the case $r = 6$ of Theorem 4.1(a).

9.1 The diagram and the butterfly chart

Diagram 6 is the *butterfly* of [AK12, §7.4]: a z-triangle and a w-triangle, called the *wings*, meet at one vertex. We use the numbering of [AK12, §8.6]. The z-wing $\{1,2,5\}$ has strokes $12, 15, 25$ and z-circles at $1, 2$; the w-wing $\{3,4,5\}$ has strokes $34, 35, 45$ and w-circles at $3, 4$. Vertex 5 is uncircled, and there is no stroke between $\{1,2\}$ and $\{3,4\}$.

The three possible circlings of the butterfly give Diagrams 6–8; 5 is uncircled in each of them [AK12, §7.4] (Fact AK4). Diagram 6 is the case in which both wings are circled. Rules 1a and 1d place the circles of the z-wing exactly at $1, 2$, and give the corresponding statement for w.

Permuting the bodies together with their masses, or exchanging $z \leftrightarrow w$, preserves (2.1). Thus the thirty coloured labels of Diagram 6, specified by a common vertex and an ordered pair of wings, are obtained from the numbering above. The exchange $z \leftrightarrow w$ composed with $(13)(24)$ preserves this numbering. It suffices to treat this case at arbitrary positive masses.

The two circled wings have opposite coordinate scales. We use a chart adapted to these scales, in which a Diagram 6 germ has a finite limit. We express the chart coordinates through rotation-invariant combinations, which allows us to fix the two triangle scales directly.

Definition 9.1. The *butterfly chart* has 21 coordinates: t, $Z_1, \dots, Z_5$, $W_1, \dots, W_5$, and D_{kl}. They are related to (2.1) by $(z_k, w_k) = (t^{-1}Z_k, tW_k)$ for $k = 1, 2$, $(z_k, w_k) = (tZ_k, t^{-1}W_k)$ for $k = 3, 4$, and $(z_5, w_5) = (tZ_5, tW_5)$. On the *crossed pairs* $13, 14, 23, 24$ we set $\delta_{kl} = tD_{kl}$; on the *wing pairs* $12, 15, 25, 34, 35, 45$ we set $\delta_{kl} = D_{kl}$.

The defining equations, also called the *rows*, are obtained as follows. For each body, we substitute these expressions into the z-equation $z_k - \sum_{l \neq k} m_l \delta_{kl}^3 (z_k - z_l) = 0$ and the corresponding equation for w. We multiply the two equations by (t, t^{-1}) for $k = 1, 2$, by (t^{-1}, t) for $k = 3, 4$, and by (t, t) for $k = 5$; the resulting rows are R_k^z, R_k^w. We also retain the ten distance equations $\delta_{kl}^2 z_{kl} w_{kl} - 1 = 0$ and impose the gauges $Z_2 - Z_1 - 1 = 0$, $W_4 - W_3 - 1 = 0$.

We denote these 22 rows by $F = 0$ and call $\mathcal{B} = F^{-1}(0) \cap \{t = 0\}$ the *boundary fibre*. The notation Z_k, W_k for chart coordinates is distinct from the force terms $Z_{kl} = z_{kl}\delta_{kl}^3$ in [AK12, §3].

Lemma 9.2. *Let $m \in \mathbb{R}^5_{>0}$.*

(i) The rows are polynomials in the 21 coordinates, of degree at most 5 in t.

(ii) Let $(z, w, \delta) \in A_m$ (so $z_{12}w_{34} \neq 0$), choose t with $t^2 = (z_{12}w_{34})^{-1}$ and put $\nu = (tz_{12})^{-1}$. The chart point of the rotated solution $(\nu z, \nu^{-1}w, \delta)$ has $Z_k = z_k/z_{12}$, $W_k = w_k z_{12}$ $(k = 1, 2)$, $Z_k = z_k w_{34}$, $W_k = w_k/w_{34}$ $(k = 3, 4)$, $Z_5 = z_5 w_{34}$, $W_5 = w_5 z_{12}$, $D_{kl} = \delta_{kl}/t$ on crossed pairs and $D_{kl} = \delta_{kl}$ on wing pairs, and satisfies $F = 0$: each coordinate row is $t^{\pm 1}\nu^{\pm 1}$ times an equation of (2.1), the distance rows are those of (2.1), and both gauges hold.

(iii) At $t = 0$ the crossed distance rows read $D_{kl}^2 Z_k W_l = -1$, and the wing distance rows read $D_{12}^2(Z_2 - Z_1)(W_2 - W_1) = 1$, $-D_{15}^2 Z_1(W_5 - W_1) = 1$, $-D_{25}^2 Z_2(W_5 - W_2) = 1$, $D_{34}^2(Z_4 - Z_3)(W_4 - W_3) = 1$, $-D_{35}^2(Z_5 - Z_3)W_3 = 1$, $-D_{45}^2(Z_5 - Z_4)W_4 = 1$. So every D_{kl} is nonzero on $\mathcal{B}$.

(iv) Identically, $m_1 R_1^z + m_2 R_2^z + m_5 R_5^z + t^2(m_3 R_3^z + m_4 R_4^z) = m_1 Z_1 + m_2 Z_2 + t^2(m_3 Z_3 + m_4 Z_4 + m_5 Z_5)$, and likewise after $z \leftrightarrow w$, $1 \leftrightarrow 3$, $2 \leftrightarrow 4$. Hence on $\mathcal{B}$: $Z_1 = -m_2/(m_1 + m_2)$, $Z_2 = m_1/(m_1 + m_2)$, $W_3 = -m_4/(m_3 + m_4)$, $W_4 = m_3/(m_3 + m_4)$.

(v) *At $t=0$ the rows $R^z_1, R^z_2, R^z_5, R^w_1, R^w_2$, the distance rows of $12, 15, 25$ and the first gauge involve only the z-*wing coordinates $Z_1, Z_2, W_1, W_2, W_5, D_{12}, D_{15}, D_{25}$*; the remaining rows except the crossed distance rows involve only the w-*wing coordinates $Z_3, Z_4, Z_5, W_3, W_4, D_{34}, D_{35}, D_{45}$*; and $D_{13}, D_{14}, D_{23}, D_{24}$ occur only in the crossed distance rows.*

Proof. Put $I=\{1,2\}$ and $J=\{3,4\}$, and let i' be the other element of I. Substitution in the defining equations gives

$$\begin{aligned}
R^z_i &= Z_i - m_{i'}D^3_{ii'}(Z_i - Z_{i'}) - m_5 D^3_{i5}(Z_i - t^2 Z_5) - t^3 \sum_{j\in J} m_j D^3_{ij}(Z_i - t^2 Z_j),\\
R^w_i &= W_i - m_{i'}D^3_{ii'}(W_i - W_{i'}) - m_5 D^3_{i5}(W_i - W_5) - t \sum_{j\in J} m_j D^3_{ij}(t^2 W_i - W_j),\\
R^z_5 &= t^2 Z_5 - \sum_{i\in I} m_i D^3_{i5}(t^2 Z_5 - Z_i) - t^2 \sum_{j\in J} m_j D^3_{j5}(Z_5 - Z_j).
\end{aligned}$$

The other coordinate rows follow by exchanging the wings and z, w. The distance rows for ii', $i5$, $j5$, and ij are, respectively,

$$\begin{aligned}
D^2_{ii'}(Z_i - Z_{i'})(W_i - W_{i'}) - 1, \qquad & D^2_{i5}(Z_i - t^2 Z_5)(W_i - W_5) - 1,\\
D^2_{j5}(Z_j - Z_5)(W_j - t^2 W_5) - 1, \qquad & D^2_{ij}(Z_i - t^2 Z_j)(t^2 W_i - W_j) - 1,
\end{aligned}$$

and the 34 row is the wing exchange of the 12 row. These formulas prove (i), (iii), and (v). For (ii), the displayed change of variables is precisely the substitution defining these rows, with $t^2 z_{12} w_{34} = 1$ and $\nu t z_{12} = 1$; the two gauges therefore hold. Finally, in the weighted sum of the original z-equations every pair force cancels with the force of the reversed pair. Multiplying this identity by t gives (iv). At $t=0$, its first identity and the first gauge give $Z_1 = -m_2/(m_1+m_2)$ and $Z_2 = m_1/(m_1+m_2)$; the other wing is identical. □

9.2 Limits in the butterfly chart

The diagram rules determine the orders of the coordinate differences. We use these orders to prove that every germ reaches the boundary fibre with nonzero inverse-distance coordinates.

Proposition 9.3 (Limits in the chart)**.** *Let $m \in \mathbb{R}^5_{>0}$ and let $\gamma(s)$, $s \to 0$, be a Diagram-6 germ, in the numbering above, on a curve in A_m. Along γ define t and ν as in Lemma 9.2(ii), and let $\tilde\gamma(s)$ be the chart point of the rotated solution. Then $t \to 0$, every chart coordinate is bounded along $\tilde\gamma$, and $\tilde\gamma(s)$ converges to a point $p \in \mathcal{B}$ at which Z_1, Z_2, W_3, W_4 and all ten D_{kl} are nonzero.*

Proof. We normalize as in [AK12, p. 547], with the largest components of $\mathcal{Z}$ and $\mathcal{W}$ both having modulus ϵ^{-2}. Lemma 3.3 realizes this normalization along γ, with ϵ comparable to a positive power of s. We write ord for order in ϵ; thus $\operatorname{ord} f > \alpha$ means $f\epsilon^{-\alpha} \to 0$. Since the chart coordinates are invariant under $z \mapsto az$, $w \mapsto a^{-1}w$, we may compute their orders in this frame.

The circles give $\operatorname{ord} z_1 = \operatorname{ord} z_2 = \operatorname{ord} w_3 = \operatorname{ord} w_4 = -2$. The uncircled coordinates $z_3, z_4, z_5, w_1, w_2, w_5$ have order > -2, so $\operatorname{ord} z_{15} = \operatorname{ord} z_{25} = -2$. A z-stroke kl means $\operatorname{ord}(z_{kl}\delta^3_{kl}) = \operatorname{ord}(z_{kl}^{-1/2} w_{kl}^{-3/2}) = -2$, or equivalently $\operatorname{ord} z_{kl} + 3\operatorname{ord} w_{kl} = 4$. Thus $\operatorname{ord} w_{15} = \operatorname{ord} w_{25} = 2$, and hence $\operatorname{ord} w_{12} \ge 2$. The stroke 12 then gives $\operatorname{ord} z_{12} \le -2$. Combining

this with $\operatorname{ord} z_{12} \geq -2$ (Estimate 1, Fact AK2), we obtain $\operatorname{ord} z_{12} = -2$, $\operatorname{ord} w_{12} = 2$. Consequently $r_{kl}^2 = z_{kl} w_{kl}$ and δ_{kl} have order 0 on all three z-wing pairs; these are the maximal edges with $r \approx 1$ in Estimate 2. The same argument applies to the w-wing.

On a crossed pair, both z_{kl} and w_{kl} are differences of a circled and an uncircled coordinate. Both therefore have order -2, and $\operatorname{ord} \delta_{kl} = 2$.

We also need bounds on the small coordinate of each outer vertex. In the w-equation $w_1 = \sum_{l \neq 1} m_l \delta_{1l}^3 (w_1 - w_l)$, the terms have orders 2 for $l = 2, 5$ and 4 for $l = 3, 4$. It follows that $\operatorname{ord} w_1 \geq 2$. Similarly, $\operatorname{ord} w_2, \operatorname{ord} z_3, \operatorname{ord} z_4 \geq 2$, while $w_5 = w_1 + w_{15}$ and $z_5 = z_3 + z_{35}$ have order ≥ 2. These are the estimates $A_w, B_z, m_1 w_1 + m_2 w_2, w_5, B_w, A_z \preccurlyeq \epsilon^2$ of [AK12, §8.6].

Finally, $\operatorname{ord} t = -\frac{1}{2}(\operatorname{ord} z_{12} + \operatorname{ord} w_{34}) = 2$ and $\operatorname{ord} \nu = -\operatorname{ord} t - \operatorname{ord} z_{12} = 0$. Lemma 9.2(ii) now shows that Z_1, Z_2, W_3, W_4 and all ten D_{kl} have order exactly 0; for the crossed pairs the order is $2 + \frac{1}{2}(-2-2) = 0$. The remaining coordinates $W_1, W_2, W_5, Z_3, Z_4, Z_5$ have order ≥ 0. Thus all chart coordinates are bounded. Their Puiseux series in s converge, giving $\tilde{\gamma}(s) \to p \in \mathcal{B}$. Coordinates of order exactly 0 have nonzero limits. □

Consequently $D_{12} = \delta_{12}$ and $D_{34} = \delta_{34}$ have finite nonzero limits. The torque identity can therefore be evaluated at the boundary point supplied by Proposition 9.3.

9.3 A necessary mass condition

The torque identity imposes a condition on the masses before any local calculation at the boundary points. Its leading crossed contribution is $\Sigma = \sum_{k \in \{1,2\},\, l \in \{3,4\}} m_k m_l D_{kl}^3 Z_k W_l$.

Lemma 9.4. *There is a polynomial Q in the chart coordinates, with coefficients polynomial in m, such that identically*

$$\sum_{k=1,2} m_k \big(W_k R_k^z - Z_k R_k^w \big) + m_5 W_5 R_5^z = -t\,\Sigma + t^2 Q. \tag{9.1}$$

Proof. Let $I = \{1,2\}$ and $J = \{3,4\}$. The polynomial in the statement can be chosen as

$$\begin{aligned} Q = t^3 \sum_{i \in I,\, j \in J} m_i m_j D_{ij}^3 W_i Z_j + m_5 W_5 Z_5 \\ + m_5 Z_5 \sum_{i \in I} m_i D_{i5}^3 (W_i - W_5) - m_5 W_5 \sum_{j \in J} m_j D_{j5}^3 (Z_5 - Z_j). \end{aligned}$$

Indeed, the terms for the pair 12 cancel in $\sum_{i \in I} m_i (W_i R_i^z - Z_i R_i^w)$. Each crossed pair contributes $m_i m_j D_{ij}^3 (-t Z_i W_j + t^5 W_i Z_j)$, and each pair $i5$ contributes $m_i m_5 D_{i5}^3 (t^2 W_i Z_5 - Z_i W_5)$. The terms $-m_i m_5 D_{i5}^3 Z_i W_5$ cancel with the constant part of $m_5 W_5 R_5^z$. The remaining terms are exactly $-t\Sigma + t^2 Q$. □

To relate this identity to [AK12, §8.6], let ρ_k^z, ρ_k^w be the left-hand sides of (2.1). The pair torque $\sum_{k=1,2} m_k (w_k \rho_k^z - z_k \rho_k^w)$ is $\sum_{k=1,2} \sum_{l=3,4,5} m_k m_l \delta_{kl}^3 (w_k z_l - z_k w_l)$, since the terms with $l \in \{1,2\}$ cancel. It is $2i$ times (31) of that paper. In the chart, this torque becomes the first sum in (9.1). The terms with $l = 3, 4$ contribute $-t\Sigma + t^5 \sum m_k m_l D_{kl}^3 W_k Z_l$, and those with $l = 5$ contribute $\sum_{k=1,2} m_k m_5 D_{k5}^3 (t^2 W_k Z_5 - Z_k W_5)$. The additional row in the lemma removes the latter contribution at order zero, because $R_5^z - (m_1 D_{15}^3 Z_1 + m_2 D_{25}^3 Z_2)$ is divisible by t^2.

Proposition 9.5 (Necessary mass condition)**.** *Let $m \in \mathbb{R}^5_{>0}$, let γ be a Diagram-6 germ in the numbering above, and let p be its limit (Proposition 9.3). Then:*

(i) $\Sigma(p) = 0$, *and* $\Sigma(p) = -(m_1m_3D_{13} + m_1m_4D_{14} + m_2m_3D_{23} + m_2m_4D_{24})(p)$;

(ii) $m_1 = m_2$ *and* $m_3 = m_4$; *this is relation (32) of [AK12, §8.6];*

(iii) $(D_{13}, D_{14}, D_{23}, D_{24})(p) = (2i\eta_{13},\, 2\eta_{14},\, 2\eta_{23},\, 2i\eta_{24})$ *with signs* $\eta_{kl} \in \{\pm 1\}$ *satisfying* $\eta_{14} = -\eta_{23}$ *and* $\eta_{13} = -\eta_{24}$.

In another numbering, wings $\{a, b, c\}$ and $\{d, e, c\}$ force $m_a = m_b$ and $m_d = m_e$.

Proof. All rows vanish along $\tilde{\gamma}$, so (9.1) gives $t\Sigma = t^2Q$, and $t \neq 0$ gives $\Sigma = tQ$. By Proposition 9.3, Q is bounded and $t \to 0$, so $\Sigma(p) = 0$. By Lemma 9.2(iii), $D^2_{kl}Z_kW_l = -1$ at p for the crossed pairs, so $\Sigma(p) = \sum m_km_lD_{kl}(D^2_{kl}Z_kW_l) = -\sum m_km_lD_{kl}$; this is (i).

By Lemma 9.2(iv), $Z_1W_3 = m_2m_4/s_0^2$, $Z_1W_4 = -m_2m_3/s_0^2$, $Z_2W_3 = -m_1m_4/s_0^2$ and $Z_2W_4 = m_1m_3/s_0^2$ at p, where $s_0 > 0$, $s_0^2 = (m_1 + m_2)(m_3 + m_4)$. Write $m_k = \mu_k^2$ with $\mu_k > 0$. Then $D^2_{kl} = -1/(Z_kW_l)$ gives

$$D_{13} = \frac{i\eta_{13}s_0}{\mu_2\mu_4}, \quad D_{14} = \frac{\eta_{14}s_0}{\mu_2\mu_3}, \quad D_{23} = \frac{\eta_{23}s_0}{\mu_1\mu_4}, \quad D_{24} = \frac{i\eta_{24}s_0}{\mu_1\mu_3}$$

with signs $\eta_{kl} \in \{\pm 1\}$, and

$$\Sigma(p) = -\frac{s_0}{\mu_1\mu_2\mu_3\mu_4}\big(P_1 + iP_2\big), \qquad P_1 = \eta_{14}\mu_1^3\mu_4^3 + \eta_{23}\mu_2^3\mu_3^3, \quad P_2 = \eta_{13}\mu_1^3\mu_3^3 + \eta_{24}\mu_2^3\mu_4^3.$$

Up to the factor $\pm i$, $P_1 + iP_2$ is the left-hand side of (12) of [AK12, §5.1] written in the μ_k, and each of its eight sign choices arises from exactly two of the sixteen sign classes η. Since the μ_k and s_0 are real, $\Sigma(p) = 0$ forces $P_1 = P_2 = 0$. If $\eta_{14} = \eta_{23}$, then $P_1 \neq 0$, and if $\eta_{13} = \eta_{24}$, then $P_2 \neq 0$; this excludes twelve classes. In the other four, $P_1 = \eta_{14}(x-y)(x^2+xy+y^2)$ with $x = \mu_1\mu_4$, $y = \mu_2\mu_3$ and $x^2 + xy + y^2 > 0$, so $\mu_1\mu_4 = \mu_2\mu_3$; likewise $P_2 = 0$ gives $\mu_1\mu_3 = \mu_2\mu_4$. Then $\mu_3(\mu_1\mu_4 - \mu_2\mu_3) - \mu_4(\mu_1\mu_3 - \mu_2\mu_4) = \mu_2(\mu_4 - \mu_3)(\mu_4 + \mu_3)$ vanishes, so $\mu_3 = \mu_4$ and then $\mu_1 = \mu_2$: this is (ii). With $m_1 = m_2$ and $m_3 = m_4$ the displayed values of D_{kl} become those of (iii). □

We have thus reduced the boundary limits to the masses $m_1 = m_2$, $m_3 = m_4$ and the four sign classes in (iii). This is relation (32) of [AK12, §8.6]. We next solve the remaining boundary equations on this mass locus.

9.4 The boundary points

We henceforth write $m = (a, a, b, b, c)$ with $a, b, c > 0$. At the boundary, the equations separate into the two wings and the crossed distance equations. We first solve the wing equations explicitly. This will give a finite list of boundary points at each positive mass.

Lemma 9.6. *The points of $\mathcal{B}$ are exactly the points with $Z_1 = W_3 = -\frac{1}{2}$, $Z_2 = W_4 = \frac{1}{2}$,*

$$(D_{12}, D_{15}, D_{25}) = (d, \lambda d, \omega\lambda d),$$
$$W_1 = -d(a + 2c\lambda), \quad W_2 = d(a + 2c\omega\lambda), \quad W_5 = ad\big(4\lambda^{-2} - 1\big),$$

$$(D_{34}, D_{35}, D_{45}) = (e, \kappa e, \omega'\kappa e),$$
$$Z_3 = -e(b + 2c\kappa), \quad Z_4 = e(b + 2c\omega'\kappa), \quad Z_5 = be\big(4\kappa^{-2} - 1\big),$$
$$(D_{13}, D_{14}, D_{23}, D_{24}) = (2i\eta_{13},\, 2\eta_{14},\, 2\eta_{23},\, 2i\eta_{24}), \quad \eta_{kl} \in \{\pm 1\},$$

where

$$\omega^3 = 1, \quad \lambda^2 = 2(1+\omega), \quad (2a + c\lambda^3)d^3 = 1, \qquad \omega'^3 = 1, \quad \kappa^2 = 2(1+\omega'), \quad (2b + c\kappa^3)e^3 = 1.$$

For each ω both roots λ are nonzero. The scale denominator $2a + c\lambda^3$ vanishes only for $(\omega, \lambda) = (1, -2)$ and $a = 4c$: for $\omega = 1$, $\lambda = \pm 2$; for $\omega \neq 1$, $\lambda^6 = -8$ and the two denominators have product $4a^2 + 8c^2$. Hence a wing has 18 classes (ω, λ, d) if $a \neq 4c$ and 15 if $a = 4c$, and $\mathcal{B}$ has $18 \cdot 18 \cdot 16 = 5184$ points if $a \neq 4c \neq b$, 4320 if exactly one of a, b equals $4c$, and 3600 if $a = b = 4c$.

Proof. By Lemma 9.2(iv),(v), the boundary equations separate into the two wing systems and an independent square-root choice on each crossed pair. On the z-wing, $Z_1 = -1/2$, $Z_2 = 1/2$. Write $d = D_{12}$, $s = D_{15}$, $q = D_{25}$; all three are nonzero. The equation $R_5^z = 0$ gives $q^3 = s^3$, whereas $R_1^z = 0$ gives $2ad^3 + cs^3 = 1$. The distance equations give

$$W_2 - W_1 = d^{-2}, \qquad W_5 - W_1 = 2s^{-2}, \qquad W_2 - W_5 = 2q^{-2}.$$

Thus, with $\lambda = s/d$ and $\omega = q/s$,

$$\omega^3 = 1, \qquad \lambda^2 = 2(1+\omega), \qquad (2a + c\lambda^3)d^3 = 1.$$

The equation $R_1^w = 0$ now reads $W_1 + ad + 2cs = 0$. Consequently

$$W_1 = -d(a + 2c\lambda),$$
$$W_2 = W_1 + d^{-2} = d(a + 2c\omega\lambda),$$
$$W_5 = W_1 + 2s^{-2} = ad(4\lambda^{-2} - 1),$$

where the second equality uses $\lambda^3 - 2\lambda = 2\omega\lambda$. Conversely, these formulas satisfy the three distance equations, $R_1^w = 0$, and $R_2^w = W_2 - ad - 2cq = 0$. They also satisfy $R_5^z = 0$ and $R_1^z = R_2^z = 0$ by the two cubic equations above, as well as the gauge. This accounts for all nine wing equations. Exchanging the wings gives the second system. Each crossed equation has the two square roots displayed in the statement, and the four choices are independent.

For $\omega = 1$ the two ratios are $\lambda = \pm 2$; only $\lambda = -2$ can have zero scale denominator, precisely at $a = 4c$. For $\omega \neq 1$, $\lambda^6 = -8$, so the product of the two scale denominators is $4a^2 + 8c^2 > 0$. Every nonzero denominator gives three distinct cube-root scales. Counting these choices gives the stated number of boundary points, including those on the exceptional mass faces. □

The classification in Lemma 9.6 gives the points at which we must test continuation into $t \neq 0$.

Definition 9.7. A *label* specifies a point of $\mathcal{B}$ by the choices (ω, λ, d), (ω', κ, e) and η of Lemma 9.6.

We classify the ratios (ω, λ) and (ω', κ) on each wing by

$$\omega = 1 \iff \lambda = \pm 2, \qquad \omega \neq 1 \iff \lambda^4 - 2\lambda^2 + 4 = 0.$$

Thus each wing has two real ratios and four nonreal ratios. The same classification applies to the primed parameters on the second wing. A label is called *corank-one* when $\eta_{14} = -\eta_{23}$ and $\eta_{13} = -\eta_{24}$, and *regular* otherwise. When $a \neq 4c \neq b$, there are $12 \cdot 324 = 3888$ regular labels and $4 \cdot 324 = 1296$ corank-one labels.

The symmetries reduce the number of symbolic calculations required. At masses (a, a, b, b, c), $F^{-1}(0)$ is preserved by exchanging 1 and 2 together with $t \mapsto it$, $(Z_1, Z_2, W_1, W_2, W_5) \mapsto (-Z_2, -Z_1, -W_2, -W_1, -W_5)$ and $D_{kl} \mapsto -iD_{kl}$ on crossed pairs. It is also preserved by $(t, D_{13}, D_{14}, D_{23}, D_{24}) \mapsto -(t, D_{13}, D_{14}, D_{23}, D_{24})$ and by complex conjugation. The *wing exchange* ($z \leftrightarrow w$, $1 \leftrightarrow 3$, $2 \leftrightarrow 4$) maps the chart at (a, a, b, b, c) to the chart at (b, b, a, a, c).

A *state* records the ratios (ω, λ), (ω', κ) and η, but leaves the two cube-root scales unspecified. The 576 states form 29 symmetry orbits: 21 regular and 8 corank-one (Appendix A). In the computations, we express the masses through the scales as $a = \frac{1}{2}(d^{-3} - c\lambda^3)$ and $b = \frac{1}{2}(e^{-3} - c\kappa^3)$. One symbolic calculation for a state therefore covers its nine choices of (d, e) at every positive (a, b, c).

It remains to exclude the corank-one labels allowed by the torque condition. The next lemma does this by solving twenty equations locally and showing that a remaining equation has a nonzero leading Taylor coefficient. The determinant and coefficient calculations are computer-assisted and use exact arithmetic.

Lemma 9.8 (Computer-assisted isolation)**.** *Every corank-one label is an isolated point of* $F^{-1}(0)$.

wing ratios (z, w)	labels	k	c_k is a nonzero multiple of
real, real	144	4	$Q_{2,2} = -\frac{2a+2b+c}{60(a+4c)(b+4c)}$, $Q_{2,-2} = \frac{2a+2b+c}{4(a+4c)(b-4c)}$, $Q_{-2,-2} = \frac{2a+2b+c}{68(a-4c)(b-4c)}$ for $(\lambda, \kappa) = (2, 2), (2, -2), (-2, -2); (-2, 2)$ by wing exchange
one real, one nonreal	576	3	$P = A\kappa^3 - 4A\kappa + B\kappa^3 - B\kappa^2 - 4B\kappa + B$ for $\lambda = \pm 2$, κ nonreal; product over the nonreal κ: $9(8A^2 + 16AB + 9B^2)^2$
nonreal, nonreal: orbit of (λ_0, λ_0)	288	2	$1 - 2\kappa^3$; product over the nonreal κ: 1089
nonreal, nonreal: orbits of $(\lambda_0, -\lambda_0)$	288	2	$2B/(2B + \kappa^3)$; product over the nonreal κ: $B^4/(B^2 + 2)^2$

Table 3: Order k and leading coefficient of R_2^w along C_p at the corank-one labels (Lemma 9.8), at one representative state of each orbit. Counts are for $a \neq 4c \neq b$; $A = a/c$, $B = b/c$; $\lambda_0 = \frac{1}{2}(\sqrt{6} + i\sqrt{2})$ is a nonreal ratio for $\omega = \frac{1}{2}(-1 + i\sqrt{3})$. In the first row R_4^w has the same order.

Proof. We first work at a representative state of each of the eight corank-one orbits. At each such representative label p, exact symbolic computation gives:

(i) the 20 rows other than R_2^w and R_4^w, together with t, have invertible derivative, so their common zeros form an analytic graph C_p over t;

(ii) on this graph, $R_2^w = c_k t^k + O(t^{k+1})$, where $c_k \neq 0$ and the order and coefficient are listed in Table 3.

The coefficient field is $\mathbb{Q}(i)(c, d, e)$ when both wing ratios are real; otherwise it is extended by a root κ of $\kappa^4 - 2\kappa^2 + 4$. The computations are given in Appendix A.

For (i), eliminating the diagonal derivative block of the distance rows reduces the determinant to a Schur complement. When both wing ratios are real, its determinant, after multiplying the

rows by d^3e^3, is $82944\,d^{61}e^{59}(cd^3-\frac{\lambda}{16})(ce^3-\frac{\kappa}{16})$. The first factor is nonzero because $cd^3-\frac{\lambda}{16}=-\frac{\lambda a}{8(2a+c\lambda^3)}\neq 0$; the second is treated by exchanging the wings.

In the other cases, after removing the unit row for t, the complement is block diagonal with one 5×5 block for each wing (Lemma 9.2(v)). The computation checks the vanishing of an off-diagonal block and evaluates the two diagonal blocks. For one real and one nonreal wing ratio, the determinants are nonzero multiples of $8cd^3-\frac{\lambda}{2}=-\lambda A/(2A+\lambda^3)$ and of a polynomial in ce^3 and κ. The product of the latter polynomial over the nonreal κ is $(\frac{9}{16})^4\cdot 64B^4(B^4-8B^2+64)/(B^2+2)^4$. Its nonvanishing follows from $B^4-8B^2+64=(B^2-4)^2+48$. This product, and those in Table 3, are evaluated using resultants with $\kappa^4-2\kappa^2+4$.

For (ii), the Taylor coefficients of C_p are found by solving the 20 rows in (i) successively in powers of t. Substitution into all 22 rows checks the resulting expansions and gives the first nonzero coefficient listed in Table 3.

Near a representative label p, (i) gives $F^{-1}(0)\subset C_p$. By (ii), R_2^w vanishes on C_p only at $t=0$, so p is isolated. The chart symmetries described above are invertible and map the full zero set to the corresponding full zero set, with the wing masses exchanged when required. Complex conjugation has the same property. Each symmetry therefore carries an isolated point to an isolated point. Since the representative states cover all corank-one labels, this proves the assertion at every label. □

The lemma excludes all boundary limits allowed by Proposition 9.5(iii).

9.5 Exclusion

We now combine the coverage, mass restriction and isolation results. Every Diagram 6 germ would converge to an isolated boundary point, which forces its chart parameter to vanish identically.

Theorem 9.9. *Let $m\in\mathbb{R}^5_{>0}$. No curve in A_m has a Diagram-6 germ, for any numbering of the bodies and either colouring. In particular the case $r=6$ of Theorem 4.1(a) holds, at the complex level.*

Proof. The symmetries at the beginning of the section allow us to use the numbering above. Suppose that $\gamma(s)$ is a Diagram-6 germ on a curve in A_m. Proposition 9.3 gives a chart germ $\tilde\gamma(s)$ with $t(s)\to 0$, $t(s)\neq 0$ for $s\neq 0$, bounded coordinates, and a limit $p\in\mathcal{B}$.

Proposition 9.5 forces $m=(a,a,b,b,c)$ and $\eta_{14}=-\eta_{23}$, $\eta_{13}=-\eta_{24}$ at p. Lemma 9.6 therefore identifies p as a label of corank one. The computer-assisted Lemma 9.8 shows that this point is isolated in $F^{-1}(0)$.

On the other hand, the coordinates of $\tilde\gamma$ are bounded Puiseux series in s. After a substitution $s=\sigma^N$, they are convergent power series in σ. Thus $\sigma\mapsto\tilde\gamma(\sigma^N)$ is a holomorphic map from a disc into $F^{-1}(0)$, taking the value p at $\sigma=0$. Since p is isolated, the map is constant near $\sigma=0$. This contradicts $t\not\equiv 0$. □

10 Diagram 7

We exclude Diagram 7 on a curve through a real configuration by showing that an unbounded four-product forces the inertia to tend to zero. The key step is a decomposition of the inertia into terms whose orders are positive. Diagram 7 is the butterfly without circles in [AK12, §7.4]; we use its two triangular clusters to obtain this decomposition.

Theorem 10.1. *Let $m \in \mathbb{R}^5_{>0}$. No Diagram 7 germ with an unbounded four-product lies on a curve through a real point: if an irreducible curve $\Gamma \subset A_m$ carries such a germ, then Γ contains no real point.*

This proves part (b) of Theorem 4.1 for $r = 7$. For the orders a, b introduced in Proposition 10.2, the proof has two cases:

$$\begin{aligned} a + b \geq 0 \quad &\Longrightarrow \quad p_{ij} \longrightarrow 0 \quad \text{for every } i < j, \\ a + b < 0 \quad &\Longrightarrow \quad I \longrightarrow 0 \quad \Longrightarrow \quad I|_\Gamma = 0. \end{aligned}$$

The first implication follows from the distance orders (Lemma 10.5); the second follows from the inertia identity (Lemma 10.4) and its constancy on Γ. At a real point, $I = \sum_k m_k |z_k|^2 > 0$, which gives the required contradiction.

10.1 The diagram and the nested chart

In the numbering of [AK12, §8.7], the *upper wing* $\{1, 2, 5\}$ is an isolated triangle of z-edges without z-circles, the *lower wing* $\{3, 4, 5\}$ is an isolated triangle of w-edges without w-circles, and the four *crossed* pairs kj ($k \in \{1, 2\}$, $j \in \{3, 4\}$) are not edges. Since Diagram 7 has no circles, Rule 1e shows that neither wing has a maximal stroke (see the proof of Proposition 10.2(i)). By Definition 3.4 and the convention after Theorem 4.1, we may use this numbering and colouring.

We first normalize the orders as in Lemma 3.3. After renormalization, Z and W have a common order. Choosing a power t of the Puiseux parameter makes this order -2. Thus t plays the role of ϵ in [AK12, §3], and ord denotes order in t. The renormalizations $(z, w) \mapsto (\lambda z, \lambda^{-1} w)$ preserve the diagram, the distance constraints, I, U and every $r_{kl}^2 = z_{kl} w_{kl}$.

To describe the two clusters at their different scales, we use the nested chart with exponents P, Q, R and coordinates A_1, A_2, C, u_3, u_4, B_3, B_4, D, v_1, v_2 and D_{kl} $(k < l)$:

$$\begin{aligned} z_1 = A_1, \quad z_2 = A_2, \qquad & w_3 = t^{-Q} B_3, \quad w_4 = t^{-Q} B_4, \\ z_j = C + t^P m_5 u_j \quad (j = 3, 4), \qquad & z_5 = C - t^P (m_3 u_3 + m_4 u_4), \\ w_k = t^{-Q} D + t^R m_5 v_k \quad (k = 1, 2), \qquad & w_5 = t^{-Q} D - t^R (m_1 v_1 + m_2 v_2), \end{aligned}$$

and $\delta_{kl} = t^{e_{kl}} D_{kl}$, where $e_{kl} = -R/2$ on the upper wing, $Q/2$ on the crossed pairs and $(Q - P)/2$ on the lower wing (D is the scaled cluster centre). Here C and $t^{-Q} D$ are the centres of mass of the z-cluster $\{3, 4, 5\}$ and the w-cluster $\{1, 2, 5\}$. The exponents P and R describe the internal separations of these clusters, while $-Q$ describes the common w-scale. The choices of e_{kl} make the two sides of all ten distance constraints $\delta_{kl}^2 z_{kl} w_{kl} = 1$ have the same order. For the pairs 12 and 34, the distance constraints become

$$m_5 D_{12}^2 (A_2 - A_1)(v_2 - v_1) = 1, \qquad m_5 D_{34}^2 (u_4 - u_3)(B_4 - B_3) = 1.$$

The *open cone* is $P > Q > 0$, $R > 0$.

We retain the rotational freedom in this chart and use the equations of (2.1) preceding the gauge $z_{12} = w_{12}$. The distance constraints give $(A_2 - A_1)(B_4 - B_3) \neq 0$. The transformations $(z, w) \mapsto (\lambda z, \lambda^{-1} w)$ preserve these equations, I and U, so the inertia calculation can be carried out directly in the chart.

10.2 Orders of the germs and the chart

We now determine the orders of all separations from the diagram rules. This isolates the order range relevant to an unbounded four-product and shows that the rescaled coordinates have finite limits.

Proposition 10.2 (Representation in the nested chart)**.** *Let γ be a Diagram 7 germ. There are a, b with $-2 < a < 1$ and $-2 < b < 1$ such that the following hold.*

(i) On the upper wing $\operatorname{ord} z_{kl} = a$ *and* $\operatorname{ord} w_{kl} = (4-a)/3$*; on the lower wing* $\operatorname{ord} w_{kl} = b$ *and* $\operatorname{ord} z_{kl} = (4-b)/3$*; on the crossed pairs* $\operatorname{ord} z_{kl} = a$ *and* $\operatorname{ord} w_{kl} = b$*. Moreover* $\operatorname{ord} z_k \geq a$ *and* $\operatorname{ord} w_k \geq b$ *for every k.*

(ii) After the rotation $(z, w) \mapsto (t^{-a}z, t^a w)$, γ is a germ of the nested chart with exponents

$$P = \frac{4-b}{3} - a, \qquad Q = -(a+b), \qquad R = \frac{4+2a}{3};$$

all chart coordinates have nonnegative order, so they converge, and every D_{kl} has order zero. Moreover $R = \frac{2}{3}(a+2) > 0$ and $P - Q = \frac{2}{3}(b+2) > 0$, so (P, Q, R) lies in the open cone if and only if $a + b < 0$.

Proof. (i) We use Fact AK2. By Rule 2e [AK12, p. 552] the three z-separations of a wing have one order, and so do its three w-separations; let a be the z-order of the upper wing and b the w-order of the lower wing. As $\delta_{kl}^2 z_{kl} w_{kl} = 1$, $\operatorname{ord} Z_{kl} = -(\operatorname{ord} z_{kl} + 3 \operatorname{ord} w_{kl})/2$ and $\operatorname{ord} W_{kl} = -(3 \operatorname{ord} z_{kl} + \operatorname{ord} w_{kl})/2$; since the strokes have order -2, this gives the other two wing orders. The upper wing is an isolated component of the z-diagram without z-circles, so it has no maximal z-stroke (Rule 1e, p. 546), and along a z-stroke Estimate 2 (p. 548) gives $-2 \leq \operatorname{ord} z_{kl} \leq 1$, with -2 exactly for a maximal z-edge and 1 exactly for a zw-edge. Hence $-2 < a < 1$, and dually $-2 < b < 1$. As $z_{13} = z_{15} + z_{53}$ with $\operatorname{ord} z_{53} = (4-b)/3 > 1 > a$, and $w_{13} = w_{15} + w_{53}$ with $\operatorname{ord} w_{15} = (4-a)/3 > 1 > b$, the crossed orders are a and b; likewise for 14, 23, 24. Finally $\sum_k m_k z_k = \sum_k m_k w_k = 0$ on A_m (proof of Lemma 3.1), so each z_k is a combination of z-differences, all of order at least a; likewise $\operatorname{ord} w_k \geq b$.

(ii) After the rotation the z-differences have order 0 on the upper wing and the crossed pairs and P on the lower wing, the w-differences have order R on the upper wing and $-Q$ elsewhere, and $\operatorname{ord} z_k \geq 0$, $\operatorname{ord} w_k \geq -Q$. Inverting the chart map, $A_k = z_k$, $B_j = t^Q w_j$, C and $t^{-Q}D$ are the centres of the two clusters, $m_5 t^P u_j = z_j - C$ and $m_5 t^R v_k = w_k - t^{-Q}D$ are combinations of differences inside them, and $D_{kl} = t^{-e_{kl}}\delta_{kl}$, where $\operatorname{ord} \delta_{kl} = -(\operatorname{ord} z_{kl} + \operatorname{ord} w_{kl})/2 = e_{kl}$. So every coordinate has nonnegative order and $\operatorname{ord} D_{kl} = 0$. □

Thus $a + b < 0$ is equivalent to $P > Q > 0$ and $R > 0$. These inequalities make each term of the inertia decomposition decay. For $a + b \geq 0$, we will instead apply the distance orders directly to the four-products in Lemma 10.5.

10.3 The inertia identity

We express I through the internal separations of the two clusters and their crossed forces. The chart then converts each summand into a positive power of the parameter times a bounded function.

Write

$$F_k^z = z_k - \sum_{l\neq k} m_l \delta_{kl}^3 (z_k - z_l), \qquad F_k^w = w_k - \sum_{l\neq k} m_l \delta_{kl}^3 (w_k - w_l)$$

for the coordinate rows of (2.1) and $T_{kl} = \delta_{kl}^2 z_{kl} w_{kl} - 1$ for the distance constraints. Then $I - U = \sum_k m_k w_k F_k^z + \sum_{k<l} m_k m_l \delta_{kl} T_{kl}$, so $U = I$ on A_m [AK12, p. 541].

Lemma 10.3 (Inertia identity)**.** *Let*

$$S_5 = \sum_{j=3,4} \sum_{l=1,2} m_j m_l \delta_{jl}^3 \big((z_j - z_5)(w_l - w_5) - (w_j - w_5)(z_l - z_5) \big).$$

Then, as polynomials in the masses and in (z, w, δ),

$$I = \sum_{k=1,2} m_k z_k (w_k - w_5) + \sum_{j=3,4} m_j w_j (z_j - z_5) + S_5 \\ - \sum_{j=3,4} m_j \big((z_j - z_5) F_j^w - (w_j - w_5) F_j^z \big) + w_5 \sum_{k=1}^{5} m_k F_k^z.$$

In particular $I = \sum_{k=1,2} m_k z_k (w_k - w_5) + \sum_{j=3,4} m_j w_j (z_j - z_5) + S_5$ on A_m.

Proof. Put $\hat z_k = z_k - z_5$ and $\hat w_k = w_k - w_5$. By antisymmetry of the force terms, $\sum_k m_k F_k^z = \sum_k m_k z_k$. Since $z_j \hat w_j - w_j \hat z_j = z_5 w_j - w_5 z_j$,

$$\begin{aligned} I &= \sum_k m_k z_k \hat w_k + w_5 \sum_k m_k z_k \\ &= \sum_{k=1,2} m_k z_k \hat w_k + \sum_{j=3,4} m_j \big(w_j \hat z_j + z_5 w_j - w_5 z_j \big) + w_5 \sum_k m_k F_k^z. \end{aligned}$$

For $j \in \{3,4\}$, using $z_j - z_l = \hat z_j - \hat z_l$ and $w_j - w_l = \hat w_j - \hat w_l$,

$$\hat z_j F_j^w - \hat w_j F_j^z = (w_5 z_j - z_5 w_j) + \sum_{l\neq j} m_l \delta_{jl}^3 (\hat z_j \hat w_l - \hat w_j \hat z_l).$$

In $\sum_{j=3,4} m_j (\hat z_j F_j^w - \hat w_j F_j^z)$ the terms with $l = 5$ vanish, those with $l \in \{3,4\}$ cancel in pairs, and those with $l \in \{1,2\}$ add up to S_5. Hence $\sum_{j=3,4} m_j (z_5 w_j - w_5 z_j) = S_5 - \sum_{j=3,4} m_j (\hat z_j F_j^w - \hat w_j F_j^z)$, and substituting this into the display gives the identity. □

The term S_5 is the torque about body 5 of the crossed forces on the lower wing; compare [AK12, (31)].

Lemma 10.4. *Let γ be a Diagram 7 germ on A_m with $a + b < 0$ (Proposition 10.2). Then $I(\gamma(t)) \to 0$ as $t \to 0$. More precisely, in the nested chart of Proposition 10.2(ii) the three terms $\sum_{k=1,2} m_k z_k (w_k - w_5)$, $\sum_{j=3,4} m_j w_j (z_j - z_5)$ and S_5 are sums of powers t^e times polynomials in the chart coordinates, with exponents*

$$e = R; \qquad e = P - Q; \qquad e \in \big\{ \tfrac{Q}{2},\ P + \tfrac{Q}{2},\ \tfrac{3Q}{2} + R,\ P + \tfrac{3Q}{2} + R \big\},$$

respectively.

Proof. On A_m the last two sums in Lemma 10.3 vanish, and the three remaining terms are invariant under the rotations, so we may evaluate them on the rotated germ in the chart. There

$$w_k - w_5 = t^R\big((m_k + m_5)v_k + m_{k'}v_{k'}\big), \qquad z_j - z_5 = t^P\big((m_j + m_5)u_j + m_{j'}u_{j'}\big),$$
$$w_j - w_5 = t^{-Q}(B_j - D) + t^R(m_1v_1 + m_2v_2),$$
$$z_l - z_5 = A_l - C + t^P(m_3u_3 + m_4u_4),$$

where $\{k, k'\} = \{1, 2\}$, $\{j, j'\} = \{3, 4\}$ and $l \in \{1, 2\}$, and $\delta^3_{jl} = t^{3Q/2}D^3_{jl}$. Substituting, together with $z_k = A_k$ and $w_j = t^{-Q}B_j$, gives the stated exponents. Since $a+b < 0$, Proposition 10.2(ii) puts (P, Q, R) in the open cone, so every exponent is positive, and all chart coordinates are bounded. Hence the three terms, and with them I, tend to zero. □

10.4 Germs outside the open cone

We complete the order analysis by treating $a+b \geq 0$. Every squared distance then has nonnegative order, and each four-product contains at least one squared distance of positive order.

Lemma 10.5. *Let γ be a Diagram 7 germ with $a + b \geq 0$ (Proposition 10.2). Then every four-product tends to zero along γ.*

Proof. By Proposition 10.2(i), $r^2_{kl} = z_{kl}w_{kl}$ has order $(4+2a)/3 > 0$ on the upper wing, $(4+2b)/3 > 0$ on the lower wing and $a + b \geq 0$ on the crossed pairs. The four-product p_{ij} contains the three squared sides of the triangle $\{k, l, m\}$ complementary to $\{i, j\}$, and this triangle contains a wing pair: if $5 \in \{k, l, m\}$, both pairs through 5 are wing pairs; otherwise $\{k, l, m\} \subset \{1, 2, 3, 4\}$ contains $\{1, 2\}$ or $\{3, 4\}$ (the crossed pairs form the bipartite graph $K_{2,2}$). Hence $\operatorname{ord} p_{ij} \geq \min\{(4 + 2a)/3, (4 + 2b)/3\} > 0$. □

10.5 Conclusion of the Diagram 7 argument

We combine the two order estimates with the constancy of the inertia to exclude an unbounded four-product on a curve through a real point.

Proof of Theorem 10.1. Let γ be a Diagram 7 germ with an unbounded four-product on an irreducible curve $\Gamma \subset A_m$, numbered as above, and let a, b be as in Proposition 10.2. Some four-product has negative order along γ (Definition 3.4), so $a + b < 0$ by Lemma 10.5, and $I(\gamma(t)) \to 0$ by Lemma 10.4. By Lemma 2.1, $I = U$ is constant on Γ, with value $\lim_{t\to 0} I(\gamma(t)) = 0$. At a real point $w_k = \bar{z}_k$, so $I = \sum_k m_k|z_k|^2$, which is positive because $m_k > 0$ and the distance constraint for the pair 12 forces $z_1 \neq z_2$. Hence Γ contains no real point. □

11 Diagram 8

We exclude Diagram 8 by comparing the force equations with the exact distance constraints. The limiting equations determine a mass relation; the first nonzero terms reduce the admissible masses to a one-parameter family, where two quadratic balances are incompatible.

Diagram 8 is the butterfly of [AK12, §7.4] with exactly one circled wing (Fact AK4). We use the numbering of [AK12, §8.8]: the upper wing $\{1, 2, 5\}$ is an isolated triangle of z-edges without

z-circles, the lower wing $\{3,4,5\}$ is a triangle of w-edges with w-circles at 3 and 4 (vertex 5 is never circled [AK12, §7.4], and Rules 1a, 1d place the circles of a circled wing at its other two vertices), and the crossed pairs 13, 14, 23, 24 are not edges. The other circling becomes this one after exchanging z and w and renumbering $1 \leftrightarrow 3$, $2 \leftrightarrow 4$; both operations are symmetries of (2.1) (the second permutes the masses), so it suffices to treat this numbering for every $m \in \mathbb{R}^5_{>0}$. The mass condition of the diagram is relation (40) of [AK12, §8.8],

$$m_5^2(m_1+m_2)^4 = m_1^3 m_2^3, \tag{11.1}$$

as stated in [AK12, p. 578]. For the other circling it reads $m_5^2(m_3+m_4)^4 = m_3^3 m_4^3$, which is their (39). The proof begins with a chart in which the lower wing has a finite limit. Every Diagram 8 germ converges to the boundary of this chart (Proposition 11.3), and its limiting equations imply (11.1) (Proposition 11.4). We then use the exact distance equations on the upper wing and the coefficients of order nine to restrict the possible germs to one curve of masses. A second-order expansion excludes the remaining germs, including complex ones (Theorem 11.8).

We shall also use two symmetries to normalize the masses and coordinates. Scaling the masses by $\lambda > 0$ and (z, w, δ) by $(\lambda^{1/3}, \lambda^{1/3}, \lambda^{-1/3})$ maps A_m onto $A_{\lambda m}$. Every coordinate and every component of $\mathcal{Z}$, $\mathcal{W}$ is multiplied by a nonzero constant, so the orders and diagrams are preserved (Lemma 3.3). The equations of (2.1), apart from $z_{12} = w_{12}$, are also invariant under $z \mapsto cz$, $w \mapsto c^{-1}w$. This changes the z-orders by $\operatorname{ord} c$ and the w-orders by $-\operatorname{ord} c$, preserving the diagram [AK12, p. 547].

11.1 A chart in distance ratios

We normalize one separation in each wing and introduce distance ratios on the upper wing. In the resulting coordinates the force and distance equations extend polynomially to the limiting configurations. Let γ be a Diagram 8 germ. We rotate it by z_{12}^{-1}, so that $z_{12} = 1$ along γ. The chart has two parameters t, h and nineteen coordinates, the positions and shapes $A_1, A_2, C, B_3, B_4, D, u, v, R_{12}, R_{25}, P, Q$ and the inverse distances $D_{13}, D_{14}, D_{23}, D_{24}, D_{34}, D_{35}, D_{45}$; we write x for all of them. It is given by

$$\begin{gathered}
z_1 = A_1, \quad z_2 = A_2, \quad z_3 = C + m_5 t^6 u, \quad z_4 = C + m_5 t^6 v, \\
z_5 = C - t^6(m_3 u + m_4 v), \quad w_3 = t^{-6} B_3, \quad w_4 = t^{-6} B_4, \\
w_1 = t^{-6} D + m_5 t^4 h^{-2} P, \quad w_2 = t^{-6} D + m_5 t^4 h^{-2} Q, \\
w_5 = t^{-6} D - t^4 h^{-2}(m_1 P + m_2 Q), \quad \delta_{15} = t^{-2} h, \\
\delta_{12} = t^{-2} h R_{12}, \quad \delta_{25} = t^{-2} h R_{25}, \\
\delta_{bc} = t^3 D_{bc} \ \ (b \in \{1,2\},\ c \in \{3,4\}), \qquad \delta_{kl} = D_{kl} \ \ (k,l \in \{3,4,5\}).
\end{gathered}$$

On the gauge slice defined below, we take a local sixth root t of $(w_4 - w_3)^{-1}$. Once this root is fixed and $th \neq 0$, the inverse chart map is uniquely determined: C and $t^{-6}D$ are the centres of mass of z_3, z_4, z_5 and of w_1, w_2, w_5, $h = t^2\delta_{15}$, $R_{12} = \delta_{12}/\delta_{15}$, $R_{25} = \delta_{25}/\delta_{15}$, $u = (z_3 - C)/(m_5 t^6)$, $P = h^2(w_1 - t^{-6}D)/(m_5 t^4)$, and the remaining coordinates are recovered in the same way. Along a germ, a finite ramification allows all these roots to be chosen consistently. The rows of the chart are the gauges $E_{20} := A_2 - A_1 - 1$ and $E_{21} := B_4 - B_3 - 1$ (that is, $z_{12} = 1$ and $w_4 - w_3 = t^{-6}$) and $E_0, \dots, E_{19}$, obtained by substituting the chart into the twenty equations of (2.1) other than

$z_{12} = w_{12}$ and multiplying each by a monomial in t and $h^{1/2}$: E_{2k-2} and E_{2k-1} come from the z- and the w-equation of body k, and $E_{10}, \dots, E_{19}$ from the distance constraints of the pairs $12, 13, 14, 15, 23, 24, 25, 34, 35, 45$. In the corner coordinates

$$a := th^{-1/2}, \qquad \eta := t^2 h^{1/2} \qquad (\text{so } t^3 = a\eta,\ h^{3/2} = \eta\, a^{-2})$$

we use the monomials

$$\mu := a^6 = t^6 h^{-3}, \qquad \nu := a^2\eta^2 = t^6, \qquad T := a\eta = t^3, \qquad \eta^2 = t^4 h,$$

and we write $X := A_1 - C$, $Y := A_2 - C$, $M_{12} := m_1 + m_2$, $M := m_3 + m_4 + m_5$, and $C_5 := C - \nu(m_3 u + m_4 v)$, which is z_5.

The chart parameters may be complex. In asymptotic estimates we therefore interpret sums of scales as sums of absolute values: $O(\mu + \rho^2)$ means $O(|\mu| + |\rho|^2)$, and $o(\mu + \rho^2)$ means $o(|\mu| + |\rho|^2)$. Products inside O and o are interpreted in the same way. Algebraic identities and equations between leading coefficients retain their usual complex meaning.

Lemma 11.1. *The chart equations have the following polynomial identities, valid for all masses.*

(i) Every E_j is a polynomial in a, η and x, and its restriction $f_j := E_j|_{a=\eta=0}$ to the boundary is

$$\begin{aligned}
f_0 &= -\big(m_2(X-Y)R_{12}^3 + m_5 X\big), & f_8 &= m_1 X + m_2 Y R_{25}^3,\\
f_2 &= -\big(m_1(Y-X)R_{12}^3 + m_5 Y R_{25}^3\big), & f_{20} &= A_2 - A_1 - 1,\\
f_{10} &= m_5(X-Y)(P-Q)R_{12}^2 - 1, & f_{13} &= X\big((m_1+m_5)P + m_2 Q\big) - 1,\\
f_{16} &= Y R_{25}^2\big(m_1 P + (m_2+m_5)Q\big) - 1, & f_4 &= f_6 = C, \quad f_1 = f_3 = D,\\
f_{11} &= X(D - B_3)D_{13}^2 - 1, & f_{12} &= X(D - B_4)D_{14}^2 - 1,\\
f_{14} &= Y(D - B_3)D_{23}^2 - 1, & f_{15} &= Y(D - B_4)D_{24}^2 - 1,\\
f_5 &= B_3 + m_4(B_4 - B_3)D_{34}^3 - m_5(B_3 - D)D_{35}^3, & f_{21} &= B_4 - B_3 - 1,\\
f_7 &= B_4 - m_3(B_4 - B_3)D_{34}^3 - m_5(B_4 - D)D_{45}^3,\\
f_9 &= D + m_3(B_3 - D)D_{35}^3 + m_4(B_4 - D)D_{45}^3,\\
f_{17} &= m_5(B_3 - B_4)(u - v)D_{34}^2 - 1,\\
f_{18} &= (B_3 - D)\big((m_3+m_5)u + m_4 v\big)D_{35}^2 - 1,\\
f_{19} &= (B_4 - D)\big(m_3 u + (m_4+m_5)v\big)D_{45}^2 - 1.
\end{aligned}$$

(ii) $m_1 E_0 + m_2 E_2 + m_5 E_8 + a^6(m_3 E_4 + m_4 E_6) = a^6(m_1 A_1 + m_2 A_2 + MC)$.

(iii) $E_1 = D + \eta^2\beta_1 + T^3\tau_1 + \sum_{i\ge 6,\, j\ge 2} a^i\eta^j(\cdots)$, *and E_3 is the same with β_3, τ_3, where*

$$\begin{aligned}
\beta_1 &= -m_5\big(m_2(P-Q)R_{12}^3 + (m_1+m_5)P + m_2 Q\big),\\
\beta_3 &= -m_5\big(m_1(Q-P)R_{12}^3 + (m_1 P + (m_2+m_5)Q)R_{25}^3\big),\\
\tau_1 &= m_3(B_3 - D)D_{13}^3 + m_4(B_4 - D)D_{14}^3,\\
\tau_3 &= m_3(B_3 - D)D_{23}^3 + m_4(B_4 - D)D_{24}^3;
\end{aligned}$$

the monomials of $E_1 - E_3$ are η^2, T^3, $a^6\eta^2$ and $a^9\eta^5$.

(iv) *For* $k \in \{0,2,8,10,13,16,20\}$, $E_k = f_k(\mathcal{I}) + \sum_{i\geq 6} a^i \eta^j(\cdots)$, *where* f_k *is read as a function of* $(X, Y, R_{12}, R_{25}, P, Q)$ *and* $\mathcal{I} := (A_1 - C_5,\, A_2 - C_5,\, R_{12},\, R_{25},\, P,\, Q)$.

(v) $E_4 = C + \nu L_3 + T^3 F_3 + T^5 G_3$ *exactly, where* $L_3 = -m_5\big(m_4(u-v)D^3_{34} + ((m_3+m_5)u + m_4 v)D^3_{35} - u\big)$, $F_3 = -\big(m_1(C-A_1)D^3_{13} + m_2(C-A_2)D^3_{23}\big)$ *and* $G_3 = -m_5 u\big(m_1 D^3_{13} + m_2 D^3_{23}\big)$.

Proof. We write out the chart equations to identify both their limiting terms and the monomials that enter the later order comparisons. Put $I = \{1,2\}$, $J = \{3,4\}$, $P_1 = P$, $P_2 = Q$, $u_3 = u$, $u_4 = v$, and set

$$q = R_{12}, \quad s = R_{25}, \quad V = m_3 u + m_4 v, \quad S = m_1 P + m_2 Q, \quad \gamma = \mu\eta^2 = \nu a^4.$$

The unscaled positions are

$$z_i = A_i, \quad z_j = C + \nu m_5 u_j, \quad z_5 = C_5 = C - \nu V,$$

whereas their scaled w-coordinates are

$$\nu w_i = D + \gamma m_5 P_i, \qquad \nu w_j = B_j, \qquad \nu w_5 = D - \gamma S.$$

The inverse-distance cubes are $\mu^{-1}q^3, \mu^{-1}, \mu^{-1}s^3$ on $12, 15, 25$, respectively, $T^3 D^3_{ij}$ on crossed pairs, and D^3_{kl} on the lower wing. The chart equations are obtained by multiplying the z-equations of $1, 2, 5$ by μ, leaving those of $3, 4$ unscaled, and multiplying all w-equations by ν. Writing $X' = A_1 - C_5$ and $Y' = A_2 - C_5$, the resulting upper z-rows are

$$\begin{aligned}
E_0 &= \mu A_1 - m_2(A_1 - A_2)q^3 - m_5 X' - \mu T^3 \sum_{j\in J} m_j (A_1 - C - \nu m_5 u_j) D^3_{1j},\\
E_2 &= \mu A_2 - m_1(A_2 - A_1)q^3 - m_5 Y' s^3 - \mu T^3 \sum_{j\in J} m_j (A_2 - C - \nu m_5 u_j) D^3_{2j},\\
E_8 &= \mu C_5 + m_1 X' + m_2 Y' s^3 + \mu\nu \sum_{j\in J} m_j (V + m_5 u_j) D^3_{j5}.
\end{aligned}$$

If j' denotes the other element of J, the lower z-rows are

$$\begin{aligned}
E_{2j-2} = C &+ \nu m_5 u_j - \nu m_{j'} m_5 (u_j - u_{j'}) D^3_{jj'} - \nu m_5 (V + m_5 u_j) D^3_{j5}\\
&- T^3 \sum_{i\in I} m_i (C - A_i + \nu m_5 u_j) D^3_{ij}.
\end{aligned}$$

This formula at $j = 3$ proves (v).

The upper w-rows are

$$\begin{aligned}
E_1 &= D + \eta^2\beta_1 + T^3\tau_1 + \gamma m_5 P - \gamma T^3 m_5 P \sum_{j\in J} m_j D^3_{1j},\\
E_3 &= D + \eta^2\beta_3 + T^3\tau_3 + \gamma m_5 Q - \gamma T^3 m_5 Q \sum_{j\in J} m_j D^3_{2j}.
\end{aligned}$$

Thus (iii) follows, including its complete list of monomials. The lower and central w-rows are

$$E_{2j-1} = B_j - m_{j'}(B_j - B_{j'}) D^3_{jj'} - m_5 (B_j - D) D^3_{j5} - \gamma m_5 S D^3_{j5}$$

$$- T^3 \sum_{i\in I} m_i(B_j - D)D_{ij}^3 + \gamma T^3 m_5 \sum_{i\in I} m_i P_i D_{ij}^3,$$
$$E_9 = D + \sum_{j\in J} m_j(B_j - D)D_{j5}^3 + \eta^2\beta_5 + \gamma S\Big(-1 + \sum_{j\in J} m_j D_{j5}^3\Big),$$

where

$$\beta_5 = m_1((m_1+m_5)P + m_2Q) + m_2 s^3(m_1P + (m_2+m_5)Q).$$

Finally the distance rows are obtained by subtracting 1 from the following expressions:

$$\begin{array}{ll} 12 & m_5(A_1 - A_2)(P-Q)q^2 \\ 15 & X'((m_1+m_5)P + m_2Q) \\ 25 & Y's^2(m_1P + (m_2+m_5)Q) \\ ij & (A_i - C - \nu m_5 u_j)(D - B_j + \gamma m_5 P_i)D_{ij}^2 \\ 34 & m_5(u-v)(B_3 - B_4)D_{34}^2 \\ j5 & (V + m_5 u_j)(B_j - D + \gamma S)D_{j5}^2. \end{array}$$

These formulas account for all ten distance rows; the last two rows are $E_{20} = A_2 - A_1 - 1$ and $E_{21} = B_4 - B_3 - 1$. Every expression is polynomial. Setting $a = \eta = 0$ gives the boundary equations listed in (i). The three upper distance rows and the gauge are already exact functions of $\mathcal{I}$; in the upper z-rows all other terms have a factor μ. This proves (iv). Finally, the mass-weighted sum of the original z-equations has no force terms, since the two terms contributed by each pair cancel. Multiplying this sum by μ gives (ii), because the weighted position of the lower cluster is MC. □

Definition 11.2. For a germ in this chart with $t \to 0$, x convergent, $\operatorname{ord} a > 0$ and $\operatorname{ord}\eta > 0$, we define the order ratios

$$r := \frac{\operatorname{ord}\eta}{\operatorname{ord} a} \in (0,\infty), \qquad p := \frac{\operatorname{ord} h}{\operatorname{ord} t}.$$

Since $t^3 = a\eta$ and $h^{3/2} = \eta a^{-2}$, $p = (2r-4)/(r+1)$, an increasing bijection of $(0,\infty)$ onto $(-4,2)$. In particular,

$$r = \tfrac{1}{2}, 2, 3 \quad\Longleftrightarrow\quad p = -2, 0, \tfrac{1}{2},$$

respectively. The boundary components $\{a = 0\}$ and $\{\eta = 0\}$ correspond to $p = -4$ and $p = 2$.

11.2 Representation of all Diagram 8 germs

The diagram rules determine the orders of both wings and the crossed separations. We use them to prove convergence of the chart coordinates and to identify the full range of the order ratios.

Proposition 11.3 (Representation in the chart)**.** *Every Diagram 8 germ converges in the chart with* $\operatorname{ord} a > 0$, $\operatorname{ord}\eta > 0$ *and* $p \in (-4,2)$*; its lower-wing inverse distances have finite nonzero limits. Precisely: let* γ *be a Diagram 8 germ, rotated so that* $z_{12} = 1$*, and put* $t := (w_4 - w_3)^{-1/6}$ *and* $h := t^2\delta_{15}$ *along* γ*. Let* θ *be the order of* z_{12} *before the rotation, in the normalization* $Z = W = \epsilon^{-2}$ *with orders in* ϵ*. Then* $-2 < \theta < 1$*; every coordinate in* x *converges along* γ*, and* R_{12}*,* R_{25}*,* $D_{13}, D_{14}, D_{23}, D_{24}$*,* D_{34}, D_{35}, D_{45} *have nonzero limits; and*

$$\operatorname{ord} t = \tfrac{2-\theta}{6}, \qquad \operatorname{ord} h = -\tfrac{2\theta}{3}, \qquad p = -\tfrac{4\theta}{2-\theta},$$
$$\operatorname{ord} a = \tfrac{2+\theta}{6}, \qquad \operatorname{ord}\eta = \tfrac{2-2\theta}{3}, \qquad r = \tfrac{4(1-\theta)}{2+\theta}.$$

Proof. By Fact AK2 and Lemma 3.3 the rules and estimates of [AK12, §3] hold for the orders along γ. We begin with the lower wing. The triangle $\{3,4,5\}$ is an isolated component of the w-diagram with w-circles, so it has a maximal w-stroke [AK12, Rule 1e, p. 546]. By Rule 2e its three w-separations have one order, so all three w-strokes are maximal, and the right equality case of Estimate 2 [AK12, p. 548], in the colour w, gives $\operatorname{ord} w_{kl} = -2$, $\operatorname{ord} z_{kl} = 2$, $\operatorname{ord} r_{kl} = 0$ for $kl \in \{34, 35, 45\}$. The upper wing $\{1,2,5\}$ is an isolated component of the z-diagram without circles. It has no maximal z-stroke by Rule 1e and no w-stroke, hence no zw-edge. Thus neither equality case of Estimate 2 holds, and by Rule 2e the three z-separations have one order θ with $-2 < \theta < 1$. The z-strokes have $\operatorname{ord} Z_{kl} = -2$, so $\operatorname{ord} r_{kl} = (\theta+2)/3$ and $\operatorname{ord} w_{kl} = (4-\theta)/3$. These wing orders determine the crossed separations. For $b \in \{1,2\}$, $c \in \{3,4\}$, $z_{bc} = z_{b5} + z_{5c}$ has order θ (as $\theta < 2$) and $w_{bc} = w_{b5} + w_{5c}$ has order -2 (as $(4-\theta)/3 > -2$), so $\operatorname{ord} r_{bc} = (\theta-2)/2$. We next recover the orders of the positions. Every w_k has order ≥ -2, with equality for $k = 3, 4$ (w-circles). Let C be the centre of mass of z_3, z_4, z_5. Then $(m_1+m_2+M)C = -m_1(z_1 - C) - m_2(z_2 - C)$, and $z_b - C = z_{5b} + (z_5 - C)$ with $\operatorname{ord} z_{5b} = \theta$ and $\operatorname{ord}(z_5 - C) \geq 2$; so C, z_1, z_2 have order $\geq \theta$, and $z_k - C$ has order ≥ 2 for $k = 3, 4, 5$. After the rotation, all z-orders decrease by θ and all w-orders increase by θ. Consequently $\operatorname{ord}(w_4 - w_3) = \theta - 2$ and $\operatorname{ord} t = (2-\theta)/6 > 0$. The inverse chart formulas now give the orders of the new coordinates: A_1, A_2, C have order ≥ 0; u, v have order $\geq (2-\theta) - 6\operatorname{ord} t = 0$; B_3, B_4 have order 0 and D order ≥ 0; $\operatorname{ord} h = 2\operatorname{ord} t - \operatorname{ord} r_{15} = -2\theta/3$; R_{12}, R_{25} have order 0; P, Q have order $\geq 2\operatorname{ord} h + \frac{4+2\theta}{3} - 4\operatorname{ord} t = 0$, because $w_1 - t^{-6}D = (m_2 w_{21} + m_5 w_{51})/(m_1+m_2+m_5)$ and similarly for w_2; $D_{bc} = t^{-3}\delta_{bc}$ has order $\frac{2-\theta}{2} - 3\operatorname{ord} t = 0$; and $D_{kl} = \delta_{kl}$ has order 0. Hence every coordinate converges (Lemma 3.3), the inverse-distance coordinates to nonzero limits, and $p = \operatorname{ord} h / \operatorname{ord} t = -4\theta/(2-\theta) \in (-4, 2)$. Finally $\operatorname{ord} a = \operatorname{ord} t - \frac{1}{2}\operatorname{ord} h = (2+\theta)/6$ and $\operatorname{ord} \eta = 2\operatorname{ord} t + \frac{1}{2}\operatorname{ord} h = (2-2\theta)/3$, both positive. □

Thus every germ has a finite limit at which all inverse-distance coordinates are nonzero. We next solve the equations at these limits.

11.3 The mass relation and the limiting configuration

We combine the limiting force equations with the closure of the upper triangle. This gives the mass relation and determines the coordinates needed for the subsequent coefficient comparisons.

Proposition 11.4 (Mass relation and limiting coordinates)**.** *At positive masses every Diagram 8 germ satisfies* (11.1)*: $m_5^2(m_1+m_2)^4 = m_1^3 m_2^3$. Moreover, in the chart of Proposition 11.3 its limit x_0 has $A_1 = -m_2/M_{12}$, $A_2 = m_1/M_{12}$, $C = 0$, $R_{25} = 1$, $R_{12} = -m_5 M_{12}/(m_1 m_2)$, $(P,Q) = (P_0, Q_0)$ the unique solution of $f_{10} = f_{13} = 0$, $D = 0$, $B_3 = -m_4/(m_3+m_4)$, $B_4 = m_3/(m_3+m_4)$, and, with $L := (M_{12}(m_3+m_4))^{1/2} > 0$ and signs $s_{bc} \in \{\pm 1\}$,*

$$D_{13} = \frac{i s_{13} L}{\sqrt{m_2 m_4}}, \qquad D_{14} = \frac{s_{14} L}{\sqrt{m_2 m_3}},$$
$$D_{23} = \frac{s_{23} L}{\sqrt{m_1 m_4}}, \qquad D_{24} = \frac{i s_{24} L}{\sqrt{m_1 m_3}}.$$

We write $U := (A_1, A_2, C, R_{12}, R_{25}, P, Q)$, U_0 for its limit, and $\mathcal{I}_0 := (X_0, Y_0, R_{12}^0, 1, P_0, Q_0)$, with $X_0 = -m_2/M_{12}$, $Y_0 = m_1/M_{12}$ and $R_{12}^0 = -m_5 M_{12}/(m_1 m_2)$, for the limit of $\mathcal{I}$ (note $C_5 \to C \to 0$).

Proof. Let $m \in \mathbb{R}^5_{>0}$ and let γ be a Diagram 8 germ, in the chart of Proposition 11.3, with limit x_0. By Lemma 11.1 (i) and Proposition 11.3, $E_j = f_j(x) + O(|a| + |\eta|)$ along γ, so $f_j(x_0) = 0$ for all j. The centre-of-mass identity in Lemma 11.1 (ii), together with $a \neq 0$, gives $m_1A_1 + m_2A_2 + MC = 0$ along γ. At the limit this identity, $f_4 = C$ and f_{20} form a linear system with determinant M_{12}. It follows that $A_1 = -m_2/M_{12}$, $A_2 = m_1/M_{12}$ and $C = 0$ at x_0. Thus $X = A_1$, $Y = A_2$ and $X - Y = -1$ there.

At these positions the upper force equations reduce to $f_8 = \frac{m_1m_2}{M_{12}}(R_{25}^3 - 1)$ and $f_0 = \frac{m_2}{M_{12}}F$, where $F := M_{12}R_{12}^3 + m_5$. Hence $R_{25}^3 = 1$ and $F = 0$.

The distance equations impose a further relation, expressing the closure of the w-triangle 125. The identity

$$\begin{aligned}(X - Y)R_{12}^2 X f_{16} - (X - Y)R_{12}^2 Y R_{25}^2 f_{13} + XYR_{25}^2 f_{10}\\ = (X - Y)R_{12}^2(YR_{25}^2 - X) - XYR_{25}^2,\end{aligned}$$

eliminates P, Q. At x_0 its right-hand side is $-G/M_{12}^2$, where $G := M_{12}R_{12}^2(m_1R_{25}^2 + m_2) - m_1m_2R_{25}^2$, so $G = 0$.

To combine this relation with the force equations, set $H := M_{12}(m_1 + m_2R_{25})R_{12}^2 - m_1m_2$ and $\Phi(s) := m_5^2M_{12}(m_1 + m_2s)^3 - (m_1m_2)^3$. The identities

$$\begin{aligned}H &= R_{25}G + (1 - R_{25}^3)\, m_1(M_{12}R_{12}^2 - m_2),\\ \Phi(R_{25}) &= H\big(H^2 + 3m_1m_2H + 3m_1^2m_2^2\big) - M_{12}(m_1 + m_2R_{25})^3F(F - 2m_5),\end{aligned}$$

give $H = 0$ and $\Phi(R_{25}) = 0$ at x_0.

Positivity of the masses excludes the nonreal cube roots. Indeed, if $R_{25} \neq 1$, then $R_{25}^2 + R_{25} + 1 = 0$ and $\Phi(R_{25}) = \Phi_0 + \Phi_1R_{25}$ with real Φ_0, Φ_1. Since $R_{25} \notin \mathbb{R}$, both coefficients must vanish: $\Phi_0 = \Phi_1 = 0$. However, $\Phi_1 = 3m_1m_2m_5^2M_{12}(m_1 - m_2)$ forces $m_1 = m_2$, after which $\Phi_0 = -m_1^4(m_1^2 + 2m_5^2) < 0$. Thus $R_{25} = 1$, and $\Phi(1) = 0$ is precisely (11.1).

The remaining limit coordinates are now determined successively. With $R_{25} = 1$, the equations $F = 0$ and $H = M_{12}^2R_{12}^2 - m_1m_2 = 0$ give $R_{12} = R_{12}^3/R_{12}^2 = -m_5M_{12}/(m_1m_2)$. The equations f_{10}, f_{13} are linear in (P, Q) with determinant $m_2m_5(m_1 + m_2 + m_5)R_{12}^2/M_{12} \neq 0$, so their solution is unique. Next, $f_1 = D$ gives $D = 0$. The identity $m_1f_1 + m_2f_3 + m_3f_5 + m_4f_7 + m_5f_9 = (m_1 + m_2 + m_5)D + m_3B_3 + m_4B_4$ and f_{21} determine B_3, B_4. Finally, the crossed distance equations $f_{11}, f_{12}, f_{14}, f_{15}$ yield $D_{13}^2 = -M_{12}(m_3 + m_4)/(m_2m_4)$, $D_{14}^2 = M_{12}(m_3 + m_4)/(m_2m_3)$, $D_{23}^2 = M_{12}(m_3 + m_4)/(m_1m_4)$ and $D_{24}^2 = -M_{12}(m_3 + m_4)/(m_1m_3)$, which give the stated square roots. □

11.4 Exclusion

We first use the distance equations on the upper wing to exclude the range $0 < p < 2$. For the remaining order ratios, we compare the coefficients of $t^9 = T^3$ in E_1 and E_3. When $-4 < p < \frac{1}{2}$, this comparison forces the coefficient Ξ defined below to vanish. A second-order expansion then excludes the masses and signs on the locus $\Xi = 0$.

Lemma 11.5. *No Diagram 8 germ at positive masses has* $0 < p < 2$.

Proof. Suppose that such a germ exists. Its order ratio satisfies $r > 2$, so $\nu = o(\mu)$, $T^3 = o(\mu)$ and $T^5 = o(\mu)$. The exact lower row $E_4 = C + \nu L_3 + T^3 F_3 + T^5 G_3 = 0$ gives $C = o(\mu)$ and hence $C_5 = o(\mu)$. The centre of mass and the gauge then give

$$A_1 = X_0 + o(\mu), \quad A_2 = Y_0 + o(\mu), \quad X := A_1 - C_5 = X_0 + o(\mu), \quad Y := A_2 - C_5 = Y_0 + o(\mu),$$

where $X_0 = -m_2/M_{12}$ and $Y_0 = m_1/M_{12}$. Write $q = R_{12}$, $s = R_{25}$ and $q_0 = R_{12}^0 \neq 0$. In the upper force equations, the crossed forces contribute $O(\mu T^3)$, giving

$$\begin{aligned} 0 = E_0 &= m_2 q^3 - m_5 X + \mu A_1 + O(\mu T^3), \\ 0 = E_2 &= -m_1 q^3 - m_5 Y s^3 + \mu A_2 + O(\mu T^3). \end{aligned}$$

The first row gives

$$q^3 - q_0^3 = \frac{\mu}{M_{12}} + o(\mu), \qquad q - q_0 = \frac{\mu}{3M_{12}q_0^2} + o(\mu).$$

In the second row the terms $-m_1\mu/M_{12}$ and μY_0 cancel, so $s^3 - 1 = o(\mu)$ and therefore $s - 1 = o(\mu)$.

The three exact upper distance equations give

$$P - Q = -\frac{1}{m_5 q^2}, \qquad (m_1 + m_5)P + m_2 Q = X^{-1}, \qquad m_1 P + (m_2 + m_5)Q = (Ys^2)^{-1}.$$

Subtracting the last two equations yields the closure identity

$$q^{-2} = (Ys^2)^{-1} - X^{-1}.$$

Subtracting the limiting closure identity yields

$$-\frac{2\mu}{3M_{12}q_0^5} + o(\mu) = o(\mu), \qquad \text{hence} \qquad -\frac{2}{3M_{12}q_0^5} = 0,$$

contradicting $M_{12} > 0$ and $q_0 \neq 0$. □

Lemma 11.6. *Let $m \in \mathbb{R}^5_{>0}$ satisfy (11.1); let x_0, $\mathcal{I}_0$, U_0 and the signs s_{bc} be as in Proposition 11.4, $\beta := \beta_1 - \beta_3$, $b := (f_0, f_2, f_{10}, f_{13}, f_{16}, f_{20})$ as a function of $(X, Y, R_{12}, R_{25}, P, Q)$, and $B := \partial b/\partial \mathcal{I}$ at $\mathcal{I}_0$.*

(a) $\beta(\mathcal{I}_0) = 0$.

(b) If $m_1 \neq m_2$, then B is invertible. If $m_1 = m_2$ $(= 4m_5)$, then $\ker B$ is spanned by a vector k_0, equal to $(\frac{27}{272}, \frac{27}{272}, \frac{9}{272}, -\frac{9}{68}, -\frac{19}{17}, 1)$ at $(m_1, m_2, m_5) = (1, 1, \frac{1}{4})$; $d\beta(\mathcal{I}_0)\, k_0 = 0$; and the Jacobian of the eight functions $f_0, f_2, f_{10}, f_{13}, f_{16}, f_{20}$ (evaluated at $X = A_1 - C$, $Y = A_2 - C$), C and $m_1 A_1 + m_2 A_2 + MC$ with respect to U at U_0 has rank 7. (These three facts are proved at $(m_1, m_2, m_5) = (1, 1, \frac{1}{4})$ with m_3, m_4 free, which suffices by the mass scaling.)

(c) With $\Xi := s_{14}(m_1 m_4)^{3/2} + s_{23}(m_2 m_3)^{3/2} + i\big(s_{13}(m_1 m_3)^{3/2} + s_{24}(m_2 m_4)^{3/2}\big)$,

$$(\tau_1 - \tau_3)(x_0) = \frac{m_3 m_4}{m_3 + m_4}\, L^3\, (m_1 m_2 m_3 m_4)^{-3/2}\, \Xi.$$

Consequently, if a Diagram 8 germ at the masses m has $-4 < p < \frac{1}{2}$, then $\Xi = 0$, that is, $m_1 = m_2 = 4m_5$, $m_3 = m_4$, $s_{13} = -s_{24}$ and $s_{14} = -s_{23}$.

Proof of (a)–(c). On the zero set of the three upper distance equations at the boundary, writing $q = R_{12}$ and $s = R_{25}$, the same subtraction as above gives

$$q^{-2} = (Ys^2)^{-1} - X^{-1}, \qquad \beta = M_{12}q - m_5(X^{-1} - s/Y).$$

The second identity follows by replacing $P - Q$ in the definition of β by $-1/(m_5q^2)$ and the other two linear combinations by their distance values. At the boundary point it gives $\beta = 0$, proving (a).

To prove (b), the gauge eliminates Y by $Y = X + 1$. The equations $f_{10} = f_{13} = 0$ determine P, Q analytically, because their linear coefficient matrix has nonzero determinant proportional to $m_5q^2X(M_{12} + m_5)$. After this elimination the remaining three equations are equivalent to

$$m_2q^3 - m_5X = 0, \qquad -m_1q^3 - m_5Ys^3 = 0, \qquad q^{-2} - (Ys^2)^{-1} + X^{-1} = 0.$$

After scaling, we may take $m_2 = 1$, $m_1 = \alpha^2$. The mass relation then gives $m_5 = \alpha^3/(\alpha^2 + 1)^2$; we set $N = \alpha^2 + 1$. At the boundary point, $(X, Y, q, s) = (-1/N, \alpha^2/N, -\alpha/N, 1)$. The linearizations of the first two equations give

$$\delta q = \frac{\alpha}{3}\,\delta X, \qquad \delta s = -\frac{N^2}{3\alpha^2}\,\delta X.$$

Substitution in the linearization of the third gives

$$\frac{N^2(1 - \alpha^4)}{3\alpha^4}\,\delta X = 0.$$

Thus B is invertible for $\alpha \neq 1$. For $\alpha = 1$, its kernel is one-dimensional; choosing $\delta s = 1$ gives

$$(\delta X, \delta Y, \delta q, \delta s, \delta P, \delta Q) = (-\tfrac{3}{4}, -\tfrac{3}{4}, -\tfrac{1}{4}, 1, \tfrac{76}{9}, -\tfrac{68}{9}).$$

Multiplication by $-9/68$ gives precisely the vector k_0 in (b). Differentiating the displayed formula for β on this tangent gives $d\beta = 2\delta q + \frac{1}{2}\delta s = 0$. Finally C fixes the translation variable and the centre-of-mass condition is nonzero on the remaining tangent, since $\delta(X + Y) = -3/2$. The eight functions in (b) therefore have rank 7.

For (c), $D = 0$, $B_3 = -m_4/(m_3 + m_4)$ and $B_4 = m_3/(m_3 + m_4)$ give

$$(\tau_1 - \tau_3)(x_0) = \frac{m_3m_4}{m_3 + m_4}(-D_{13}^3 + D_{23}^3 + D_{14}^3 - D_{24}^3).$$

Substituting the four square roots from Proposition 11.4 gives the following imaginary and real terms, respectively:

$$iL^3s_{13}(m_2m_4)^{-3/2}, \quad iL^3s_{24}(m_1m_3)^{-3/2},$$
$$L^3s_{23}(m_1m_4)^{-3/2}, \quad L^3s_{14}(m_2m_3)^{-3/2}.$$

Factoring the common denominator gives exactly the stated formula involving Ξ. □

Proof of the last assertion. Let γ be such a germ. Along γ, $\mathcal{I} \to \mathcal{I}_0$ and $b(\mathcal{I}) = O(a^6)$ by Lemma 11.1 (iv). We claim

$$\beta(\mathcal{I}) = O(a^6 + a^4\eta^4). \tag{11.2}$$

When $m_1 \neq m_2$, the invertibility in (b) and the inverse function theorem give $\mathcal{I} - \mathcal{I}_0 = O(a^6)$. Since (a) gives the vanishing of the constant term, this proves (11.2).

For $m_1 = m_2$, we use the eight functions in (b) to control the direction left undetermined by the upper equations. These functions are $O(a^6 + \nu + T^3)$ along γ. For the first six, this follows from $f_k(A_1 - C, A_2 - C, \dots) - f_k(\mathcal{I}) = O(\nu)$; the seventh is controlled by Lemma 11.1 (v). The last is identically zero by Lemma 11.1 (ii), since $a \neq 0$. Their rank therefore gives the bounds $U - U_0$ and $\mathcal{I} - \mathcal{I}_0$ of order $O(a^6 + \nu + T^3)$.

Taylor expansion also gives $B(\mathcal{I} - \mathcal{I}_0) = O(a^6 + |\mathcal{I} - \mathcal{I}_0|^2)$, so the component of $\mathcal{I} - \mathcal{I}_0$ transverse to k_0 is $O(a^6 + |\mathcal{I} - \mathcal{I}_0|^2)$. Because $\beta(\mathcal{I}_0) = 0$ and $d\beta(\mathcal{I}_0)k_0 = 0$, the uncontrolled direction has no linear contribution to the difference of the upper equations. Consequently $\beta(\mathcal{I}) = O(a^6 + |\mathcal{I} - \mathcal{I}_0|^2) = O(a^6 + \nu^2 + T^6)$. This is (11.2), since $\nu^2 = a^4\eta^4$ and $T^6 = a^6\eta^6$.

Now $E_1 = E_3 = 0$ along γ, so Lemma 11.1 (iii) and (11.2) give $T^3(\tau_1 - \tau_3)(x) = O(a^6\eta^2 + a^4\eta^6)$, that is, $(\tau_1 - \tau_3)(x) = O(a^3\eta^{-1} + a\eta^3)$. As $-4 < p < \frac{1}{2}$ means $r < 3$, $\operatorname{ord}(a^3\eta^{-1}) = (3 - r)\operatorname{ord} a > 0$; so $(\tau_1 - \tau_3)(x_0) = 0$, and $\Xi = 0$ by (c). Since all $m_k > 0$, the real part $s_{14}(m_1m_4)^{3/2} + s_{23}(m_2m_3)^{3/2}$ of Ξ vanishes only if $s_{14} = -s_{23}$ and $m_1m_4 = m_2m_3$, and its imaginary part only if $s_{13} = -s_{24}$ and $m_1m_3 = m_2m_4$. These give $m_3 = m_4$ and $m_1 = m_2$, and then (11.1) gives $m_1 = 4m_5$. □

The order-nine coefficient vanishes precisely on the mass curve $m = (4, 4, y, y, 1)$, $y > 0$, up to scaling, with the four sign classes $s_{13} = -s_{24}$, $s_{14} = -s_{23}$. On this locus the next obstruction comes from the quadratic terms. We now show that the upper force and distance equations give incompatible leading coefficients.

Lemma 11.7. *Let $m_1 = m_2 = 4m_5$ and $m_3 = m_4$, scaled to $(1, 1, m, m, \frac{1}{4})$ with $m > 0$. No Diagram 8 germ at these masses has a limit with $s_{13} = -s_{24}$ and $s_{14} = -s_{23}$.*

Proof. Suppose, for contradiction, that such a germ exists, and let x_0 be its limit in the chart of Proposition 11.3. We shall use the distance equations to reduce the upper-wing equations to two scalar relations, then compare their second-order terms.

We first show that the limiting equations on the lower wing have an invertible Jacobian. Put $d = D_{34}$, $g = D_{35}$ and $q = D_{45}$. At a common zero of the lower boundary equations, $f_1 = 0$ gives $D = 0$. The identity

$$mf_5 + mf_7 + \tfrac{1}{4}f_9 = m(B_3 + B_4) + \tfrac{1}{4}D$$

and $f_{21} = 0$ give $B_3 = -1/2$, $B_4 = 1/2$. Then $f_9 = 0$ is $q^3 = g^3$, and $f_5 = 0$ is $8md^3 + g^3 = 4$. The equations $f_{18} = f_{19} = 0$ read

$$(m + \tfrac{1}{4})u + mv = -2g^{-2}, \qquad mu + (m + \tfrac{1}{4})v = 2q^{-2}.$$

Their coefficient determinant is $\frac{1}{4}(2m + \frac{1}{4}) > 0$. Subtracting them and using $f_{17} = 0$ gives $2d^2(g^{-2} + q^{-2}) = 1$. Writing $q = \omega g$ and $d = \lambda g$, we have thus obtained

$$\omega^3 = 1, \qquad 2\lambda^2(1 + \omega^{-2}) = 1, \qquad g^3(8m\lambda^3 + 1) = 4.$$

The representation in the chart ensures that all inverse distances are nonzero. The preceding elimination gives an invertible Jacobian for the eight lower boundary equations: f_1, the displayed

weighted sum, and f_{21} determine D, B_3, B_4; the next three successively determine ω, λ, g, with derivatives

$$3\omega^2, \qquad 4\lambda(1+\omega^{-2}), \qquad 3g^2(8m\lambda^3+1) = 12/g,$$

all nonzero; and the last linear system determines u, v.

We now turn to the upper wing and write its deviations from the limit as

$$X = A_1 - C_5 = -\tfrac{1}{2} + x, \qquad Y = A_2 - C_5 = \tfrac{1}{2} + x, \qquad R_{12} = -\tfrac{1}{2} + v_*, \qquad R_{25} = 1 + h_*.$$

The equality $Y - X = 1$ follows from the gauge. The symbols v_*, h_* denote upper-wing deviations and are distinct from the chart coordinates v, h. Write $r_* = R_{12}$ and $s_* = R_{25}$. The three upper distance equations hold along the whole germ:

$$P - Q = -4r_*^{-2}, \qquad \tfrac{5}{4}P + Q = X^{-1}, \qquad P + \tfrac{5}{4}Q = (Ys_*^2)^{-1}.$$

Consequently

$$r_*^{-2} = (Ys_*^2)^{-1} - X^{-1}. \tag{11.3}$$

They determine P, Q analytically once this closure equation holds. Also, direct grouping of the upper forces gives

$$\begin{aligned} E_0 &= r_*^3 - \tfrac{1}{4}X + \mu A_1 + O(\mu T^3), \\ E_2 &= -r_*^3 - \tfrac{1}{4}Ys_*^3 + \mu A_2 + O(\mu T^3), \end{aligned}$$

because every crossed force in an upper z-row has factor μT^3. Here and below all estimates are along the germ, whose chart coordinates are bounded.

Expanding (11.3) and $E_2 = 0$ to second order gives

$$\begin{aligned} 0 &= 16v_* + 4h_* + 48v_*^2 - 16x^2 - 8xh_* - 6h_*^2 + O\big((|x| + |v_*| + |h_*|)^3\big), \\ \tfrac{1}{2}\mu &= \tfrac{1}{4}x + \tfrac{3}{4}v_* + \tfrac{3}{8}h_* - \tfrac{3}{2}v_*^2 + \tfrac{3}{4}xh_* + \tfrac{3}{8}h_*^2 + O\big((|x| + |v_*| + |h_*|)^3\big) + o(\mu). \end{aligned}$$

The derivative of these two equations with respect to (x, v_*) is invertible at zero. Solving the linear and quadratic terms therefore gives

$$x = -\tfrac{3}{4}h_* + 2\mu + o(\mu + h_*^2), \qquad v_* = -\tfrac{1}{4}h_* + \tfrac{3}{8}h_*^2 + o(\mu + h_*^2). \tag{11.4}$$

The analytic implicit solution has remainder $O(h_*^3 + \mu h_* + \mu^2) + o(\mu)$ before substitution of the other bounded chart coordinates, which yields the displayed estimates. Substituting in $E_0 = 0$ now gives

$$-\mu + \tfrac{3}{16}h_*^2 = o(\mu + h_*^2). \tag{11.5}$$

The same exact distance equations simplify the difference $\beta = \beta_1 - \beta_3$ to

$$\beta = 2r_* - \tfrac{1}{4}(X^{-1} - s_*/Y).$$

Its quadratic expansion is $2v_* + \frac{1}{2}h_* + 4x^2 - xh_* + O((|x| + |v_*| + |h_*|)^3)$. By (11.4),

$$\beta = \tfrac{15}{4}h_*^2 + o(\mu + h_*^2). \tag{11.6}$$

Finally, direct substitution in the two upper w-equations gives

$$E_1 - E_3 = \eta^2\beta + T^3(\tau_1 - \tau_3) + \tfrac{1}{4}\mu\eta^2(P - Q) + O(\mu\eta^2 T^3). \tag{11.7}$$

In particular, $P_0 - Q_0 = -16$.

To compare the remaining terms, we need to control the lower and crossed coordinates. Their deviations from the limits are

$$O(\nu + \mu + |h_*|). \tag{11.8}$$

Indeed, in the eight lower rows, the terms beyond the limiting equations carry factors T^3, $\mu\eta^2$, $\mu\eta^2 T^3$, or $\eta^2\beta_j$; the last kind occurs in the upper or central w-row. Its coefficient vanishes at the upper boundary point, where $(P, Q, r_*, s_*) = (-8, 8, -1/2, 1)$. For the central row this coefficient is $-4(\beta_1 + \beta_3)$. Thus these errors are $O(\nu + \mu + |h_*|)$, since $T^3 = o(\nu)$ and (11.4) bounds the upper deviations by $O(\mu + |h_*|)$. The crossed distance rows have only the additional factors ν, $\mu\eta^2$ and $\nu\mu\eta^2$; each crossed boundary equation has derivative $2/D_{bc} \neq 0$ in its own inverse distance. The implicit function theorem and the lower nondegeneracy proved above give (11.8).

Put $M = 2m + \frac{1}{4}$ and

$$\Psi = L_3 + \frac{2m(u + v)}{2 + M}.$$

The centre-of-mass identity and the exact row $E_4 = C + \nu L_3 + T^3 F_3 + T^5 G_3 = 0$ give

$$X + Y = (2 + M)(\nu\Psi + T^3 F_3 + T^5 G_3).$$

Since $X + Y = 2x = -\frac{3}{2}h_* + O(\mu + h_*^2)$, first $h_* = O(\nu + T^3 + \mu)$ and then, by (11.8),

$$h_* = h_\nu \nu + O(T^3 + \nu^2 + \mu), \qquad h_\nu = -\tfrac{2}{3}(2 + M)\Psi(x_0).$$

If $h_\nu = 0$, then $h_*^2 = O(T^6 + \nu^4 + \mu^2) = o(\mu)$, contradicting (11.5). If $h_\nu \neq 0$, then $h_*^2 = h_\nu^2\nu^2 + o(\mu + \nu^2)$. Equation (11.5) forces $\operatorname{ord}\eta = \frac{1}{2}\operatorname{ord} a$ and gives

$$-a_0^2 + \tfrac{3}{16}h_\nu^2\eta_0^4 = 0, \tag{11.9}$$

where $a_0, \eta_0 \neq 0$ are the leading coefficients.

The crossed signs in the statement give $(\tau_1 - \tau_3)(x_0) = 0$. By (11.8), $T^3(\tau_1 - \tau_3) = O(T^3(\nu + \mu + |h_*|))$. Since $\operatorname{ord}\eta = \frac{1}{2}\operatorname{ord} a$, we have

$$T^3(\nu + \mu + |h_*|) = o(a^6\eta^2).$$

Equations (11.6) and (11.7) therefore give the second balance

$$-4a_0^2 + \tfrac{15}{4}h_\nu^2\eta_0^4 = 0.$$

Together with (11.9), this gives

$$\begin{pmatrix} -1 & 3/16 \\ -4 & 15/4 \end{pmatrix}\begin{pmatrix} a_0^2 \\ h_\nu^2\eta_0^4 \end{pmatrix} = 0, \qquad \det\begin{pmatrix} -1 & 3/16 \\ -4 & 15/4 \end{pmatrix} = -3.$$

Thus $a_0^2 = 0$, contradicting $a_0 \neq 0$ and excluding the germ. □

Theorem 11.8. *For $m \in \mathbb{R}^5_{>0}$, no curve in A_m carries a Diagram 8 germ, real or complex. In particular, Theorem 4.1 (a) holds for $r = 8$.*

Proof. By the symmetries at the beginning of this section we may use the numbering of [AK12, §8.8]. Let γ be a Diagram 8 germ at $m \in \mathbb{R}^5_{>0}$. By Proposition 11.3 it converges in the chart with order ratio $r \in (0, \infty)$, and by Proposition 11.4 the masses satisfy (11.1) and the limit has signs s_{bc}. The case $r > 2$, equivalently $0 < p < 2$, is excluded by Lemma 11.5. If $r \leq 2$, equivalently $-4 < p \leq 0$, Lemma 11.6 gives $m_1 = m_2 = 4m_5$, $m_3 = m_4$, $s_{13} = -s_{24}$ and $s_{14} = -s_{23}$. These masses and signs are excluded by Lemma 11.7. The two ranges exhaust $r > 0$, proving the theorem. □

A Computational statements used in the proof

Table 4 lists the verification functions used for the computer-assisted steps in the proofs of Diagrams 2, 3 and 6.

Computational step	Verification functions
Lemma 6.2, Definition 6.3	`verify_chart_and_boundary_fibre`, `verify_shape_resultant`, `verify_mass_resultants`
Lemma 6.12	`verify_L4_exclusion`, `verify_K_and_L5_exclusions`
Lemma 6.13: jets, minors and boundary-point orbits	`verify_ordinary_seed`, `verify_fold_seed`, `verify_equal_mass_labels`
Lemmas 6.14, 6.15	`verify_mass_rays`, `verify_ray_responses`
Lemma 6.16	`verify_K1_branches`, `verify_K4_arcs`, `verify_K5_mass_ray`, `verify_all_K5_labels`
Lemmas 6.17, 6.18	`verify_S5_response`, `verify_regular_X`, `verify_regular_Euler`, `verify_regular_Lagrange`, `verify_general_fold`
Lemmas 7.5, 7.7	`verify_order_three`, `verify_order_four`
Lemma 7.10: jet and product identities	`verify_fifth_order_jet`, `verify_nine_root_norm`
Lemma 7.11: 144 excluded boundary points	`verify_modular_order_six`
Diagram 6 atlas and orbit coverage	`verify_atlas_substitution_and_counts`, `verify_atlas_inverse_elimination`, `verify_symmetry_orbits`
Lemma 9.8: rank and leading jets	`verify_selected_rank`, `verify_leading_jets`, `verify_mixed_norm`, `verify_same_shape_norm`, `verify_opposite_norms`

Table 4: Functions used for computational verification.

Verification code

The source code and instructions for running these functions are available at https://github.com/hitchenzhe/smale-n5-paper.

Declaration of AI use

This manuscript was prepared with assistance from ChatGPT (OpenAI, GPT-6), including computational verification and the development of Lean 4 proofs. We provide a Lean 4 formalization of key algebraic and analytic components of the argument, comprising 20 modules and 121 machine-checked auxiliary theorems. The formalization is publicly available at https://github.com/ZhichengTong/smale5-lean. The second author has completed a manual verification of the manuscript.

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References

[ACS12] A. Albouy, H. E. Cabral, A. A. Santos, Some problems on the classical n-body problem. Celestial Mech. Dynam. Astronom. 113 (2012), no. 4, pp. 369–375. https://doi.org/10.1007/s10569-012-9431-1

[AFS08] A. Albouy, Y. Fu, S. Sun, Symmetry of planar four-body convex central configurations. Proc. R. Soc. A 464 (2008), pp. 1355–1365. https://doi.org/10.1098/rspa.2007.0320

[AK12] A. Albouy, V. Kaloshin, Finiteness of central configurations of five bodies in the plane. Ann. of Math. (2) 176 (2012), no. 1, pp. 535–588. https://doi.org/10.4007/annals.2012.176.1.10

[AS25] A. Albouy, J. Sun, On the eigenvalues of a central configuration. Preprint, 2025. https://doi.org/10.48550/arXiv.2512.00661

[BCR98] J. Bochnak, M. Coste, M.-F. Roy, Real Algebraic Geometry. Ergeb. Math. Grenzgeb. (3), vol. 36, Springer, 1998. https://doi.org/10.1007/978-3-662-03718-8

[CC24] K.-M. Chang, K.-C. Chen, Toward finiteness of central configurations for the planar six-body problem by symbolic computations. (I) Determine diagrams and orders. J. Symbolic Comput. 123 (2024), Article 102277. https://doi.org/10.1016/j.jsc.2023.102277

[CC25] K.-M. Chang, K.-C. Chen, Toward finiteness of central configurations for the planar six-body problem by symbolic computations. (II) Determine mass relations. SIAM J. Appl. Dyn. Syst. 24 (2025), no. 3, pp. 2369–2404. https://doi.org/10.1137/24M1716070

[CC26] K.-M. Chang, K.-C. Chen, Rhomboidal central configurations of the five-body problem. Regul. Chaotic Dyn. 31 (2026), no. 3, pp. 468–483. https://doi.org/10.1134/S1560354725560011

[CH12] K.-C. Chen, J.-S. Hsiao, Convex central configurations of the n-body problem which are not strictly convex. J. Dynam. Differential Equations 24 (2012), pp. 119–128. https://doi.org/10.1007/s10884-011-9233-2

[CH18] K.-C. Chen, J.-S. Hsiao, Strictly convex central configurations of the planar five-body problem. Trans. Amer. Math. Soc. 370 (2018), pp. 1907–1924. https://doi.org/10.1090/tran/7010

[CT26] Z. Chen, Z. Tong, Finiteness of planar central configurations of six bodies for all positive masses: Part I, exceptional-mass first limits. Preprint, 2026. https://doi.org/10.48550/arXiv.2610.23329

[DH23] Y. Deng, M. Hampton, Equilateral chains and cyclic central configurations of the planar 5-body problem. J. Nonlinear Sci. 33 (2023), Article 4. https://doi.org/10.1007/s00332-022-09864-z

[FGMS26] A. C. Fernandes, B. A. Garcia, L. F. Mello, L. R. dos Santos, Some convex central configurations in the planar five-body problem. Z. Angew. Math. Phys. 77 (2026), Article 131. https://doi.org/10.1007/s00033-026-02777-x

[Ham05] M. Hampton, Stacked central configurations: new examples in the planar five-body problem. Nonlinearity 18 (2005), no. 5, pp. 2299–2304. https://doi.org/10.1088/0951-7715/18/5/021

[Ham24] M. Hampton, "Twist vectors" for central configuration equations. Celestial Mech. Dynam. Astronom. 136 (2024), Article 50. https://doi.org/10.1007/s10569-024-10222-2

[HJ11] M. Hampton, A. N. Jensen, Finiteness of spatial central configurations in the five-body problem. Celestial Mech. Dynam. Astronom. 109 (2011), pp. 321–332. https://doi.org/10.1007/s10569-010-9328-9

[HJ15] M. Hampton, A. N. Jensen, Finiteness of relative equilibria in the planar generalized N-body problem with fixed subconfigurations. J. Geom. Mech. 7 (2015), no. 1, pp. 35–42. https://doi.org/10.3934/jgm.2015.7.35

[HM06] M. Hampton, R. Moeckel, Finiteness of relative equilibria of the four-body problem. Invent. Math. 163 (2006), pp. 289–312. https://doi.org/10.1007/s00222-005-0461-0

[HLO20] X. Hu, Y. Long, Y. Ou, Linear stability of the elliptic relative equilibrium with $(1+n)$-gon central configurations in planar n-body problem. Nonlinearity 33 (2020), no. 3, pp. 1016–1045. https://doi.org/10.1088/1361-6544/ab5927

[HOS25] X. Hu, Y. Ou, J. Sun, Quantitative linear stability analysis of elliptic relative equilibria in the planar N-body problem. Preprint, 2025. https://doi.org/10.48550/arXiv.2509.09809

[HOT23] X. Hu, Y. Ou, X. Tang, Linear stability of an elliptic relative equilibrium in the spatial n-body problem via index theory. Regul. Chaotic Dyn. 28 (2023), nos. 4–5, pp. 731–755. https://doi.org/10.1134/S1560354723040135

[HOY21] X. Hu, Y. Ou, G. Yu, An index theory for collision, parabolic and hyperbolic solutions of the Newtonian n-body problem. Arch. Ration. Mech. Anal. 240 (2021), pp. 565–603. https://doi.org/10.1007/s00205-021-01619-6

[JL25] A. N. Jensen, A. Leykin, Smale's 6th problem for generic masses. Journal of Experimental Mathematics 1 (2025), no. 2, pp. 306–314. https://doi.org/10.56994/JXM.001.002.006

[LS09] T.-L. Lee, M. Santoprete, Central configurations of the five-body problem with equal masses. Celestial Mech. Dynam. Astronom. 104 (2009), pp. 369–381. https://doi.org/10.1007/s10569-009-9219-0

[LL26a] K. Liu, Y. Liu, Finiteness of planar central configurations of five bodies with positive masses. Preprint, 2026. https://doi.org/10.48550/arXiv.2610.28168

[LL26b] K. Liu, Y. Liu, Finiteness of Dziobek central configurations in the Newtonian five-body problem. Preprint, 2026. https://doi.org/10.48550/arXiv.2610.32751

[LX26] Y. Liu, Z. Xie, Symmetric central configurations in the concave 4-body problem with two pairs of equal masses. Discrete Contin. Dyn. Syst. 55 (2026), pp. 243–262. https://doi.org/10.3934/dcds.2026085

[MZ19] M. Moczurad, P. Zgliczyński, Central configurations in planar n-body problem with equal masses for $n = 5, 6, 7$. Celestial Mech. Dynam. Astronom. 131 (2019), Article 46. https://doi.org/10.1007/s10569-019-9920-6

[MZ26] M. Moczurad, P. Zgliczyński, Central configurations with unequal masses: finiteness in several exceptional cases of five bodies. Preprint, 2026. https://doi.org/10.48550/arXiv.2601.01165

[Moe01] R. Moeckel, Generic finiteness for Dziobek configurations. Trans. Amer. Math. Soc. 353 (2001), no. 11, pp. 4673–4686. https://doi.org/10.1090/S0002-9947-01-02828-8

[Mou10] F. R. Moulton, The straight line solutions of the problem of n bodies. Ann. of Math. (2) 12 (1910), pp. 1–17. https://doi.org/10.2307/2007159

[NWW25] J. Natrup, Q. Wang, Y. Wang, Finiteness of non-degenerate central configurations of the planar n-body problem with a homogeneous potential. Preprint, 2025. https://doi.org/10.48550/arXiv.2502.20228

[OP25] Y. Ou, A. Portaluri, Morse index for homothetic motions in the gravitational n-body problem. J. Fixed Point Theory Appl. 27 (2025), Article 10. https://doi.org/10.1007/s11784-025-01161-y

[OS23] Y. Ou, J. Sun, Linear stability of the elliptic relative equilibrium with $(1 + 7)$-gon central configuration in planar N-body problem. Discrete Contin. Dyn. Syst. 43 (2023), no. 2, pp. 559–567. https://doi.org/10.3934/dcds.2022162

[OW25] Y. Ou, Y. Wang, Spectral instability of the regular n-gon elliptic relative equilibrium in the planar n-body problem. Preprint, 2025. https://doi.org/10.48550/arXiv.2510.26211

[Rob99] G. E. Roberts, A continuum of relative equilibria in the five-body problem. Phys. D 127 (1999), pp. 141–145. https://doi.org/10.1016/S0167-2789(98)00315-7

[Rob25] G. E. Roberts, On kite central configurations. Nonlinearity 38 (2025), no. 7, Article 075001. https://doi.org/10.1088/1361-6544/add834

[San21a] M. Santoprete, On the uniqueness of co-circular four body central configurations. Arch. Ration. Mech. Anal. 240 (2021), pp. 971–985. https://doi.org/10.1007/s00205-021-01626-7

[San21b] M. Santoprete, On the uniqueness of trapezoidal four-body central configurations. Nonlinearity 34 (2021), pp. 424–437. https://doi.org/10.1088/1361-6544/abbe61

[Shu71] M. Shub, Appendix to Smale's paper: Diagonals and relative equilibria. In: N. H. Kuiper (ed.), Manifolds—Amsterdam 1970, Lecture Notes in Math., vol. 197, Springer, 1971, pp. 199–201. https://doi.org/10.1007/BFb0068619

[Sma98] S. Smale, Mathematical problems for the next century. Math. Intelligencer 20 (1998), no. 2, pp. 7–15. https://doi.org/10.1007/BF03025291

[SXY23] S. Sun, Z. Xie, P. You, On the uniqueness of convex central configurations in the planar 4-body problem. Regul. Chaotic Dyn. 28 (2023), nos. 4–5, pp. 512–532. https://doi.org/10.1134/S1560354723520076

[SXY25] S. Sun, Z. Xie, P. You, Degeneracy of planar central configurations in the N-body problem. Preprint, 2025. https://doi.org/10.48550/arXiv.2510.25649

[TZ24] Z. Tang, S. Zhu, Perturbing masses: A study of centered co-circular central configurations in power-law n-body problems. Phys. D 461 (2024), Article 134115. https://doi.org/10.1016/j.physd.2024.134115

[Tom26] T. S. Tomar, Smale's sixth problem for five bodies in the plane, with a proof formalized in Lean. Preprint, 2026. https://doi.org/10.48550/arXiv.2610.13100

[Win41] A. Wintner, The Analytical Foundations of Celestial Mechanics. Princeton Math. Ser., vol. 5, Princeton University Press, 1941.

[Xia91] Z. Xia, Central configurations with many small masses. J. Differential Equations 91 (1991), no. 1, pp. 168–179. https://doi.org/10.1016/0022-0396(91)90137-X

[YZ26] X. Yu, S. Zhu, On the finiteness of four-body central configurations. Ann. Sc. Norm. Super. Pisa Cl. Sci. (5), to appear. https://doi.org/10.2422/2036-2145.202501_008